

ANSI/IES LS-1-22

Nomenclature and Definitions for Illuminating Engineering

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PREFACE

(This preface is not part of American National Standard ANSI/IESNA LS-1-22.)

One of the first undertakings of the Illuminating Engineering Society was to clarify and standardize the nomenclature in its field. The Society's Committee on Nomenclature and Standards published a provisional set of definitions in 1910. This subsequently was revised and enlarged nine times during the following 32 years. The reports of 1922, 1925, 1932, and 1942 were approved as American Standards by the American Standards Association, and the IESNA was designated as sponsor for future revisions under the procedure of the Association.

The last American National Standard *Nomenclature and Definitions for Illuminating Engineering* (RP-16) was completed and approved by the Society and by the American National Standards Institute (ANSI) in 2010 and reaffirmed in 2017.

Since the last revisions, advances in lighting technology have given birth to new lighting terminology. Better measurement techniques have led to more international agreement in fundamental units and constants used in basic laws of physics. There is greater use of SI units today in illuminating engineering. This standard reflects the above with several new terms and definitions and revisions in existing definitions submitted through the continuous maintenance process.

The IES Nomenclature Committee, which has been primarily responsible for this revision, acknowledges the material contributions of the technical committees of the Illuminating Engineering Society in providing suggested terms and definitions pertinent to their specialized activities.

1.0 INTRODUCTION

The terms defined in this Recommended Practice are in alphabetical order *within each section*, in the sequence listed in the **Table of Contents**. Individual terms may be found by using the Index [online only].

Throughout this document, where the terms (*CIE*) and *Commission Internationale de l'Eclairage* are used, they all refer to the International Commission on Illumination.

In context of this document, *SI* refers to the International System of Units (abbreviated SI from the French *Le Système International d'Unités*) – a measurement system used throughout the world and commonly referred to as the *metric system*. Public Law 100-418 (US) designated the metric system as the preferred system of weights and measures for the United States.

1.1 Scope

Illuminating engineering, strictly speaking, comprises the production, measurement and application of light, or radiant energy within the limits of the visual spectrum. Since, however, most light sources furnish radiant energy also in the adjoining infrared and ultraviolet regions, and since many lamp-type devices are used for the production of radiant energy in these regions, it is customary to include the infrared and ultraviolet within the province of the illuminating engineer.

1.2 Spectral Parameters

There are four spectral parameters that are widely used in different disciplines. The spectral parameter used in illuminating engineering is almost always wavelength, λ , in units of nanometer ($\text{nm} = 10^{-9} \text{ m}$), micrometer ($\mu\text{m} = 10^{-6} \text{ m}$) or Ångström* ($\text{Å} = 10^{-10} \text{ m}$). The name *wavelength* and symbol λ will be used as the spectral parameter in this publication except where otherwise specifically indicated. The other spectral parameters, their symbols, and equations for computing them from wavelengths in nanometers are as follows:

Frequency, [†] ν (s^{-1} , or Hz)	$\nu = c \times 10^9 / \lambda$
Wavenumber, σ (cm^{-1})	$\sigma = (\lambda)^{-1} \times 10^7$
Photon energy, q (J)	$q = h \times \nu \times 10^9 / \lambda$

where:

c = the speed of light in vacuum, 299,792,458 m/s

*Deprecated unit.

[†]Conventionally, ν is the symbol for spectral frequency, and f is the symbol for time frequency.

h = the Planck constant of action, 6.626176×10^{-34} J·s

When a parameter symbol, e.g., λ , is used as a subscript to a radiometric quantity symbol, it denotes the parameter concentration of that quantity; e.g., spectral radiant flux, $\Phi_\lambda = d\Phi/d\lambda$. To denote a distribution, it is also shown as a function of that parameter; e.g., the spectral distribution of radiant flux, $\Phi_\lambda(\lambda)$. Properties, such as reflectance or emissivity, also vary with, and may be shown as functions of, the radiation parameters; e.g., directional emissivity $\varepsilon(\theta, \phi)$, or spectral transmittance $\tau(\lambda)$.

Since they are not distributions, but only weighting functions, the use of parameter subscripts to property symbols may be misleading and is accordingly deprecated. The value of a function (distribution or weighting function) at a particular parameter value is denoted by replacing the parameter symbol in the function notation by that value, e.g., spectral flux at 555 nanometers would be: $\Phi_\lambda(555 \text{ nm})$.

Note: In addition to the spectral parameter, the other radiation parameters that together fully characterize the distribution of radiant flux in the domain of ray optics are: the spatial parameters of position (x, y for a point on a surface or x, y, z for a point in space) and direction (θ, ϕ), where θ = the angle from a reference axis and ϕ = the azimuth angle from a reference direction perpendicular to that axis); time (t) or frequency ($f < \nu$) of fluctuation or modulation; and polarization (e.g., the Stokes parameters¹).

2.0 ELECTROMAGNETIC RADIATION*

2.1 optical radiation

Electromagnetic radiation having wavelengths between approximately 100 nanometers and one millimeter.

2.2 electromagnetic spectrum

A continuous range of electric and magnetic radiation encompassing all wavelengths (or frequencies). (See *regions of the electromagnetic spectrum*.)

2.3 radiant energy, Q

Energy traveling in the form of electromagnetic waves. It is measured in units of energy such as joules or kilowatt-hours.

*See **Table 13.2** for a summary of the units, symbols, and defining equations.

2.3.1 spectral radiant energy

$$Q_\lambda = dQ/d\lambda$$

Radiant energy per unit wavelength interval; e.g., joules per nanometer. $Q_\lambda(\lambda) = dQ/d\lambda$ at wavelength λ .

2.4 photon

A quantum of radiant energy (see **Section 1.2 Spectral Parameters**).

2.5 radiant energy density

$$w = dQ/dV$$

Radiant energy per unit volume: e.g., joules per cubic meter.

2.6 radiant flux (radiant power)

$$\Phi = dQ/dt$$

The time rate of flow of radiant energy. It is expressed preferably in watts.

2.6.1 spectral radiant flux

$$\Phi_\lambda = d\Phi/d\lambda$$

Radiant flux per unit wavelength interval at wavelength λ ; e.g., watts per nanometer.

2.6.2 radiant efficiency

(of a source of radiation) (η_e, η)

Ratio of the radiant flux of the emitted radiation to the power consumed by the source.²

Note: It must be specified whether or not the power dissipated by any auxiliary equipment such as ballasts, if any, is included in the power consumed by the source.

2.6.3 total spectral radiant flux

Spectral radiant flux emitted from a source, integrated over all directions (4π sr).

Units: W/nm

2.6.4 radiant flux maintenance

The ratio of the radiant flux of an electric light source at a given time in its operational life to its initial radiant flux.

Notes:

1. Radiant flux maintenance is usually expressed as a percentage.
2. For the purpose of measurement, the electric light source is assumed to be operated under specified controlled conditions. In field measurements, the value may vary due to uncontrolled conditions.

2.7 radiant flux areal density

$d\Phi/dA$, (at a point on a surface)

The quotient of the radiant flux incident on or emitted by an element of surface area at the point, by the area of the element. Radiant flux density emitted from a surface has been called *emittance*.^{*} The preferred term for radiant flux density leaving a surface is *exitance*, (M). Radiant flux density incident on a surface is *irradiance*, (E).

2.7.1 spectral radiant exitance, (M_λ), and irradiance, (E_λ)

Spectral concentration of radiant exitance, $M_\lambda = dM/d\lambda$

Spectral concentration of irradiance, $E_\lambda = dE/d\lambda$

2.7.2 radiant exitance

(See Section 2.7 radiant flux areal density.)

2.7.3 radiant flux density at a surface

(See Section 2.7 radiant flux areal density.)

2.8 radiant intensity

$I = d\Phi/d\omega$ (in a given direction)

The radiant flux proceeding from a source per unit solid angle in the given direction; e.g., watts per steradian.

Note: Mathematically, a solid angle must have a point at its apex; the definition of *radiant intensity*, therefore, applies strictly only to a point source. In practice, however, radiant energy emanating from a source whose dimensions are negligible in comparison with the distance from which it is observed may be considered as coming from a point. Specifically, this implies that with change of distance, 1) the variation in solid angle subtended by the source at the receiving point approaches $1/(\text{distance})^3$; and 2) the average radiance of the projected source area as seen from the receiving point does not vary

^{*} Deprecated.

appreciably.

2.8.1 spectral radiant intensity

$$I_{\lambda} = dI/d\lambda$$

Radiant intensity per unit wavelength interval; e.g., watts per steradian-nanometer.

2.9 radiance, L

$$L = d^2\Phi/[d\omega(dA \cos\theta)] = dI/(dA \cos\theta) *$$

(in a given direction at a point on the surface of a source, of a receiver, or of any other real or virtual surface)

The quotient of the radiant flux leaving, passing through, or arriving at an element of the surface surrounding the point, and propagated in directions defined by an elementary cone containing the given direction, by the product of the solid angle of the cone and the area of the orthogonal projection of the element of the surface on a plane perpendicular to the given direction.

Note: In the defining equation, θ is the angle between the normal to the element of the surface and the given direction.

2.9.1 spectral radiance, L_{λ}

Spectral concentration of radiance.

$$L_{\lambda} = d^3\Phi/[d\omega(dA \cos\theta)d\lambda]$$

2.10 radiant fluence, $F(\Delta t)$ (for a time interval Δt , at a point in space)

The omnidirectional radiant energy externally incident on an elementary sphere about the point, per unit cross-sectional area of the sphere.

$$F(\Delta t) = dQ/da = \int_{\Delta t} \int_{4\pi} L d\omega dt,$$

where da is the cross-sectional area of the sphere.

2.10.1 spectral radiant fluence, $F_{\lambda}(\Delta t)$

Spectral concentration of fluence.

* Properly, this should be a second partial derivative since area and solid angle are independent variables; however, the symbol "d" is used due to the convenience in printing and typing. For this specific use, no possible errors or confusion are foreseen. This practice is in accordance with the International Lighting Vocabulary (CIE S 017/E 2020) and the practice of the national Institute of Standards and Technology.

$$F_{\lambda}(\Delta t) = dF(\Delta t)/d\lambda$$

2.11 radiant fluence rate, F_t

(at a point in space)

The omnidirectional radiant flux externally incident on an elementary sphere, per unit cross-sectional area of the sphere.

$$F_t = d\Phi/dA = \int_{4\pi} L d\omega,$$

where dA is the cross-sectional area of the sphere.

2.11.1 spectral radiant fluence rate, $F_{t,\lambda}$

Spectral concentration of fluence rate.

$$F_{t,\lambda} = dF_t/d\lambda$$

2.12 radiant steriscent, $L^*(x)$

(at a point along a ray path)

Rate of increase in radiance, per unit path length, at the point and in the direction of the ray, due to "generated" (emitted or scattered) radiance, or the "generated" radiant intensity per unit volume, at the point and in the direction of the ray, by which a distributed source can be characterized.

$$L^* = dL_g/dx = dI_g/dV,$$

where dx is an element of distance along the ray path, dV is an element of volume at the point, and the subscript g denotes a "generated" quantity.

Note: This quantity is useful in dealing with radiative transfer through a region where significant emission or scattering into the ray, as well as attenuation by absorption or scattering out of the ray, occurs along a ray path; also for evaluating the intensity of an extended volume of emitting and partially transmitting (transparent) material. This is discussed in more detail on pp. 41-45 of Reference 3.

2.12.1 Spectral radiant steriscent, L^*_{λ}

Spectral concentration of steriscent.

$$L^*_{\lambda} = dL^*/d\lambda.$$

2.13 radiator

An emitter of radiant energy.

2.13.1 temperature radiator

An ideal radiator whose radiant flux density (radiant exitance) is determined by its temperature and the material and character of its surface, and is independent of its previous history.

2.13.2 blackbody

A temperature radiator of uniform temperature whose radiant exitance in all parts of the spectrum is the maximum obtainable from any temperature radiator at the same temperature. Such a radiator is called a blackbody because it will absorb all the radiant energy that falls upon it. All other temperature radiators may be classed as non-blackbodies. They radiate less in some or all wavelength intervals than a blackbody of the same size and the same temperature.

Note: The blackbody is practically realized over limited solid angles in the form of a cavity with opaque walls at a uniform temperature and with a small opening for observation purposes. It is variously called a *standard radiator*, an *ideal radiator*, or a *complete radiator*.

2.13.3 Stefan-Boltzmann law

The statement that the radiant exitance (M) or radiance (L) of a blackbody radiator is proportional to the fourth power of its absolute temperature; that is, $M = \sigma T^4$ or $L = \sigma_L T^4$.

Note: The currently recommended value⁴ of σ is $5.67032 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$ and that of σ_L is:

$$1.80492 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{sr} \cdot \text{K}^4)$$

2.13.4 Planck radiation law

An expression representing the spectral radiance of a blackbody as a function of the wavelength and temperature. This law commonly is expressed by the formula:

$$L_\lambda = dI_\lambda/dA' = c_{1L} \lambda^{-5} [e^{(c_2/\lambda T)} - 1]^{-1},$$

where:

L_λ = the spectral radiance

dI_λ = the spectral radiant intensity

dA' = the projected area ($dA \cos \theta$) of the aperture of the blackbody

e = the base of natural logarithms (approx. 2.71828)

T = absolute temperature

c_{1L} and c_2 are constants designated as the first and second radiation constants.

Note: The designation c_{1L} is used to indicate that the equation in the form given here refers to

the radiance L , or to the intensity I per unit projected area A' , of the source. Numerical values are commonly given not for c_{1L} but for c_I , which applies to the total flux radiated from a blackbody aperture (that is, in a hemisphere [2π steradians]) so that, with the Lambert cosine law taken into account, $c_I = \pi c_{1L}$.

The currently recommended value of c_I is $3.741832 \times 10^{-16} \text{ W}\cdot\text{m}^2$, or $3.741832 \times 10^{-12} \text{ W}\cdot\text{cm}^2$.

Then c_{1L} is $1.191062 \times 10^{-16} \text{ W}\cdot\text{m}^2/\text{sr}$, or $1.191062 \times 10^{-12} \text{ W}\cdot\text{cm}^2/\text{sr}$.

If, as is more convenient, wavelengths are expressed in micrometers and area in square centimeters, $c_{1L} = 1.191062 \times 10^4 \text{ W}\cdot\mu\text{m}^4/(\text{cm}^2\cdot\text{sr})$, with L_λ being given in $\text{W}/(\text{cm}^2\cdot\text{sr}\cdot\mu\text{m})$.

The currently recommended value⁵ of c_2 is $1.438786 \times 10^{-2} \text{ m}\cdot\text{K}$.⁴

The Planck law in the following form gives the energy radiated from the blackbody in a given wavelength interval $(\lambda_1 - \lambda_2)$:

$$Q = \int_{\lambda_1}^{\lambda_2} Q_\lambda d\lambda$$

$$Q = Atc_1 \int_{\lambda_1}^{\lambda_2} \lambda^{-5} [e^{(c_2/\lambda T)} - 1]^{-1} d\lambda$$

If A is the area of the radiation aperture or surface in square centimeters, t is time in seconds, λ is wavelength in micrometers, and $c_I = 3.741832 \times 10^4 \text{ W}\cdot\mu\text{m}^4/\text{cm}^2$, then Q is the total energy in watt-seconds (joules) emitted from this area (that is, in the solid angle 2π) in time t within the wavelength interval $(\lambda_1 - \lambda_2)$.

2.13.5 Wien radiation law

An expression representing approximately the spectral radiance of a blackbody as a function of its wavelength and temperature. It is commonly expressed by the formula:

$$L_\lambda = \frac{dI_\lambda}{d'A} = c_{1L} \lambda^{-5} e^{-(c_2/\lambda T)}$$

This formula is accurate to one percent or better for values of λT less than 3,000 micrometer-kelvins.

2.13.6 Wien displacement law

An expression representing, in a functional form, the spectral radiance of a blackbody as a function of the wavelength and the temperature.

$$L_\lambda = dI_\lambda/dA' = c_{1L}(\lambda)^{-5} \cdot f(\lambda T)$$

or

$$L_{\lambda} T^{-5} = c_{1L} (\lambda T)^{-5} \cdot f(\lambda T)$$

$$f(\lambda T) = e^{-(c_2/\lambda T)}$$

The two principal corollaries of this law are:

$$\lambda_m T = b$$

$$L_m / T^5 = b'$$

These show how the maximum spectral radiance L_m and the wavelength λ_m at which it occurs are related to the absolute temperature T .

Note: The currently recommended value of b is $2.8978 \times 10^{-3} \text{ m}\cdot\text{K}$, or $2.8978 \times 10^{-1} \text{ cm}\cdot\text{K}$. From the Planck radiation law, and with the use of the value of b , c_1 , and c_2 as given in that definition, b' is found to be:

$$4.0956 \times 10^{-14} \text{ W/cm}^2\cdot\text{sr}\cdot\mu\text{m}\cdot\text{K}^5$$

2.13.7 graybody

A temperature radiator whose spectral emissivity is less than unity and the same at all wavelengths.

2.13.8 emissivity, ϵ

2.11.8.1 spectral-total directional emissivity, $\epsilon(\theta, \phi, T)$

(at a point on the surface of a thermal radiator and in a given direction)

The ratio of the radiance of the thermal radiator at temperature T , at the point and in the given direction, to that of a blackbody at the same temperature T .

$$\epsilon(x, y, \theta, \phi, T) = L(x, y, \theta, \phi, T) / L_{bb}(T),$$

where:

x, y = the coordinates of the point

θ, ϕ defines the direction

2.11.8.2 spectral-total hemispherical emissivity, $\epsilon(x, y, 2\pi, T)$

(at a point on the surface of a thermal radiator)

The ratio of the radiant exitance of the thermal radiator at temperature T , at the given point on the surface, to that of a blackbody at the same temperature T .

$$\varepsilon(x, y, 2\pi, T) = (1/\pi) \int_{2\pi} \varepsilon(x, y, \theta, \varphi, T) \cos \theta \, d\omega$$

$$= (1/\pi) \int_0^\infty \int_{2\pi} \varepsilon(\lambda, x, y, \theta, \varphi, T) \cos \theta \, d\omega \, d\lambda$$

$$\equiv M(x, y, T) / M_{bb}(T)$$

2.13.8.3 spectral-directional emissivity, $\varepsilon(\lambda, \theta, \phi, T)$

(of an element of surface of a temperature radiator at a given wavelength and in a given direction)

The ratio of its spectral radiance at that wavelength and in the given direction to that of a blackbody at the same temperature and wavelength:

$$\varepsilon(\lambda, \theta, \phi, T) = L_\lambda(\lambda, \theta, \phi, T) / L_{\lambda, \text{blackbody}}(\lambda, T)$$

2.13.8.4 spectral-hemispherical emissivity, $\varepsilon(\lambda, T)$

(of an element of surface of an opaque temperature radiator that has an optically smooth surface)

The ratio of its spectral radiant exitance to that of a blackbody at the same temperature:

$$\varepsilon(\lambda, T) = (1/\pi) \int_{2\pi} \varepsilon(\lambda, \theta, \varphi, T) \cos \theta \, d\omega$$

$$\equiv M_\lambda(\lambda, T) / M_{\lambda, \text{blackbody}}(\lambda, T)$$

Note: Hemispherical emissivity is frequently called "total" emissivity. "Total" by itself is ambiguous, and should be avoided since it may also refer to "spectral-total" (all wavelengths) as well as "directional-total" (all directions). (See [Section 2.13.8.2](#).)

2.13.8.5 total emissivity

(See [Section 2.13.8.1 spectral-total directional emissivity](#) or [Section 2.13.8.2 spectral-total hemispherical emissivity](#).)

2.13.9 emittance, ε

The ratio of radiance in a given direction (for directional emittance) or radiant exitance (for hemispherical emittance) of a sample of a thermal radiator to that of a blackbody radiator at the same temperature.

2.13.10 ideal radiator

(See **Section 2.13.2 blackbody.**)

2.14 units of wavelength

The distance between two successive points of a periodic wave in the direction of propagation, in which the oscillation has the same phase. The three commonly used units are:

Name	Symbol	Value
micrometer (micron)*	μm	$1\ \mu\text{m} = 10^{-6}\ \text{m}$
nanometer	nm	$1\ \text{nm} = 10^{-9}\ \text{m}$
Ångstrom*	Å	$1\ \text{Å} = 10^{-10}\ \text{m}$

2.15 regions of the electromagnetic spectrum

For convenience of reference, the electromagnetic spectrum is arbitrarily divided as follows:

Vacuum Ultraviolet

Extreme ultraviolet	10 to 100 nm
Far ultraviolet	100 to 200 nm
Middle ultraviolet	200 to 300 nm
Near ultraviolet	300 to 400 nm
<i>Visible</i>	380 to 770 nm
<i>Infrared</i>	
Near (short wavelength) infrared	770 to 1,400 nm
Intermediate infrared	1,400 to 5,000 nm
Far (long wavelength) infrared	5,000 to 1,000,000 nm

Note: The spectral limits indicated above have been chosen as a matter of practical convenience. There is a gradual transition from region to region without sharp delineation. Also, the division of the spectrum is not unique. In various fields of science, the classifications may differ due to the phenomena of interest.

*Deprecated

2.15.1 ultraviolet subdivisions

Common subdivisions in the ultraviolet range between 100 nm and 400 nm as given by the International Commission on Illuminance (CIE)⁶ are:

UV-A	315 to 400 nm
UV-B	280 to 315 nm
UV-C	100 to 280 nm

2.16 photoeffect

An effect caused by optical radiation.

2.17 far field

Region of space sufficiently far from a radiating source that the inverse-square law can be applied with negligible error.

2.18 near field

Region of space too close to a radiating source for the inverse-square law to be applied.

2.19 irradiance (E)

(See Section 2.7 radiant flux areal density.)

2.20 nanometer (nm)

(See Section 1.2 spectral parameters.)

2.21 Mie scatter

Dispersion of electromagnetic radiation by particles that are larger than the wavelength of the radiation. Mie scatter is nearly wavelength independent (i.e., all wavelengths are scattered equally).

In Mie scatter, the amount of forward scatter (in the direction of propagation) is much greater than the amount of backward scatter. In the human eye, most of the scattered light is from Mie Scatter.

Also known as *large-particle scatter*.

2.22 Rayleigh scatter

Dispersion of electromagnetic radiation by particles much smaller than the wavelength of the radiation. The amount of scatter varies inversely as the fourth power of the wavelength, resulting in short wavelengths being scattered far more than longer wavelengths.

In Rayleigh scatter, the amounts of forward scatter (in the direction of propagation) and backward scatter are nearly equal.

Also known as *small-particle scatter*.

2.23 photon flux (Φ_p)

The number of photons, dN_p , emitted, transmitted, or received in an element of time, dt , divided by that element of time:

$$\Phi_p = \frac{dN_p}{dt}$$

Units: s^{-1}

Note 1: The photon flux emitted by a source in the wavelength interval from λ_1 to λ_2 can be calculated from the spectral radiant flux:

$$\Phi_p = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{hc} \Phi_\lambda(\lambda) d\lambda,$$

where:

h is Planck's constant

c is the speed of light

$\Phi_\lambda(\lambda)$ is the spectral radiant flux, measured in W/nm

Note 2: To convert to moles of photons, divide by Avogadro's number (N_A):

$$\Phi_p = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{N_A hc} \Phi_\lambda(\lambda) d\lambda$$

The unit for photon flux, Φ_p , is mol/s and is usually expressed in micromoles per second ($\mu\text{mol/s}$).

2.23.1 photon flux maintenance

The ratio of the photon flux of an electric light source at a given time in its operational life to its initial photon flux.

Notes:

1. Photon flux maintenance is usually expressed as a percentage.
2. For the purpose of measurement, the electric light source is assumed to be operated under specified controlled conditions. In field measurements, the value may vary due to uncontrolled conditions.

3.0 LIGHT*

3.1 light

Radiant energy that is capable of exciting the retina and producing a visual sensation in humans. The visible portion of the electromagnetic spectrum extends from about 380 to 780 nanometers.

Note: The subjective impression produced by stimulating the retina is sometimes designated as light. Visual sensations are sometimes arbitrarily defined as sensations of light, and in line with this concept it is sometimes said that light cannot exist until an eye has been stimulated. Electrical stimulation of the retina or the visual cortex is described as producing flashes of light. In illuminating engineering, however, light is a physical entity—radiant energy weighted by the luminous efficiency function. (See **Section 3.10.2** and **Section 3.10.1**.) It is a physical stimulus that can be applied to the retina.

3.1.1 phot-

Combining form meaning, broadly, within the visible spectrum. Combining forms are often used to build new compound words in science and technology.

3.2 luminous flux Φ

The time rate of flow of radiant energy, evaluated in terms of a standardized visual response.

$$\Phi_v = K_m \int \Phi_{e,\lambda} V(\lambda) d\lambda,$$

where:

Φ_v = lumens

$\Phi_{e,\lambda}$ = watts per nanometer

λ = nanometers

$V(\lambda)$ = the spectral luminous efficiency function

K_m = the maximum spectral luminous efficacy in lm/W

Unless otherwise indicated, the luminous flux is defined for photopic vision. For scotopic vision, the corresponding spectral luminous efficiency function, $V'(\lambda)$, and the corresponding maximum spectral luminous efficacy, K'_m , are substituted in the above equation. K_m and K'_m are derived from the basic SI definition of luminous intensity and have the values 683 lm/W and 1,700 lm/W, respectively. (Also see

*Symbols for radiometric terms are identified by the subscript *e*, and those for photometric (luminous) terms by the subscript *v*. In the event that no confusion will occur, the subscripts may be omitted. **Table T-2** summarizes the units, symbols, and defining equations.

Sections 3.4.1, 3.10, 3.10.2, and 3.10.3.)

3.2.1 lumen, lm

SI unit of luminous flux. Radiometrically, it is determined from the radiant power (see Section 3.2). Photometrically, it is the luminous flux emitted within a unit solid angle (one steradian) by a point source having a uniform luminous intensity of one candela.

3.3 luminous flux density at a surface, $d\Phi/dA$

The luminous flux per unit area at a point on a surface.

Note: This need not be a physical surface; it may also be a mathematical plane.

(See also Sections 3.3.1 and 3.3.2.)

3.3.1 Illuminance, $E = d\Phi/dA$

The areal density of the luminous flux *incident* at a point on a surface.

3.3.1.1 illumination

An alternative, but deprecated, term for *illuminance*. It is sometimes used because illuminance is subject to confusion with luminance and illuminants, especially when not clearly pronounced.

Note: The term *illumination* also is commonly used in a qualitative or general sense to designate the act of illuminating or the state of being illuminated. Usually, the context will indicate which meaning is intended, but occasionally it is desirable to use the expression *level of illumination* to indicate that the quantitative meaning is intended.

3.3.1.2 lux, lx

The SI unit of illuminance. One lux is one lumen per square meter (lm/m^2). (See Table T-3 for conversion values.)

3.3.1.3 footcandle, fc

A unit of illuminance. One footcandle is one lumen per square foot (lm/ft^2).

3.3.1.4 phot, ph

A unit of illuminance equal to one lumen per square centimeter. The use of this unit is deprecated.

3.3.1.5 level of illumination

(See Section 3.3.1 illuminance.)

3.3.2 luminous exitance, $M = d\Phi/dA$

The areal density of luminous flux leaving a surface at a point. Formerly, *luminous emittance* (deprecated).

Note: This is the total luminous flux emitted, reflected, and transmitted from the surface and is independent of direction.

3.3.2.1 Units of luminous exitance

Lumens per square meter (lm/m²) and lumens per square foot (lm/ft²) are preferred practice for the SI and U.S. Customary System (USCS, formerly English) systems respectively.

3.3.3 exitance

(See Section 3.3.2 luminous exitance and Section 2.7.1 spectral radiant exitance and irradiance.)

3.4 luminous intensity, I

(of a point source of light in a given direction)

$$I = d\Phi/d\omega$$

The luminous flux per unit solid angle in the direction in question. Hence, it is the luminous flux on a small surface centered on and normal to that direction, divided by the solid angle (in steradians) that the surface subtends at the source. Luminous intensity may be expressed in candelas (cd) or in lumens per steradian (lm/sr).

Note: Mathematically, a solid angle must have a point as its apex; the definition of luminous intensity, therefore, applies strictly only to a point source. In practice, however, light emanating from a source whose dimensions are negligible in comparison with the distance from which it is observed may be considered as coming from a point. Specifically, this implies that with change of distance: 1) the variation in solid angle subtended by the source at the receiving point approaches $1/(\text{distance})^2$, and 2) the average luminance of the projected source area as seen from the receiving point does not vary appreciably. (For extended sources, see Section 3.4.3.)

The word *intensity* as defined above is used to designate *luminous intensity* (or *candlepower*; see Section 3.4.2). It is also widely used in other ways, either informally or formally in other disciplines. *Stimulus intensity* may be used to designate the retinal illuminance of a proximal stimulus (see Section 4.1.2) or the luminance of a distal stimulus (see Section 4.1.1). *Intensity* is used in the same sense with

other modulates, such as *audition*. *Intensity* has been used to designate the level of illuminance on a surface, or the flux density in the cross-section of a beam of light. In physical optics, *intensity* usually refers to the square of the wave amplitude.

3.4.1 candela, cd

The SI unit of luminous intensity. One candela is one lumen per steradian (lm/sr). Formerly, *candle*.

Note: The fundamental luminous intensity definition in the SI is the candela.* The candela is the luminous intensity in a given direction of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian. The candela so defined is the base unit applicable to photopic quantities, scotopic quantities, and quantities to be defined in the mesopic domain.

From 1909 until January 1, 1948, the unit of luminous intensity in the United States, as well as in France and Great Britain, was the *international candle*, which was maintained by a group of carbon-filament vacuum lamps. For the present unit as defined above, the internationally accepted term is *candela*. The difference between the candela and the old international candle is so small that only measurements of high precision are affected.⁷

From 1948 to 1979, the unit of luminous intensity was defined in terms of a complete (blackbody) radiator. From this relationship, K_m and K'_m , and consequently the lumen, were determined (see **Section 3.2**). One candela was defined as the luminous intensity of 1/600,000 of one square meter of projected area of a blackbody radiator operating at the temperature of solidification of platinum and at a pressure of 101,325 newtons per square meter ($\text{N/m}^2 = \text{Pa}$).

3.4.2 candlepower, cp

Luminous intensity expressed in candelas (cd).

3.4.3 equivalent luminous intensity of an extended source at a specified distance

The intensity of a point source that would produce the same illuminance at that distance. Formerly, apparent luminous intensity of an extended source.

3.4.4 mean spherical luminous intensity, I_{ms}

Average value of the luminous intensity in all directions for a light source. Also, the quotient of

*Adopted at the *Seizième Conférence Générale des Poids et Mesures* on October 11, 1979 (the Sixteenth Conference on Weights and Measures).

the total emitted luminous flux of the source by 4π .

$$I_{ms} = \frac{1}{4\pi} \int_0^{4\pi} I \, d\omega = \frac{\Phi_{\text{total}}}{4\pi}$$

3.4.5 mean horizontal intensity (candlepower)*

The average intensity (candelas) of a lamp in a plane perpendicular to the axis of the lamp that passes through the luminous center of the lamp.

3.4.6 spherical reduction factor*

The ratio of the mean spherical luminous intensity to the mean horizontal intensity.

3.4.7 mean zonal candlepower

The average intensity (candelas) of a symmetrical luminaire or lamp at an angle to the luminaire axis or lamp axis that is in the middle of the zone under consideration.

3.4.8 solid angle, ω

A measure of that portion of space about a point bounded by a conic surface whose vertex is at the point. It is defined as the ratio of intercepted surface area of a sphere centered on that point to the square of the sphere's radius. It is expressed in steradians, sr.

3.4.9 intensity

(See **Section 3.4 luminous intensity** or **Section 2.8 radiant intensity**.)

3.5 luminance, L

(in a direction and at a point on a real or imaginary surface)

$$L = d^2\Phi / (d\omega \, dA \cos\theta)$$

The quotient of the luminous flux at an element of the surface surrounding the point, and propagated in directions defined by an elementary cone containing the given direction, by the product of the solid angle of the cone and the area of the orthogonal projection of the element of the surface on a plane perpendicular to the given direction. The luminous flux may be leaving, passing through, and/or arriving at the surface. Formerly, *photometric brightness*. (See **Figure 1**.)

*This term has been deprecated; it is retained here for reference of literature.

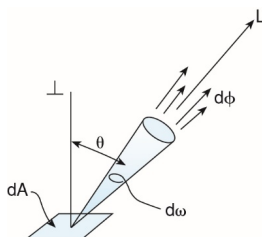


Figure 1.

By introducing the concept of luminous intensity (see **Section 3.4**), luminance may be expressed as:

$$L = dI / (dA \cos \theta)$$

Here, luminance at a point on a surface in a given direction is interpreted as the quotient of luminous intensity in the given direction produced by an element of the surface surrounding the point, by the area of the orthogonal projection of the element of the surface on a plane, perpendicular to the given direction.

Luminance may be measured at a receiving surface by using (see **Figure 2**):

$$L = dE / (d\omega \cos \theta)$$

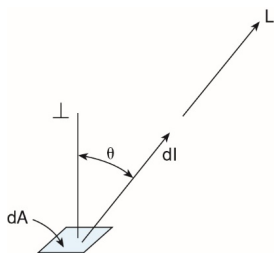


Figure 2.

This value may be less than the luminance of the emitting surface due to the attenuation by the transmitting media.) (See **Figure 3**.)

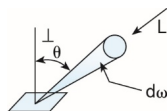


Figure 3.

Note: In common usage, the term *brightness* usually refers to the strength of sensation that results from viewing surfaces or spaces from which light comes to the eye. This sensation is determined in part by

the definitely measurable luminance defined above and in part by conditions of observation, such as the state of adaptation of the eye. (See **Section 5.9.10.**)

In much of the literature, the term *brightness*, used alone, refers to both luminance and sensation. The context usually indicates which meaning is intended. Previous usage notwithstanding, neither the term *brightness*, nor the term *photometric brightness* should be used to denote the concept of luminance.

3.5.1 luminance difference

The difference in luminance between two areas, such as the detail of a visual task and its immediate background, in which case it is quantitatively equal to the numerator in the formula for luminance contrast. (See note under **Section 3.5.**)

3.5.2 luminance threshold

The minimum perceptible difference in luminance for a given state of adaptation of the eye.

3.5.3 SI unit of luminance

Candela per square meter (cd/m^2); also, lumen per steradian-square meter [$\text{lm}/(\text{sr}\cdot\text{m}^2)$]. This unit is sometimes called the *nit*; the use of *nit* is deprecated.

3.5.4 U.S. Customary System (USCS, formerly English) unit of luminance

Candela per square foot (cd/ft^2). Also, lumen per steradian-square foot [$\text{lm}/(\text{sr}\cdot\text{ft}^2)$]. Another units include the candela per square inch (cd/in^2), and the lumen per steradian-square inch [$\text{lm}/(\text{sr}\cdot\text{in}^2)$]. (See *footlambert* [**Section 3.5.6.2.**])

Note: All of these terms have been deprecated. The SI units for luminance are candelas per square meter (cd/m^2), sometimes called *nits*.

3.5.5 stilb

A CGS (centimeter-gram-second) unit of luminance. One stilb equals one candela per square centimeter. The use of this term is deprecated.

3.5.6 Lambertian units of luminance

The luminance of a surface in a specified direction also has been expressed in terms of the luminous exitance the surface would have if the luminances in all directions, within the hemisphere on the side of the surface being considered, were the same as the luminance in the specified direction. In other words, *luminance* has been expressed as the luminous exitance of a Lambertian surface (see **Section 7.1.1.1**) whose luminance equals the luminance of the surface in question and in the specified direction.

Lambertian Luminance Unit		Lambertian Luminous Exitance		Equivalent Luminance
1 asb	=	1 lm/m ²	=	(1/π)cd/m ²
1 fL	=	1 lm/ft ²	=	(1/π)cd/ft ²
			=	(1/144π)cd/in ²
1 L	=	1 lm/cm ²	=	(1/π)cd/m ²

Note: The Lambertian units of luminance are numerically equal to the corresponding units of luminous exitance. Thus, the luminance (in Lambertian units) of a surface could be determined, and the numerical value directly used in the equation for a Lambertian reflecting surface relating illuminance and luminous exitance; i.e., $M = \rho E$, where ρ is the reflectance and M is the luminous exitance. In practice, no surface follows exactly the cosine formula of emission or reflection, and many do not even approximate it; hence, the luminance is not uniform but varies with the angle from which it is viewed. Since the *raison d'être* for this system was the use of a relationship of generally unknown and variable accuracy when applied to real surfaces, the use of Lambertian units of luminance has not been acceptable since 1967.⁸

3.5.6.1 apostilb, asb

A Lambertian unit of luminance equal to (1/π) candelas per square meter. This term is obsolete, and its use is deprecated.⁹

3.5.6.2 footlambert, fL

A Lambertian unit of luminance equal to (1/π) candelas per square foot. This term is obsolete, and its use is deprecated.⁹

3.5.6.3 lambert, L

A Lambertian unit of luminance equal to (1/π) candelas per square centimeter. This term is obsolete, and its use is deprecated.⁹

3.5.7 average luminance

Luminance is the property of a geometric ray. Luminance as measured by conventional meters is averaged with respect to two independent variables: area and solid angle. Both must be defined for a complete description of a luminance measurement.

3.5.8 nit (nt)

(See **Section 3.5.3 SI unit of luminance.**)

3.5.9 photometric brightness

(See Section 3.5 luminance.)

3.6 luminous fluence, $F(\Delta t)$

The omnidirectional luminous energy externally incident on an elementary sphere per unit cross-sectional area, in time Δt .

$$F(\Delta t) = dQ/da = \int_{\Delta t} \int_{4\pi} L d\omega dt ,$$

where da is the cross-sectional area of the sphere.

3.7 luminous fluence rate, F_t

The omnidirectional luminous flux externally incident on an elementary sphere per unit cross-sectional area of the sphere.

$$F_t = d\Phi/da = \int_{4\pi} L d\omega ,$$

where da is the cross-sectional area of the sphere.

3.8 luminous steriscent, $L^*(x)$

(at a point along a ray path)

Rate of increase in luminance, per unit path length, at the point and in the direction of the ray, due to "generated" (emitted or scattered) luminance; or the "generated" luminous intensity per unit volume, at the point and in the direction of the ray, by which a distributed source can be characterized.

$$L^* = dL_g/dx = dI_g/dV ,$$

where dx is an element of distance along the ray path, dV is an element of volume at the point, and the subscript g denotes a "generated" quantity.

Note: This quantity is useful in dealing with radiative transfer through a region where significant emission or scattering into the ray, as well as attenuation by absorption or scattering out of the ray, occurs along a ray path; also for evaluating the intensity of an extended volume of emitting or partially transmitting (transparent) material. (For discussion in more detail, see pp. 41-45 of NBS Technical Note 910-2.³)

3.9 quantity of light (luminous energy, Q)

$$Q = \int \Phi dt$$

The product of the luminous flux by the time it is maintained. It is the time integral of luminous flux.

3.9.1 lumen-second, lm·s

A unit of quantity of light, the SI unit of luminous energy (also called a *talbot*). It is the quantity of light delivered in one second by a luminous flux of one lumen.

3.9.2 talbot, T

[See Section 3.9.1 lumen-second.]

3.10 spectral luminous efficacy of radiant flux

$$K(\lambda) = \Phi_{v\lambda} / \Phi_{e\lambda}$$

The quotient of the luminous flux at a given wavelength by the radiant flux at that wavelength. It is expressed in lumens per watt. (See also Section 2.4 and 2.4.1.)

Note: This term formerly was called *luminosity factor*. The reciprocal of the maximum luminous efficacy of radiant flux is sometimes called “mechanical equivalent of light;” that is, the ratio between radiant and luminous flux at the wavelength of maximum luminous efficacy. The most probable value is 0.00146 W/lm, corresponding to 683 lm/W as the maximum possible luminous efficacy. For scotopic vision values (Table 13.7) the maximum luminous efficacy is 1,700 “scotopic lumens” per watt.

3.10.1 spectral luminous efficiency of radiant flux

The ratio of the luminous efficacy for a given wavelength to the value at the wavelength of maximum luminous efficacy. It is dimensionless.

Note: The term *spectral luminous efficiency* replaces the previously used terms *relative luminosity* and *relative luminosity factor*.

3.10.2 CIE photopic luminous efficiency function, $V(\lambda)$

The spectral luminous efficiency function for photopic vision. It was adopted in 1924 by the International Commission on Illumination (CIE) and in 1970 by the International Committee of Weights and Measures (CIPM).⁴² It is based on a 2° viewing field and principles of heterochromatic photometry and was incorporated into the CIE 1931 Colorimetric Observer as the $\bar{y}$ function.

Unless otherwise indicated, the values used for spectral luminous efficiency in photopic vision are the values agreed upon internationally in 1924 by the CIE, completed by interpolation and extrapolation, and recommended by the CIPM in 1972.

3.10.2.1 CIE 1988 2° modified spectral luminous efficiency function for photopic vision, $V_M(\lambda)$

A supplemental modification of $V(\lambda)$, adopted by the International Commission on

Illumination (CIE) in 1990.⁴² This function is principally of interest in the visual sciences.
(See **Table A.9.**)

3.10.3 CIE scotopic luminous efficiency function, $V'(\lambda)$

The spectral luminous efficiency function for scotopic vision, adopted by the International Commission on Illumination (CIE) in 1951. (See **Table T-7.**)

3.10.4 luminous efficacy of radiant flux

The quotient of the total luminous flux by the total radiant flux. It is expressed in lumens per watt (lm/W).

3.10.5 relative luminosity

(See **Section 3.10.1 spectral luminous efficiency of radiant flux.**)

3.10.6 relative luminosity factor

(See **Section 3.10.1 spectral luminous efficiency of radiant flux.**)

3.10.7 spectral luminous efficiency for photopic vision

$V(\lambda)$

(See **Section 3.10.2 values of spectral luminous efficiency for photopic vision.**)

3.10.8 spectral luminous efficiency for scotopic vision

$V'(\lambda)$

(See **Section 3.10.3 values of spectral luminous efficiency for scotopic vision.**)

3.11 luminous efficacy of a source of light

The total emitted luminous flux divided by the total source electrical input power; expressed in lumens per watt (lm/W).

Note 1: The term *luminous efficacy* can be applied to a single component, multiple components, or component(s) plus any associated electrical control equipment. The user will need to state the intended meaning. For example: lamp efficacy, luminaire efficacy, system efficacy (stating what the "system" is).

Note 2: Luminous efficacy as defined here is not to be confused with luminous efficacy of radiant flux.

3.11.1 efficacy

(See **Section 3.11 luminous efficacy of a source of light** and **Section 3.10 spectral luminous efficacy of radiant flux.**)

3.12 luminous energy

[See Section 3.9 quantity of light (luminous energy, Q).]

3.13 speed of light

(See Section 1.2 spectral parameters.)

4.0 COLOR

4.1 color

The characteristic of light by which a human observer can distinguish between two structure-free patches of light of the same size, shape, and brightness. (See Sections 4.4 light source color and 4.3.1.11 object color.)

4.2 physical stimuli (distal or proximal)

4.2.1 distal stimuli

An object in the field of view that forms an optical image on the retina.

4.2.2 proximal stimuli

The distribution of illuminance on the retina created from a distal stimulus.

4.3 perceived color

The proximal stimulus applied to the retina initiates color, which is perceived as a substance occupying the space in front of the observer's eyes. Color may be perceived as self-luminous or as being reflected or transmitted light. It may be perceived as confined to a point or line, or arrayed as a surface or film, or distributed in three dimensions as in the case of the perceived image of a patch of fog. A perceived image may be perceived as composed of *volume color*, as in the case of fog, or as covered by *surface color*, as in the case of a piece of chalk. In the case of the sky or a patch of color seen through an aperture where it cannot be identified as belonging to a specific object, it is called *aperture color*, and judgment is suspended as to whether the color is self-luminous or perceived by reflected or transmitted light. The color of a point source of light may be perceived and described as such. This is a special case of a self-luminous color.

4.3.1 chromatic color

Perceived color possessing a hue. In everyday speech, the word *color* is often used in this sense, in contradistinction to white, grey, or black.

4.3.1.1 hue of a perceived color

The attribute that determines whether it is, for example, red, yellow, green, or blue.

4.3.1.2 colorfulness of a perceived color

The attribute according to which it appears to exhibit more or less chromatic color.

For a stimulus of a given chromaticity, colorfulness normally increases as the absolute luminance is increased.

4.3.1.3 saturation of a perceived color

The attribute according to which it appears to exhibit more or less chromatic color judged in proportion to its brightness. In a given set of viewing conditions, and at luminance levels that result in photopic vision, a stimulus of a given chromaticity exhibits approximately constant saturation for all luminances.

4.3.1.4 perceived chroma of an area of surface color

The attribute according to which it appears to exhibit more or less chromatic color when judged in proportion to the brightness of a similarly illuminated area that appears to be white or highly transmitting. In a given set of viewing conditions, and at luminance levels that result in photopic vision, a stimulus of a given chromaticity and luminance factor exhibits approximately constant perceived chroma for all illuminance levels; but for a stimulus of a given chromaticity viewed at a given level of illuminance, the perceived chroma generally increases if the luminance factor is increased.

4.3.1.5 lightness (of a perceived patch of surface color)

The brightness of an area judged relative to the brightness of a similarly illuminated area that appears to be white or highly transmitting.

4.3.1.5.1 brightness (of a perceived aperture color)

The attribute by which an area of color of finite size is perceived to emit, transmit, or reflect a greater or lesser amount of light. No judgment is made as to whether the light comes from a reflecting, transmitting, or self-luminous object.

4.3.1.6 color comparison, or color grading

The judgment of equality, or of the amount and character of difference, of the color of two objects viewed under identical illumination.

4.3.1.7 color correction (of a photograph or printed picture)

The adjustment of a color reproduction process to improve the perceived-color conformity of the reproduction to the original.

4.3.1.8 color matching

The action of making a color appear the same as a given color.

4.3.1.9 color rendering

Effect of an illuminant on the color appearance of objects by conscious or subconscious comparison with their color appearance under a reference illuminant.⁷

4.3.1.10 color rendering index (CRI) (of a light source)

A measure of the degree of color shift that objects undergo when illuminated by the light source, as compared with the color of those same objects when illuminated by a reference source of comparable color temperature.⁹

4.3.1.10.1 general color rendering index, R_a

A measure of the average shift of eight standardized colors chosen to be of intermediate saturation and spread throughout the range of hues. If the color rendering index is not qualified as to the color samples used, R_a is assumed.

4.3.1.10.2 special color rendering index, R_i

A measure of color shift of various standardized special colors, including saturated colors, typical foliage, and "white" (European) skin. It can also be defined for other color samples when the spectral reflectance distributions are known.

4.3.1.10.3 color preference index (of a light source), R_p

A measure for appraising a light source with respect to its ability to meet the color preference for a particular application of a population of observers.

Note: Examples include Jerome's Color Flattery Index and the Color Preference Index (CPI) from WA Thornton.

4.3.1.11 object color

Color perceived as belonging to an object.⁷

4.3.1.12 colorimetric shift

The change in chromaticity and luminance factor of an object color due to change of the light source. (See **4.3.1.13 adaptive color shift** and **4.3.1.14 resultant color shift**.)

4.3.1.13 adaptive color shift

The change in the perceived object color caused solely by change of the state of chromatic adaptation. (See also **Section 4.3.1.16**).

4.3.1.14 resultant color shift

The difference between the perceived color of an object illuminated by a test source and the perceived color of the same object illuminated by the reference source, considering the state of chromatic adaptation (see **Section 4.2.1.16**) in each case i.e., the resultant of colorimetric shift and adaptive color shift.

4.3.1.15 chromatic adaptation

The process by which the chromatic properties of the visual system are modified by the observation of stimuli of various chromaticities and luminances.

4.3.1.16 state of chromatic adaptation

The condition of the chromatic properties of the visual system at a specified moment as a result of exposure to the totality of colors of the visual field currently and in the past. (See also **Section 4.2.1.15**.)

4.3.1.17 chromatic contrast threshold (color contrast threshold)

A threshold of chromaticity difference between two patches of color juxtaposed and separated only by a color contrast border, below which they cannot be perceived as different in chromatic-ness or as being separated by a contrast border. A contrast border can involve differences both in luminance and in chromaticity between the sides.

4.3.1.18 film (or aperture) color

(See **Section 4.3 perceived color**.)

4.3.1.19 gamut area index

Gamut area index (GAI),* is based on the area of the polygon formed by the eight test-color samples (TCS) in the CIE 1976 uniform color space under the light source of interest, relative to the polygon formed by the TCSs when illuminated by an equal energy spectrum (EES) light source.

* Rea MS, Freysinnier-Nova JP. Color rendering: A tale of two metrics. Color

Research and Application. 2008;33(3):192-202. DOI: 10.1002/col.20399.

4.3.1.20 chroma

The colorfulness of an area judged as a proportion of the brightness of an achromatic stimulus of the same brightness.

(Refer to CIE S 017, *ILV: International Lighting Vocabulary* for additional info.)

4.3.1.21 shade (color)

In color terms: A variation of a color produced by a pigment or dye mixture having some black in it.

(Related terms: *tint*, *tone*, and *value*.)

4.3.1.22 tint

A variation of a color produced by adding white to it, which lowers saturation and increases lightness.

(Related terms: *shade*, *tone*, and *value*.)

4.3.1.23 tone

A reduction in saturation of a color pigment produced by adding gray, with lighter shades of gray bringing the color closer to white, and darker shades of gray bringing the color closer to black.

(Related terms: *tint*, *shade*, and *value*.)

4.3.1.24 value

In color terms: The lightness or darkness of a color pigment, determined by how close it is to white or black.

(Related terms: *tint*, *shade*, and *tone*.)

4.3.1.25 color appearance model

A model characterizing at least lightness, chroma, and hue to indicate the entire effect of optical radiation on the perception of color.

Note: More complex models also characterize brightness, colorfulness and saturation.

4.3.1.26 color evaluation sample

A spectral reflectance function or a physical sample used for evaluation of light source

or surface-reflectance color characteristics.

Note: Sets of color evaluation samples are specific to an evaluation methodology. For example, the method described in ANSI/IES TM-30-20, *Technical Memorandum: IES Method for Evaluating Light Source Color Rendition* uses 99 color samples.

4.3.1.27 color fidelity

The degree to which an object's color looks similar to its appearance under a reference illuminant.

4.3.1.28 color fidelity index

A measure of the difference in appearance between one or more samples when illuminated by two different light sources, one of which is a reference source.

4.3.1.29 color preference

The subjective ranking of light sources by an observer for a particular application, based on spectral properties.

4.3.1.30 gamut shift pattern

The pattern of hue and chroma changes for all colors.

An example is the Color Vector Graphic (CVG) in ANSI/IES TM-30.

4.3.1.31 Helmholtz-Kohlrausch Effect

A phenomenon in which highly saturated colored light appears brighter to an observer than less-saturated colored light or surface of the same hue and same luminance.

4.3.1.32 Hunt Effect

A color appearance phenomenon where colorfulness of chromatic objects increases with luminance (even though chromaticity remains unchanged).

4.3.1.33 ANSI/IES TM-30 terms

4.3.1.33.1 Color Vector Graphic, CVG

A visual representation of hue and chroma shifts around the hue circle of ANSI/IES TM-30.

(Refer to ANSI/IES TM-30-20, *Technical Memorandum: IES Method for Evaluating Light Source Color Rendition* for additional information.)

4.3.1.33.2 Fidelity Index (R_f)

A measure of the average color deviation produced by a test illuminant relative to a standardized reference illuminant for the 99 color evaluation samples specified in ANSI/IES TM-30.

Note: The values range from 0 to 100, with a value of 100 indicating no deviation and a value of 0 indicating maximum deviation. Typical values: 70 to 95.

4.3.1.33.3 Gamut Index (R_g)

A measure of the average change in chroma produced by a test illuminant relative to a standardized reference illuminant for the 99 color evaluation samples specified in ANSI/IES TM-30-20, *Technical Memorandum: IES Method for Evaluating Light Source Color Rendition*.

Note: Values above 100 indicate an average increase in chroma; values less than 100 indicate an average decrease in chroma. Typical values: 80 to 120.

4.3.1.33.4 Local Chroma Shift (R_{cs,h_j})

A measure of the average change in chroma (radial deviation) produced by a test illuminant relative to a standardized reference illuminant for colors within an ANSI/IES TM-30 hue angle bin, h_j (range for j : 1 – 16).

4.3.1.33.5 Local Color Fidelity (R_{f,h_j})

A measure of the average color deviation produced by a test illuminant relative to a standardized reference illuminant within an ANSI/IES TM-30 hue angle bin, h_j (range for j : 1 – 16).

Note: The values range from 0 to 100, with a value of 100 indicating no deviation and a value of 0 indicating maximum deviation. Typical values: 70 to 95.

4.3.1.33.6 Local Hue Shift (R_{hs,h_j})

A measure of the average tangential change in hue, relative to a standardized reference illuminant, for colors within an ANSI/IES TM-30 hue angle bin, h_j (range for j : 1 – 16), referenced to the center of the hue bin angle.

Note: Positive values indicate a counterclockwise shift (e.g., red to orange), negative values indicate a clockwise shift. Possible values: Approximately -1 to 1 (varies by hue). Typical values: Approximately -0.2 to 0.2 (varies by hue).

4.3.1.33.7 Sample Color Fidelity ($R_{f,CES}$)

A measure of the color deviation produced by a test illuminant relative to a standardized reference illuminant for one of the 99 color evaluation samples specified in ANSI/IES TM-30.

4.3.2 Ostwald color system

A system of describing colors in terms of color content, white content, and black content. It is usually exemplified by color charts in triangular form, with full color, white, and black at the apexes, providing a gray scale of white and black mixtures and parallel scales of constant white content as these grays are mixed with varying proportions of the full color. Each chart represents a constant dominant wavelength (called hue), and the colors lying on a line parallel to the gray scale represent constant purity (called *shadow series*).

4.3.3 Munsell color system

A system of surface-color specification based on perceptually uniform color scales for three variables: Munsell hue, Munsell value, and Munsell chroma. For an observer of normal color vision who is adapted to daylight and is viewing a specimen that is illuminated by daylight and surrounded with a middle-gray to white background, the Munsell hue, Munsell value, and Munsell chroma of the color correlate well with the hue, lightness, and perceived chroma, respectively.

Note: A number of other color specification systems have been developed, usually for specific commercial purposes. Their description is beyond the scope of this document.

4.3.3.1 Munsell hue, H

The index of the hue of the perceived object color defined in terms of the luminance factor (Y) and chromaticity coordinates (x, y) for CIE Standard Illuminant C and the CIE 1931 Standard Observer.¹⁰

4.3.3.2 Munsell value, V

The index of the lightness of perceived object color defined in terms of the luminance factor (Y) for CIE Standard Illuminant C and the CIE 1931 Standard Observer.¹¹

Note: The exact definition gives V as a fifth-power function of Y , so that tabular or iterative methods are needed to find V as a function of Y . However, V can be estimated within ± 0.1 by $V = 11.6 (Y/100)^{1/3} - 1.6$, or within ± 0.6 by $V = Y^{1/2}$, where Y is the luminance factor expressed in percent.

4.3.3.3 Munsell chroma, C

The index of the perceived chroma of the object color defined in terms of the luminance factor (Y) and chromaticity coordinates (x, y) for CIE Standard Illuminant C and the CIE 1931 Standard Observer.

4.4 light source color

The color of the light emitted by the source.

Note: The color of a point source may be defined by its luminous intensity and chromaticity coordinates; the color of an extended source may be defined by its luminance and chromaticity coordinates.

4.4.1 perceived light source color

The color perceived to belong to a light source.

4.5 RGB color system

In the CIE RGB colorimetry system: An additive color system that uses red, green, and blue primaries mixed in various proportions to reproduce a broad array of colors.

Ref. CIE's RGB system and the 1931 CIE Standard Observer.

4.5.1 primary color

One of a set of colorants or colored lights that can be combined in varying amounts to produce a gamut of colors.

4.6 chromaticity

4.6.1 chromaticity of a color

The dominant or complementary wavelength and purity aspects of the color taken together, or of the aspects specified by the chromaticity coordinates of the color taken together.

4.6.1.1 chromaticity shift (of a light source)

The change in chromaticity of a light source between two specified operating conditions or over time.

Note: This is sometimes called color shift of a light source.

4.6.1.2 chromaticity maintenance (of a light source)

The ability of a light source to maintain, at any given time in its operational life (under specified operating conditions), its original chromaticity.

4.6.2 chromaticity coordinates of a color, x, y, z

The ratios of each of the tristimulus values of the color to the sum of the three tristimulus values (see Sections 4.65 and 4.6.5.1).

4.6.3 chromaticity diagram

A plane diagram formed by plotting one of the three chromaticity coordinates against another.

4.6.3.1 CIE standard chromaticity diagram

One in which the x and y chromaticity coordinates are plotted in rectangular coordinates.

Note: The diagram may be based on the CIE 1931 Standard Observer or on the CIE 1964 supplementary Standard Observer. (See note to Section 4.6.5.1.)

4.6.3.2 spectrum locus

The locus of points representing the colors of the visible spectrum in a chromaticity diagram.

4.6.3.3 purple boundary

The straight line drawn between the ends of the spectrum locus on a chromaticity diagram.

4.6.3.4 blackbody (Planckian) locus

The locus of points on a chromaticity diagram representing the chromaticities of blackbodies having various (color) temperatures.

4.6.3.5 Planckian locus

[See Section 4.6.3.4 blackbody (Planckian) locus.]

4.6.3.6 blackbody curve

[See Section 4.6.3.4 blackbody (Planckian) locus.]

4.6.3.7 uniform chromaticity scale diagram (UCS)

A type of chromaticity diagram that has been adjusted so that at a given luminance, equal differences in stimulus chromaticity correspond to equal perceptual differences.

4.6.3.8 D_{uv}

A measure of the distance from the blackbody locus in the CIE 1960 (u, v) uniform

chromaticity scale (UCS) diagram. A positive value is above the blackbody locus, and a negative value is below the blackbody locus.

4.6.4 distribution temperature of a light source

The absolute temperature of a blackbody whose relative spectral distribution is most nearly the same in the visible region of the spectrum as that of the light source.

4.6.4.1 color temperature of a light source

The thermodynamic temperature of a blackbody radiator having a chromaticity equal to that of the light source. Expressed in kelvins (K).

4.6.4.2 correlated color temperature of a light source

The thermodynamic temperature of a blackbody whose chromaticity most nearly resembles that of the light source. Expressed in kelvins (K).

4.6.4.3 reciprocal color temperature

Color temperature T_c expressed on a reciprocal scale ($1/T_c$).

An important use stems from the fact that a given small increment in reciprocal color temperature is approximately equally perceptible regardless of color temperature. Also, color temperature conversion filters for sources approximating graybody sources (see **Section 2.13.7**) change the reciprocal color temperature by nearly the same amount anywhere on the color temperature scale.

Note: The unit is the reciprocal megakelvin (MK^{-1}). The reciprocal color temperature expressed in this unit has the numerical value of $10^6/T_c$ when T_c is expressed in kelvins. The acronym *MIREK* (for micro-reciprocal-kelvin) occasionally has been used in the literature. The acronym *MIREC* (for micro-reciprocal-degree) is now considered obsolete as the name for this unit.

4.6.4.4 chromaticity difference thresholds

The smallest difference in chromaticity between two colors of the same luminance that makes them perceptibly different. The difference may be a difference in hue, saturation, or a combination of the two.

4.6.4.5 color difference thresholds

The difference in chromaticity and/or luminance, between two colors, that makes them just perceptibly different. The difference may be a difference in hue, saturation, brightness (lightness for surface colors) or a combination of the three.

4.6.4.6 color contrast thresholds

(See Section 4.6.4.4 chromaticity difference thresholds.)

4.6.4.7 MIRED

(See Section 4.6.4.3 reciprocal color temperature.)

4.6.5 tristimulus values of a light, X, Y, Z

The amounts of each of three specific primaries required to match the color of the light (see Section 4.6.5.1).

4.6.5.1 color matching functions (spectral tristimulus values)

$$\bar{x} = X_\lambda / \Phi_{e\lambda}; \bar{y} = Y_\lambda / \Phi_{e\lambda}; \bar{z} = Z_\lambda / \Phi_{e\lambda}$$

The tristimulus values per unit wavelength interval and unit spectral radiant flux.

Notes:

- i. Color matching functions have been adopted by the International Commission on Illumination (CIE). They are tabulated as functions of wavelength throughout the spectrum and are the basis for the evaluation of radiant energy as light and color. The standard values adopted by the CIE in 1931 are given in **Table T-5A**. The values are identical with the values of spectral luminous efficiency for photopic vision (see **Section 3.10.2** and **Table T-4**).
- ii. The $\bar{x}$, $\bar{y}$, $\bar{z}$ values for the 1931 Standard Observer are based on a two-degree bipartite field, and are recommended for predicting matches for stimuli subtending between one and four degrees. Supplementary data based on a ten-degree field were adopted in 1964 for use for angular subtends greater than four degrees (see **Table T-5B**). Tristimulus computational data for CIE standard sources A and C are given in **Table T-6**.

4.6.5.2 spectral tristimulus values

[See Section 4.6.5.1 color matching functions (spectral tristimulus values).]

4.6.5.3 standard colorimetric observer

An ideal observer whose color-matching properties correspond to a CIE color-matching function. (Examples include the CIE 1931 and 1964 standard observers for colorimetry.)

4.6.6 colorimetric purity of a light

A quantity (p_c) defined by $p_c = L_1 / (L_1 + L_2)$, where L_1 is the luminance of a single-wavelength

(monochromatic) light source, and L_2 is the luminance of the reference standard light source. These values are obtained by adjusting the reference standard light source until the additive mixture of the two match the color of the considered light source.

4.6.6.1 excitation purity of a light, p_e

The ratio of the distance on the CIE chromaticity diagram between the reference point and the light point to the distance in the same direction between the reference point and the spectrum locus or the purple boundary.

4.7 dominant wavelength (of a color stimulus)

(λ_d)

Wavelength of the monochromatic stimulus that, when additively mixed in suitable proportions with the specified achromatic stimulus, matches the color stimulus considered in the CIE 1931 x, y chromaticity diagram.

Note: In the case of a purple stimulus, the dominant wavelength is replaced by the complementary wavelength.

Source: International Commission on Illumination (CIE). CIE S 017/E:2020, ILV: International Lighting Vocabulary. Vienna: CIE; 2020.

4.7.1 complementary wavelength of a light, λ_c

The wavelength of radiant energy of a single frequency that, when combined in suitable proportion with the light, matches the color of a reference standard.

4.7.2 centroid wavelength, λ_c

The wavelength calculated as the weighted mean of a spectral function according to:

$$\lambda_c = \frac{\int_{\lambda_1}^{\lambda_2} \lambda \cdot S(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} S(\lambda) d\lambda}$$

where:

λ is the wavelength, $S(\lambda)$ is the spectral function, and λ_1 and λ_2 are the limits for the wavelength range of $S(\lambda)$.

4.8 metamers

Lights of the same color but of different spectral energy distribution.

Note: The term *metamers* is also used to denote objects which when illuminated by a given source and

viewed by a given observer produce metamerism.

4.8.1 index of illuminant metamerism

As applied to two objects that are metamerism when illuminated by a reference source: a measure of the degree of color difference between the two objects when a specified test source is substituted for the reference source.

4.8.2 index of observer metamerism

As applied to two objects that are metamerism when viewed by a reference observer: a measure of the degree of color difference between the two objects when a specified test observer is substituted for the reference observer.

4.8.3 metamerism

A condition in which spectrally different color stimuli match in appearance.

Note: This condition is affected by the observer, light level, and other factors.

4.9 standard source

A light source having the same spectral distribution as a specified standard illuminant.

4.9.1 standard source A

A tungsten filament lamp operated at a correlated color temperature of 2856 K and approximating the relative spectral power distribution of standard illuminant A.

4.9.2 standard source C

An approximation of standard illuminant C obtained by a combination of Source A and a special filter.

4.9.3 standard illuminant

A hypothetical light source of specified relative spectral power distribution.

Note: The CIE has specified spectral power distributions for standard illuminants A and C, and for several D illuminants.

4.9.3.1 standard illuminant A

A blackbody at a temperature of 2856 K, using $c_2 = 1.4388$ (see **Section 2.13.4**). It is defined by its relative spectral power distribution over the range from 300 to 830 nm.

4.9.3.2 standard illuminant C

A representation of daylight having a correlated color temperature of approximately 6800 K. It is defined by its relative spectral power distribution over the range from 320 to 770 nm.

4.9.3.3 standard illuminant D₆₅

A representation of daylight at a correlated color temperature of approximately 6500 K. It is defined by its relative spectral power distribution over the range from 300 to 830 nm.

Note: At present, no manufactured source for matching this illuminant has been recommended.

4.9.4 supplementary standard illuminant D₅₅

A representation of a phase of daylight at a correlated color temperature of approximately 5500 K.

4.9.5 supplementary standard illuminant D₇₅

A representation of a phase of daylight at a correlated color temperature of approximately 7500 K.

4.10 color space

A mathematical and/or graphical representation of colors. Color spaces can be devised that represent all perceivable colors or only those colors that can be realized on a particular display or print medium (usually called “gamut”). The loci of colors may be expressed in different formats, such as chromaticity coordinates (CIE) or perceptual descriptors (e.g., hue, saturation, brightness; or hue, chroma, luminance).

4.10.1 CIE ($L^*u^*v^*$) uniform color space

Abbreviation: CIELUV¹⁵

A transformation of CIE tristimulus values X, Y, Z into three rectangular coordinates (L^*, u^*, v^*) that define a space in which equal distances are more nearly representative of equal magnitudes of perceived color difference. This space is especially useful in cases where colored lights are mixed additively (e.g., color television).

4.10.2 CIE ($L^*a^*b^*$) uniform color space

Abbreviation: CIELAB.¹⁵

A transformation of CIE tristimulus values, X, Y, Z , into three rectangular coordinates (L^*, a^*, b^*) that define a space in which equal distances are more nearly representative of equal magnitudes of perceived color difference. This space is specially useful in cases of colorant mixtures (e.g., dye-stuffs, paints).

4.10.3 psychometric lightness

A correlate of *lightness* defined in terms of CIELUV or CIELAB. Equal scale intervals correspond approximately to equal differences in (perceived) lightness.

4.10.4 psychometric chroma

A correlate of *perceived chroma* defined in terms of CIELUV or CIELAB. Equal scale intervals correspond approximately to equal differences in (perceived) chroma.

4.10.5 psychometric saturation

A correlate of *saturation* defined in terms of CIELUV. Equal scale intervals correspond approximately to equal differences of (perceived) saturation.

Note: Psychometric saturation cannot be calculated in terms of CIELAB.

4.10.6 psychometric hue-angle

A correlate of *hue* defined in terms of CIELUV or CIELAB.

4.10.7 CIECAM02

A color appearance model defined in CIE 159:2004, *A Color Appearance Model for Colour Management Systems: CIECAM02*.

4.10.8 CAM02-UCS

A uniform color space based on the CIECAM02 color appearance model and designed to accurately predict a wide range of color difference data.

Ref.: ISO/CIE 11664-5:2016, *Colorimetry* - Part 5.

4.10.9 HSL color space

A device-dependent system used to represent colors in a cylindrical-coordinate RGB color space, based on hue, saturation, and lightness.

4.10.10 HSV color space

A device-dependent system used to represent colors in a cylindrical-coordinate RGB color space, based on hue, saturation, and value.

4.10.11 gamut area (of a light source)

That portion of a color space enclosed by a set of coordinates corresponding to a selected set of colors illuminated by the light source.

4.10.12 hue angle (h)

A numerical representation of hue, specified as the angle from a cardinal axis in a color space or color appearance model.

Note: Hue angle is often measured in the counterclockwise direction from the cardinal axis most closely related to "red."

4.10.12.1 hue-angle bin

A subset of all possible hue angles in a color space.

Note: The ANSI/IES TM-30 Color Vector Graphic (CVG) is split radially into sixteen 22.5° hue-angle bins.

4.11 spectral power distribution (SPD)

A tabular or graphical representation of the radiant power emitted by a source at each wavelength or within bands of wavelengths across the electromagnetic spectrum.

4.11.1 peak wavelength (λ_p)

The wavelength at which the maximum value occurs in a spectral power distribution.

5.0 VISUAL TERMS

5.1 accommodation

The process by which the eye changes focus from one distance to another.

5.2 adaptation

The process by which the all or part of the retina becomes accustomed to more or less light than it was exposed to during an immediately preceding period. It results in a change in the sensitivity to light. (See also Sections 4.3.1.15, 5.2.1.1, and 5.2.2.1.)

Note: The term *adaptation* is also used to refer to the final state of the process; for example, reaching a condition of adaptation to a specific level of luminance.

5.2.1 dark adaptation

The process by which the retina becomes adapted to a luminance less than about 0.034 cd/m².

5.2.2 light adaptation

The process by which the retina becomes adapted to a luminance greater than about 5 cd/m².

5.2.3 Purkinje phenomenon

The relative increase in the apparent brightness of short-wavelength lights (emitted or reflected) when the luminances are reduced in the same proportion without changing the respective spectral distributions. In passing from photopic to scotopic vision, the curve of spectral luminous efficiency changes, the wavelength of maximum efficiency being displaced toward the shorter wavelengths.

5.2.4 inhibition

Reduction in magnitude of the sensation aroused by a stimulus (or a reduction in visual sensitivity), and caused by some other stimulation that is adjacent spatially or temporally.

5.2.5 Broca-Sulzer effect

At high luminance levels, the momentary brightness shortly after the commencement of a flash appears higher than the brightness of a maintained light of the same luminance as the flash.

5.2.6 brightness enhancement

(Also known as *Bartley effect enhancement*)

If a light of relatively high brightness is first seen as a steady light and is then repetitively extinguished about four to ten times per second, its apparent brilliance increases, rather than diminishes, by a factor of up to 5. Its conspicuity increases considerably, even though the power consumption and the photometric quantity of light may be half or less. The same effect may occur at lower levels of luminance if the light is seen extra-foveally.

5.2.7 photopic vision

Vision mediated essentially or exclusively by the cones. It is generally associated with adaptation to a luminance of at least 5 cd/m².

5.2.8 mesopic vision

Vision with fully adapted eyes at luminance conditions between those of photopic and scotopic vision; i.e., between about 5 cd/m² and 0.001 cd/m².

5.2.9 scotopic vision

Vision mediated essentially or exclusively by the rods. It is generally associated with adaptation to a luminance below about 0.001 cd/m².

5.3 retina

A membrane lining the posterior part of the inside of the eye. It comprises visual photoreceptors (cones and rods) and non-visual photoreceptors (intrinsically photosensitive retinal ganglion cells) that are sensitive to light, and nerve cells that transmit signals via the optic nerve to various portions of the brain.

5.3.1 fovea

A small region at the center of the retina, subtending about two degrees, which contains cones but no rods in the central one degree, and which forms the site of the most-distinct vision.

5.3.2 cones

Retinal receptors with three distinct spectral responses that dominate the retinal response when the luminance level is high and provide the basis for high resolution visual acuity and the perception of color. Cones are highly concentrated within the fovea and less concentrated across the remainder of the retina.

5.3.3 rods

Retinal receptors that continue to respond at levels of luminance below the threshold for cones, and that provide no ability to discriminate colors. Rods are absent in the center of the fovea.

5.3.4 retinal ganglion cells

A class of neurons on the inner surface of the retina that vary significantly in their size, connections, and responses. Some receive information from neurons connected to the rods and cones and pass visual response signals to the brain. Others (the intrinsically photosensitive retinal ganglion cells, ipRGC) are photosensitive and contribute primarily to nonvisual physiological processes and the pupillary light reflex.

5.4 pupil (pupillary aperture)

The opening of the iris that admits light into the eye.

5.4.1 artificial pupil

A device or arrangement for confining the light passing through the pupil of the eye to an area smaller than that of the natural pupil.

5.5 point of observation

A single point located at the midpoint of the baseline connecting the centers of the entrance pupils of the two eyes.

5.6 point of fixation

A point or object in the visual field at which the eyes look and upon which they are focused.

5.7 primary line of sight

The line connecting the point of observation and the point of fixation. In terms of a single eye, it is the line connecting the point of fixation and the center of the entrance pupil.

5.8 visual field

The locus of objects or points in space that can be perceived when the head and eyes are kept fixed. Separate monocular fields for the two eyes may be specified or the combination of the two.

5.8.1 monocular visual field

The visual field for a single eye.

5.8.2 binocular portion of the visual field

That portion of space where the fields of the two eyes overlap.

5.8.3 central visual field

That region of the visual field that corresponds to the foveal portion of the retina.

5.8.3.1 central (foveal) vision

The seeing of objects in the central (foveal) part of the visual field, approximately two degrees in diameter. It permits seeing much finer detail than does peripheral vision.

5.8.3.2 foveal vision

[See Section 5.8.3.1 central (foveal) vision.]

5.8.4 peripheral visual field

That portion of the visual field that falls outside the region corresponding to the foveal portion of the retina.

5.8.4.1 peripheral vision

The seeing of objects displaced from the primary line of sight and outside the central visual field.

5.8.5 visual task

Conventionally designates those details and objects that must be seen for the performance of a

given activity, and includes the immediate background of the details or objects.

Note: The term *visual task* as used is a misnomer because it refers to the visual display itself and not the task of extracting information from it. In addition, the task of extracting information has to be differentiated from the overall task performed by the observer.

5.8.6 visual surround

Includes all portions of the visual field except the visual display used in performing a task.

5.9 threshold

The value of a variable of a physical stimulus (such as size, luminance, contrast or time) that permits the stimulus to be seen a specific percentage of the time or at a specific accuracy level. In many psychophysical experiments, thresholds are presented in terms of 50% accuracy, or being accurately perceived 50% of the time. However, the threshold also is expressed as the value of the physical variable that permits the object to be just barely seen. The threshold may be determined by merely detecting the presence of an object, or it may be determined by discriminating certain details of the object.

5.9.1 absolute luminance threshold

Luminance threshold for a bright object such as a disk on a totally dark background.

5.9.2 luminance difference threshold

The minimum perceptible difference in luminance between two areas.

Note: This can apply to the difference between two separated objects on a common background, or the difference between two juxtaposed patches separated by a contrast border, or the difference between a uniform small object and its background.

5.9.3 modulation threshold

In the case of sine wave grating, manipulation of luminance differences can be specified in terms of modulation, and the threshold may be called the *modulation threshold*.

$$\text{modulation} = (L_{\max} - L_{\min}) / (L_{\max} + L_{\min})$$

Periodic patterns that are not sinusoidal can be specified in terms of the modulation of the fundamental sine wave component.

5.9.4 luminance contrast

The relationship between the luminances of an object and its immediate background. It is equal to:

$$(L_1 - L_2) / L_1 \text{ or } (L_2 - L_1) / L_1 = |\Delta L / L_1|,$$

where L_1 and L_2 are the luminances of the background and object, respectively.

The form of the equation must be specified. The ratio $\Delta L/L_1$ is known as *Weber's fraction*.

Note: See note under **Section 3.5**. Because of the relationship among luminance, illumination, and reflectance, contrast often is expressed in terms of reflectance when only reflecting surfaces are involved. Thus, contrast is equal to $(\rho_1 - \rho_2)/\rho_1$, or $(\rho_2 - \rho_1)/\rho_1$, where ρ_1 and ρ_2 are the reflectances of the background and object, respectively. This method of computing contrast holds only for perfectly diffusing surfaces; for other surfaces it is only an approximation unless the angles of incidence and view are taken into consideration. (See **Section 7.3.3**.)

5.9.4.1 luminance contrast sensitivity

The ability to detect the presence of luminance differences. Quantitatively, it is equal to the reciprocal of the brightness contrast threshold. (See **Section 5.9.4.2**.)

5.9.4.2 brightness contrast threshold

When two patches of color are separated by a brightness contrast border, as in the case of a bipartite photometric field or the case of a disk-shaped object surrounded by its background, the border between the two patches is a *brightness contrast border*. The contrast that is just detectable is known as the *brightness contrast threshold*.

5.9.4.3 Weber's fraction

(See **Section 5.9.4 luminance contrast**.)

5.9.4.4 contrast

(See **Section 4.9.4 luminance contrast**.)

5.9.4.5 contrast sensitivity

[See **Section 5.9.4.1 luminance contrast sensitivity**.]

5.9.4.6 contrast threshold

[See **Section 5.9.4.2 brightness contrast threshold**.]

5.9.5 luminance ratio

The ratio between the luminances of any two areas in the visual field.

Note: See last paragraph of the note under **Section 3.5**.

5.9.6 visual angle

The angle which an object or detail subtends at the point of observation. It usually is measured

in minutes of arc.

5.9.6.1 size threshold

The minimum perceptible size of an object. It also is defined as the size that can be detected some specific fraction of the times it is presented to an observer, usually 50 percent. It usually is measured in minutes of arc.

5.9.6.2 visual acuity

A measure of the ability to distinguish fine details measured with a set of optotypes* of different sizes when not constrained by contrast. Quantitatively, it is the reciprocal of the minimum angular size in minutes of the critical detail of an object that can just be resolved.

5.9.6.3 resolving power

The ability of the eye to perceive the individual elements of a grating or any other periodic pattern with parallel elements; measured by the number of cycles per degree that can be resolved.

The *resolution threshold* is the period of the pattern that can be just resolved. The resolution threshold for a pair of points or lines is the distance between their centers when they can be distinguished as two, not one; expressed in minutes of arc.

5.9.7 reaction time

The interval between the beginning of a stimulus and the beginning of the response of an observer.

5.9.8 speed of vision

The reciprocal of the duration of the exposure time required for something to be seen.

5.9.9 flicker fusion frequency, fff

The frequency of intermittent stimulation of the eye at which flicker disappears. It also is called *critical fusion frequency (cff)* or *critical flicker frequency (cff)*.

5.9.9.1 flicker index

A measure of the cyclic variation in output of a light source, taking into account the

*Test type for determining visual acuity.

waveform of the light output. It is the ratio of the area under the light output curve that is above the average light output level to the total area under the light output curve for a single cycle.

As an example, **Figure 4** shows a curve of the light output variation from a fluorescent lamp during each cycle, with the method of calculating the flicker index and percent flicker.

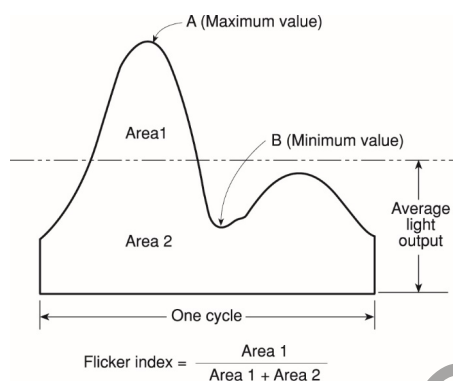


Figure 4.

5.9.9.2 percent flicker

A relative measure of the cyclic variation in output of a light source (percent modulation). It is given by the expression $(A - B)/(A + B)$, where A is the maximum and B is the minimum output during a single cycle, and is expressed as a percentage (see **Figure 4**).

5.9.10 brightness

The attribute of a visual sensation according to which an area appears to emit more light or less light.

5.9.10.1 subjective brightness

The subjective attribute of any light sensation giving rise to the percept of luminous magnitude, including the whole scale of qualities of being bright, light, brilliant, dim, or dark. (See **Section 4.3.1.3**.)

Note: The term *brightness* often is used when referring to the measurable luminance. While the context usually makes it clear as to which meaning is intended, the preferable term for the photometric quantity is *luminance*, thus reserving *brightness* for the

subjective sensation. (See **Section 3.5.**)

5.9.11 glare

The sensation produced by luminances within the visual field that are sufficiently greater than the luminance to which the eyes are adapted to cause annoyance, discomfort, or loss in visual performance or visibility.

Note: The magnitude of the sensation of glare depends upon such factors as the size, position, and luminance of a source; the number of sources; and the luminance to which the eyes are adapted.

5.9.11.1 direct glare

Glare resulting from high luminances or insufficiently shielded light sources in the field of view. It usually is associated with bright areas, such as luminaires, ceilings, and windows, that are outside the visual task or region being viewed. A direct glare source may also affect performance by distracting attention.

5.9.11.2 reflected glare

Glare resulting from reflections of high luminance in polished or glossy surfaces in the field of view. It usually is associated with reflections from within a visual task or from areas in close proximity to the region being viewed. (See **Section 5.9.11.8.**)

5.9.11.3 blinding glare

Glare that is so intense that for an appreciable length of time after it has been removed, no object can be seen.

5.9.11.4 disability glare

The effect of stray light in the eye whereby the contrast of the retinal image is reduced and, consequently, whereby visibility and visual performance may also be reduced. A direct glare source that produces discomfort may also produce disability glare by introducing a measurable amount of stray light in the eye.

5.9.11.4.1 disability glare factor, DGF

A measure of the visibility of a task in a given lighting installation in comparison with its visibility under reference lighting conditions, expressed in terms of the ratio of luminance contrasts having an equivalent effect upon task visibility. The value of DGF takes account of the equivalent veiling luminance produced in the eye by the pattern of luminances in the task surround.

5.9.11.4.2 equivalent veiling luminance

The luminance that would need to be superimposed on a scene in object space to reduce the scene's contrast by an amount equal to the added retinal illuminance from scattered light on the scene's retinal image. It is most commonly used to describe the contrast-reducing effect of a glare source in the field of view. Sometimes also called *veiling luminance* or *disability glare*.

5.9.11.4.3 threshold increment (TI)

A measure of disability glare expressed as the percentage increase in contrast required between an object and its background for the object to be seen equally well with a source of glare present.

Note: Higher values of TI correspond to greater disability glare.

5.9.11.4.4 veiling luminance

[See Section 5.9.11.4.2 equivalent veiling luminance.]

5.9.11.5 discomfort glare¹⁷

Glare that produces discomfort. It does not necessarily interfere with visual performance or visibility.

5.9.11.5.1 borderline between comfort and discomfort (BCD)

The average luminance of a source in a field of view that produces a sensation between comfort and discomfort.

5.9.11.5.2 discomfort glare factor^{*}

The numerical assessment of the capacity of a single source of brightness, such as a luminaire, in a given visual environment for producing discomfort.

5.9.11.5.3 visual comfort probability, VCP

The rating of a lighting system expressed as a percent of people who, when viewing from a specified location and in a specified direction, will be expected to find it acceptable in terms of discomfort glare. Visual comfort probability is related to Discomfort glare rating (DGR). (See Section 5.9.11.5.6.)

^{*}This term has been deprecated; it is retained here for reference of literature.

5.9.11.5.4 position index, P

A factor that represents the relative average luminance for a sensation at the borderline between comfort and discomfort (BCD) for a source located anywhere within the visual field.¹²

5.9.11.5.5 index of sensation, M

(of a source)¹³

A number that expresses the effects of source luminance (L_s), solid angle factor (Q), position index (P), and the field luminance (F) on discomfort glare rating (see Section 5.9.11.5.6), where L_s and F are expressed in cd/m^2 .

$$M = (0.5 L_s Q) / (PF^{0.44})$$

[See Section 5.9.11.5.7 for an equation defining Q .]

5.9.11.5.6 discomfort glare rating, DGR

A numerical assessment of the capacity of a number of sources of luminance, such as luminaires, in a given visual environment for producing discomfort. It is the net effect of the individual values of index of sensation M , for all luminous areas in the field of view:

$$DGR = \sum(M)^a,$$

where:

$$a = n^{-0.0914}$$

n = number of sources (ceiling elements) in the field of view.

5.9.11.5.7 solid angle factor, Q

A function of the solid angle (ω), given in steradians, that is subtended by a source. Solid angle factor is given by:

$$Q = 20.4\omega + 1.52\omega^{0.2} - 0.75$$

(See Section 5.9.11.5.5 index of sensation, M .)

5.9.11.5.8 unified glare rating (UGR)¹³

A measure of the discomfort produced by a lighting system along a psychometric scale of discomfort. UGR is used in many parts of the world as an alternative to visual comfort probability (VCP; see Section 5.9.11.5.3).

5.9.11.6 stray light (in the eye)

Light from a source that is scattered onto parts of the retina lying outside the retinal image of the source.

5.9.11.7 veiling brightness

A brightness superimposed on the retinal image that reduces its contrast. It is this “veiling” effect produced by bright sources or areas in the visual field that results in decreased visual performance and decreased visibility.

5.9.11.8 veiling reflection

Regular (specular) reflection, superimposed on diffuse reflection from an object, that partially or totally obscure the details to be seen by reducing the contrast. This sometimes is called *reflected glare* (see **Section 5.9.11.2**). Another kind of veiling reflection occurs when one looks through a pane of glass. A reflected image of a bright element or surface may be seen superimposed on what is viewed through the glass pane.

5.9.12 afterimage

A visual response that occurs after the stimulus causing it has ceased.

The effect is due to retinal photopigment regenerating from a bleaching. Afterimages can fade quickly because they are stabilized on the retina (the Troxler effect; see **Section 5.9.20**), but the reduction in sensitivity of the afterimage area continues until the photopigment has been regenerated.

5.9.13 troland

A unit of retinal illuminance that is based on the fact that retinal illuminance is proportional to the product of the luminance of the distal stimulus and the area of the entrance pupil. It is defined as the product of object luminance (candela per square meter) and pupillary aperture area (square millimeters); i.e., one troland is the retinal illuminance produced when the luminance of the distal stimulus is one candela per square meter and the area of the pupil is one square millimeter. The troland may be photopic or scotopic.

Note: The troland makes no allowance for interocular attenuation or for the Stiles-Crawford effect (see **Section 5.9.13.1**).

5.9.13.1 Stiles-Crawford effect

The reduced luminous efficiency of rays entering the peripheral portion of the pupil of the eye. This effect applies only to cones and not to rod visual cells. Hence, there is no

appreciable Stiles-Crawford effect in scotopic vision.

5.9.14 visibility

The quality or state of being perceivable by the eye. In many outdoor applications, visibility is defined in terms of the distance at which an object can be just perceived by the eye. In indoor applications, it usually is defined in terms of the contrast or size of a standard test object when observed under standardized viewing conditions and having the same threshold as the given object.

5.9.14.1 visibility reference function, VL1

A function representing a luminance contrast required at different levels of task background luminance to achieve visibility threshold for the visibility reference task consisting of a disk that subtends four minutes of arc exposed for one-fifth second.

5.9.14.2 visibility performance criteria function, VL8

A function representing the luminance contrast required to achieve 99 percent visual certainty for the same task used for the visibility reference function, including the effects of dynamic presentation and uncertainty in task location.

5.9.14.3 visibility level, VL

A contrast multiplier to be applied to the visibility reference function to provide the luminance contrast required at different levels of task background luminance in order to achieve visibility for specified conditions relating to the task and observer.

5.9.14.4 equivalent contrast, C

A numerical description of the relative visibility of a task. It is the contrast of the standard visibility reference task giving the same visibility as that of a task whose contrast has been reduced to threshold when the background luminances are the same.

This term is obsolete.

5.9.14.5 equivalent contrast, C_e

The actual equivalent contrast in a real luminous environment with non-diffuse illumination. The actual equivalent contrast C_e is less than the equivalent contrast C due to veiling reflection. $C_e = C \times CRF$ (see *contrast rendition factor*, **Section 5.9.19.2**).

This term is obsolete.

5.9.14.6 transient adaptation factor, TAF

A factor that reduces the equivalent contrast due to readaptation from one luminous background to another.

5.9.14.7 relative contrast sensitivity, RCS

The relationship between the reciprocal of the luminous contrast of a task at visibility threshold and the background luminance, expressed as a percentage of the value obtained under a very high level of diffuse task illuminance.

5.9.15 visual perception

The interpretation of impressions transmitted from the retina to the brain in terms of information about the physical world displayed before the eye.

Note: Visual perception involves any one or more of the following: recognition of the presence of something (object, aperture or medium); identifying it; locating it in space; noting its relation to other things; identifying its movement, color, brightness, or form.

5.9.16 visual performance

The quantitative assessment of the performance of a task, taking into consideration speed and accuracy.

5.9.17 color discrimination

The ability of an observer to discern a difference between colors when viewed simultaneously.

Note: The properties of the visual stimulus (e.g., the spectrum of the light source, the quantity of illumination, the observer's state of adaptation) can affect the measurement of color discrimination.

5.9.18 quality of lighting

Pertains to the distribution of luminance in a visual environment. The term is used in a positive sense and implies that all luminances contribute favorably to visual performance, visual comfort, ease of seeing, safety, and aesthetics for the specific visual tasks involved.

5.9.19 sphere illuminance

The illuminance on a task from a source providing equal luminance in all directions about the task, such as a uniformly illuminated sphere with the task located at the center.

5.9.19.1 equivalent sphere illuminance, ESI

The level of sphere illuminance that would produce task visibility equivalent to that

produced by a specific lighting environment.

This term is obsolete.

5.9.19.2 contrast rendition factor, CRF

The ratio of visual task contrast with a given lighting environment to the contrast with sphere illumination.

This term is obsolete.

5.9.19.3 lighting effectiveness factor, LEF

The ratio of equivalent sphere illumination to measured or calculated task illuminance.

5.9.20 Troxler effect

A fading of visual perception because of a stabilized retinal image. For example, the retinal blood vessels that lie in front of the retina are not normally seen, because they are fixed relative to the retinal receptors.

5.10 angle of collimation

The angle subtended by a luminaire at an irradiated surface.

This term is obsolete.

5.11 point vision

The mode of vision of a small source of light such that the sensation is determined by illuminance at the eye rather than by its size. Point vision occurs when a source is so small that its area no longer contributes to vision. This generally means less than one minute angular subtense.

5.11.1 point brilliance, E

Quantity involved in the visual observation of a source of light when viewed directly from such a distance that the apparent diameter is inappreciable. The point brilliance is measured by the illuminance produced by the source on a plane that is at the observer's eye and normal to the direction of the source.

(See also **Section 11.1.5.2 Allard's law.**)

5.11.2 visual threshold in point vision

The weakest illuminance at the eye that makes it possible to see a small source against a given background. The psychophysical definition is the point illuminance that gives 50 percent probability of detection. A widely accepted practical value for reliable vision is 2×10^{-7} lux, or

0.686 nautical-mile-candelas, for a steady light against a dark background.

5.12 vision

[See **Section 5.8.3.1 central (foveal) vision**; **Section 5.2.3.1 mesopic vision**; **Section 5.8.4.1 peripheral vision**; **Section 5.2.2.1 photopic vision**; and **Section 5.2.1.1 scotopic vision**.]

5.13 opponent channels

The three neural pathways of the human visual system, red-green, blue-yellow, and achromatic (black-white), according to the opponent-process theory of color vision.

5.14 trichromacy

The condition where three independent channels are sufficient for conveying color information.

Note: In humans, the three independent channels are represented by the three cone types.

5.15 univariance

The principle by which an infinite set of wavelength-intensity combinations can elicit the same response in a single photoreceptor of the human eye.

6.0 LIGHT SOURCES

6.1 lamp

A generic term for a manufactured source created to produce optical radiation.

Note: Through popular usage, a portable luminaire consisting of a lamp with shade, reflector, enclosing globe, housing, or other accessories is also called a "lamp." In such cases, in order to distinguish between the assembled unit and the light source within it, the latter is often called a "bulb" or "tube," if it is electrically powered (see also **Section 10.3.1**).

6.1.1 prefocus lamp

A lamp in which, during manufacture, the luminous element is accurately adjusted to a specified position with respect to the physical location element (usually the base).

6.1.2 reflector lamp

An incandescent filament lamp or electric discharge lamp in which the outer blown glass bulb is coated with a reflecting material so as to redirect the light (e.g., R- or ER-type lamps). The light transmitting region may be clear, frosted, patterned, or phosphor coated.

6.1.3 sealed-beam lamp

A reflector lamp in which the outer bulb is formed of two pressed parts that are fused or sealed together; namely, a reflectorized bowl and a cover that may be clear or patterned for optical control and having a non-replaceable light source.

Note 1: The construction of sealed-beam lamps gives a more closely controlled beam of light than ordinary reflector lamps.

Note 2: The source of light for a sealed-beam lamp may be LEDs, an incandescent filament, a halogen capsule, or an electric discharge metal halide arc tube.

Note 3: Sealed-beam lamps are sometimes called *pressed reflector lamps* or *PAR lamps* with the same meaning.

6.1.3.1 pressed reflector lamp

[See Section 6.1.3 sealed-beam lamp.]

6.1.4 projection lamp

A lamp with physical and luminous characteristics suited for projection systems (e.g., motion picture projectors, slide projectors, and microfilm viewers).

6.1.5 photoflood lamp

An incandescent filament lamp of high correlated color temperature for lighting subjects for photography or videography.

6.1.6 daylight lamp

A lamp producing a spectral distribution approximating that of a specified daylight.

6.1.7 ultraviolet lamp

A lamp that emits a significant portion of its radiative power in the ultraviolet (UV) part of the spectrum; the visible radiation is not of principal interest.

6.1.7.1 ultraviolet germicidal lamp

An ultraviolet (UV) lamp that is used for inactivation of pathogens by direct radiation.

6.1.7.2 tanning lamp

An ultraviolet (UV) lamp that emits a significant portion of its radiative power in the UV-A band (315 to 400 nm) and/or UV-B band (280 to 315 nm).

6.1.7.3 blacklight lamp

An ultraviolet (UV) lamp that emits a significant portion of its radiative power in the UV-A band (315 to 400 nm).

6.1.8 infrared lamp

A lamp that radiates predominantly in the infrared (IR); the visible radiation is not of principal interest.

6.1.9 PAR lamp

A designation that means “parabolic aluminized reflector,” most often applied to pressed hard glass, two-part (lens and reflector) reflector lamps.

Source: ANSI C78.21 (R2016), American National Standard for Electric Lamps—PAR and R Shapes. Rosslyn, VA: National Electrical Manufacturers Association; 2016.

6.1.10 spectral (spectroscopic) lamp

A discharge lamp that emits a significant portion of its radiative power in a line spectrum and that, in combination with filters, may be used to obtain monochromatic radiation.

6.1.11 laboratory reference standard

The highest-ranking order of standards at each laboratory.

6.1.12 bulb

(See **Section 6.1 lamp**.)

6.1.13 tube

(See **Section 6.1 lamp**.)

6.2 light center

The center of the smallest sphere that would completely contain the light-emitting element of the lamp.

6.2.1 light center length (LCL)

The distance from the light center to a specified reference point on the lamp.

6.3 incandescence

The self-emission of radiant energy in the visible spectrum due to the thermal excitation of atoms or molecules.

6.3.1 incandescent filament lamp

A lamp in which light is produced by a filament heated to incandescence by an electric current.

Note: Normally, the filament is of coiled or coiled-coil (doubly coiled) tungsten wire. However, it may be uncoiled wire, a flat strip, or of material other than tungsten.

6.3.1.1 vacuum lamp

An incandescent lamp in which the filament operates in an evacuated bulb.

6.3.1.2 gas-filled lamp

An incandescent lamp in which the filament operates in a bulb filled with one or more inert gases.

6.3.1.3 tungsten-halogen lamp

A gas-filled tungsten filament incandescent lamp containing halogens or halogen compounds and utilizing the halogen regenerative cycle to prevent blackening of the lamp envelope during life.

Note: The tungsten-iodine lamp (UK) and quartz-halogen lamp (USA)* belong to this category.

6.3.1.4 ribbon filament lamp

An incandescent lamp in which the luminous element is a tungsten ribbon.

Note: This type of lamp is often used as a standard in pyrometry and radiometry.

6.3.1.5 halogen infrared (HIR)

Lamp technology that improves the efficiency of a halogen filament lamp by employing a film on the halogen capsule that directs heat from the filament back onto the filament, resulting in improved lamp efficacy.

6.4 luminescence

Any emission of light not ascribable directly to incandescence.

6.4.1 fluorescence

The emission of light as the result of, and only during, the absorption of radiation of shorter

*Obsolete U.S. term.

wavelengths (time scale less than approximately 10^{-8} seconds).

6.4.2 phosphorescence

The emission of light as the result of the absorption of radiation, and continuing for a noticeable length of time after excitation (time scale longer than approximately 10^{-8} seconds).

6.4.3 electroluminescence

The emission of light from a phosphor excited by an electromagnetic field

6.5 electric-discharge lamp

A lamp in which light (or radiant energy near the visible spectrum) is produced by the passage of an electric current through a vapor or a gas.

Note: Electric-discharge lamps may be named after the filling gas or vapor that is responsible for the major portion of the radiation, e.g., mercury lamps, sodium lamps, neon lamps, and argon lamps. A second method of designating the electric-discharge lamps is by physical dimensions or operating parameters, e.g., short-arc lamps, high pressure lamps, and low pressure lamps. A third method of designating electric discharge lamps is by their application; in addition to lamps for illumination there are photochemical lamps, bactericidal lamps, blacklight lamps, sun lamps, and others.

6.5.1 gaseous discharge

The emission of light from gas atoms excited by an electric current.

6.5.2 arc discharge

An electric discharge characterized by high cathode current densities and a small voltage drop at the cathode.

Note: The cathode voltage drop is small compared with that in a glow discharge, and secondary emission plays only a small part in electron emission from the cathode.

6.5.2.1 carbon-arc lamp

An electric-discharge lamp employing an arc discharge between carbon electrodes. One or more of these electrodes may have a core of special chemicals that contribute importantly to the radiation.

6.5.2.2 hot-cathode lamp

An electric-discharge lamp whose mode of operation is that of an arc discharge. The cathodes may be heated by the discharge or by external means.

6.5.3 high intensity discharge (HID) lamp

An electric discharge lamp in which the light producing arc is stabilized by bulb wall temperature and in which the arc tube has a bulb wall loading in excess of 3 W/cm². HID lamps include groups of lamps known as mercury, metal halide, and high pressure sodium.

6.5.3.1 high pressure sodium (HPS) lamp

A high intensity discharge (HID) lamp in which light is produced by radiation from sodium vapor operating at a partial pressure about 1.33×10^4 Pa (100 Torr). Includes clear and diffuse-coated lamps.

6.5.3.2 metal halide lamp

A high intensity discharge (HID) lamp in which the major portion of the light is produced by radiation of metal halides and their products of dissociation – possibly in combination with metallic vapors such as mercury. Includes clear and phosphor-coated lamps.

6.5.3.3 mercury lamp

A high intensity discharge (HID) lamp in which the major portion of the light is produced by radiation from mercury operating at a partial pressure in excess of 10^5 Pa (approximately one atmosphere). Includes clear, phosphor-coated (mercury-fluorescent), and self-ballasted lamps.

6.5.3.3.1 mercury-fluorescent lamp (phosphor mercury lamp)

An electric discharge lamp having a high-pressure mercury arc in an arc tube, and an outer envelope coated with a fluorescing substance (phosphor), which transforms some of the ultraviolet energy generated by the arc into light.

6.5.3.3.2 phosphor mercury lamp

[See Section 6.5.3.3.1 mercury-fluorescent lamp (phosphor mercury lamp).]

6.5.3.4 hydrargyrum medium arc length iodide (HMI) lamp

An arc source utilizing a mercury vapor arc with metal halide additives to produce illumination in the range of 5000 K to 6000 K. Requires a ballast-and-ignitor system for operation.

6.5.3.5 compact source iodide (CSI) lamp

An arc source utilizing a mercury vapor arc with metal halide additives to produce

illumination with a correlated color temperature typically in the range of 5000 K to 6000 K. Requires a ballast-and-ignitor system for operation.

6.5.4 glow discharge

An electric discharge characterized by a low, approximately constant current density at the cathode (on the order of $10 \mu\text{A}/\text{mm}^2$), at low cathode temperature, and a large voltage drop (typically 50 volts or more). Secondary emission from the cathode is much greater than the thermionic emission.

Note: A distinction is made between the *normal cathode drop* (potential difference due to space charge near the cathode) that occurs when the glow does not cover the cathode completely (with constant current density) and which is independent of the discharge current, and the *abnormal cathode drop* that occurs when the glow covers the cathode completely (with increased current density) and which depends on the discharge current.

6.5.4.1 cold-cathode lamp

An electric-discharge lamp whose mode of operation is that of a glow discharge and which has electrodes so spaced that most of the light comes from the positive column between them.

6.5.4.1.1 cold-cathode fluorescent lamp

An electric-discharge lamp whose mode of operation is that of a glow discharge in mercury vapor wherein an internal fluorescing coating (phosphor) transforms ultraviolet energy from the positive column into light.

Notes:

- i. Approbation bodies in North America limit the operating current for cold cathode lamps to between 120 mA and 240 mA in specific applications.
- ii. The fluorescent cold-cathode type of lamp with a diameter greater than 20 mm is commonly used for architectural and general lighting applications, but through popular usage is commonly referred to as a *cold-cathode lamp*.
- iii. Discharge lamps used in signage and related applications may be either of the cold cathode or of the fluorescent cold-cathode type. Through popular usage, signage lamps that are less than 20 mm in diameter are frequently referred to as *neon lamps*.

6.5.4.2 glow lamp

An electric-discharge lamp whose mode of operation is that of a glow discharge and in

which light is generated in the space close to the electrodes.

6.5.5 low pressure sodium (LPS) lamp

A discharge lamp in which light is produced by radiation from sodium vapor operating at a partial pressure of 0.1 to 1.5 Pa (approximately 10^{-3} to 10^{-2} Torr).

6.5.6 low pressure mercury lamp

A discharge lamp (with or without a phosphor coating) in which the partial pressure of the mercury vapor does not exceed 100 Pa during operation.

6.5.6.1 fluorescent lamp

A low-pressure mercury electric-discharge lamp in which a fluorescing coating (phosphor) transforms some of the ultraviolet energy generated by the discharge into light.

6.5.6.1.1 preheat (switch start) fluorescent lamp

A fluorescent lamp designed for operation in a circuit requiring a manual or automatic starting switch to preheat the electrodes in order to start the arc.

6.5.6.1.2 instant-start fluorescent lamp

A fluorescent lamp designed for starting by a high voltage without preheating of the electrodes.

Note: Referred to in the UK as a *cold-start lamp*.

6.5.6.1.3 rapid-start fluorescent lamp

A fluorescent lamp designed for operation with a ballast that provides a low-voltage winding for preheating the electrodes to initiate the arc without a starting switch or the application of high voltage.

6.5.6.1.4 compact fluorescent lamp

A fluorescent lamp with a small diameter glass tube (T5 or smaller) that is folded, bent, or bridged to create a long discharge path in a small volume. The lamp designs generally include an amalgam and a cold chamber, or a cold spot, to control the mercury vapor pressure and light output.

6.5.6.1.5 electrodeless fluorescent lamp

A fluorescent lamp in which energy is coupled into the discharge vessel by means of an induced electromagnetic field without electron emitting

electrodes. It includes lamps based on induction coupling and some microwave discharges.

6.5.6.1.6 switch-start fluorescent lamp

[See **Section 6.5.6.1.1 preheat (switch start) fluorescent lamp.**]

6.5.6.2 low-pressure mercury germicidal lamp

A low-pressure mercury lamp in which the envelope has high transmittance for 254-nm radiation (see also **Section 6.1.7.1**).

6.5.7 ballast

A device used with an electric-discharge lamp to provide the necessary circuit conditions (voltage, current, and wave form) for starting and operating.

6.5.7.1 starter

A device used in conjunction with a ballast for the purpose of starting an electric-discharge lamp.

6.5.7.2 ignitor

A device, either by itself or in association with other components, that generates voltage pulses to start discharge lamps without preheating of electrodes.

6.5.8 flashtube

A tube of glass or fused quartz with electrodes at the ends and filled with a gas, usually xenon. It is designed to produce high intensity light flashes of extremely short duration.

6.5.8.1 stroboscopic lamp (strobe light)

A flash tube designed for repetitive flashing.

6.5.9 arc lamp

A discharge lamp in which the light is emitted by an arc discharge or by its electrodes.

Note: The electrodes may be either of carbon (operating in air) or of metal.

6.5.9.1 short-arc (compact-arc) lamp

An arc lamp in which the distance between the electrodes is small (on the order of 1 to 10 mm).

Note: This type of lamp (e.g., xenon or mercury) generally has an arc tube containing gas

at very high pressure.

6.5.9.2 long-arc lamp

An arc lamp in which the distance between the electrodes is large, usually greater than 10 mm.

Note: This type of lamp (e.g., xenon) generally has an arc tube containing gas at high pressure. The arc fills the discharge tube and is therefore wall-stabilized.

6.5.9.3 compact-arc lamp

[See Section 6.5.9.1 short-arc (compact-arc) lamp.]

6.5.10 amalgam

An alloy of mercury with another metal or metals. Can be used in fluorescent lamps to control the mercury vapor pressure in the discharge. The alloy absorbs or releases mercury in relation to the amalgam temperature.

6.5.11 self-ballasted lamps

Any arc discharge lamp of which the current-limiting device is an integral part.

6.5.12 electric discharge

(See Section 6.5.2 arc discharge; Section 6.5.1 gaseous discharge; or Section 6.5.4 glow discharge.)

6.6 photoflash lamp

A lamp in which combustible metal or other solid material is burned in an oxidizing atmosphere to produce light of high intensity and short duration for photographic purposes.

6.7 laser

An acronym for *light amplification by stimulated emission of radiation*. A laser produces a highly monochromatic and coherent (spatially and temporally) beam of radiation. A steady oscillation of nearly a single electromagnetic mode is maintained in a volume of an active material bounded by highly reflecting surfaces called a *resonator*. The frequency of oscillation varies according to the material used and by the methods of initially exciting or pumping the material.

6.8 light emitting diode (LED)

A p-n junction solid-state diode whose radiated output is a function of its physical construction, material used, and exciting current. The optical emission may be in the ultraviolet, visible, or infrared regions.

6.8.1 LED die

A small block of light-emitting semiconducting material on which a functional light emitting diode (LED) circuit is fabricated.

6.8.2 bin

A restricted range of LED performance characteristics used to delimit a subset of LEDs near a nominal LED performance as identified by chromaticity, photometry, and/or forward-voltage performance.

Note: As the result of small but meaningful variations in the manufacturing process of LED wafers and subsequent dies, the electrical and photometric characteristics of LEDs may vary from LED to LED, even when the dies are from the same wafer. LEDs are sorted or binned in accordance with these characteristics, but there is no existing consensus standard for binning.

6.8.3 LED power source

A transformer, power supply, battery, or other device capable of providing current, voltage, or power within its design limits. This device contains no additional control capabilities.

6.8.3.1 LED power supply

An electronic device capable of providing and controlling current, voltage, or power within its design limits.

6.8.4 LED control circuitry

Electronic components designed to control a power source by adjusting output voltage, current, or duty cycle to switch or otherwise control the amount and characteristics of the electrical energy delivered to an LED package (component) or a light emitting diode (LED) array (module). LED control circuitry does not include a power source.

6.8.4.1 LED driver

A device composed of a power source and light emitting diode (LED) control circuitry designed to operate an LED package (component), an LED array (module), or an LED lamp.

6.8.4.2 LED driver, Class II

A light emitting diode (LED) driver that operates within Class II limits as defined by the latest version of the United States National Electrical Code (NEC) and the Canadian Electrical Code (CEC).

6.8.5 LED luminaire

A complete lighting unit consisting of light emitting diode (LED)-based light emitting elements and a matched driver together with parts to distribute light, to position and protect the light emitting elements, and to connect the unit to a branch circuit. The LED-based light emitting elements may take the form of LED packages (components), LED arrays (modules), an LED light engine, or LED lamps. The LED luminaire is intended to connect directly to a branch circuit.

6.8.5.1 LED package

An assembly of one or more light emitting diode (LED) dies that includes wire bond or other type of electrical connections, possibly with an optical element and thermal, mechanical, and electrical interfaces. Power source and ANSI standardized base are not incorporated into the device. The device cannot be connected directly to the branch circuit.

6.8.5.2 LED array or module

An assembly of light emitting diode (LED) packages (components), or dies on a printed circuit board or substrate, possibly with optical elements and additional thermal, mechanical, and electrical interfaces that are intended to connect to the load side of an LED driver. Power source and ANSI standard base are not incorporated into the device. The device cannot be connected directly to the branch circuit.

6.8.5.3 LED lamp, non-integrated

An assembly composed of a light emitting diode (LED) array (module) or LED packages (components), and an ANSI standard base. The device is intended to connect to the LED driver of an LED luminaire through an ANSI standard lamp-holder (socket). The device cannot be connected directly to the branch circuit.

6.8.5.4 LED lamp, integrated

An integrated assembly composed of light emitting diode (LED) packages (components) or LED arrays (modules), as well as an LED driver, an ANSI standard base, and other optical, thermal, mechanical and electrical components. The device is intended to connect directly to the branch circuit through a corresponding ANSI standard lamp-holder (socket).

6.8.5.5 LED light engine

An integrated assembly composed of light emitting diode (LED) packages (components) or LED arrays (modules), as well as an LED driver and other optical, thermal, mechanical and electrical components. The device is intended to connect directly to the branch

circuit through a custom connector compatible with the LED luminaire for which it was designed. It does not use an ANSI standard base.

6.8.5.6 hybrid LED luminaire

A luminaire that incorporates LED-based light-emitting elements and another type of light source, such as incandescent or fluorescent lamps.

6.9 rated lamp life

The life value assigned to a particular type lamp. This is commonly a statistically determined estimate of median operational life.

6.10 median life

The total operating time at which, under normal operating conditions, 50 percent of any large group of initially installed lamps is expected to still be operating.

6.11 organic light emitting diode (OLED)

A thin-film light emitting diode composed of carbon-based material layered between two electrode surfaces and which serves as a diffuse area source.

6.12 solid state lighting (SSL)

Lighting that uses light-emitting diodes (LED), organic light-emitting diodes (OLED), or polymer light-emitting diodes (PLED) as sources of illumination.

6.13 excimer lamp

A discharge source of primarily ultraviolet radiation produced by excited dimer* ("excimer") molecules as they transition from their excited state to their ground state.

Notes:

1. The wavelength of UV radiation emitted depends on the composition of the working excimer molecule (often a combination of a noble gas and a halogen gas).
2. One example is the Krypton-chloride excimer (chemical symbol: KrCl*) lamp, which emits primarily around 222 nm, with lesser emissions at longer UV wavelengths. (The asterisk in the symbol indicates that the molecule is an excimer.)

* A dimer is a molecule or molecular complex consisting of two identical molecules linked together.

7.0 SURFACES AND MEDIA FOR LIGHT CONTROL

7.1 diffusing surfaces and media

Those surfaces and media that redistribute at least some of the incident flux by scattering.

7.1.1 perfect diffusion

That diffusion in which flux is uniformly scattered in accord with Lambert's cosine law (see **Section 7.1.1.1**).

7.1.1.1 Lambert's cosine law

$$(I_{\theta} = I_0 \cos \theta)$$

The law stating that the luminous intensity in any direction from an element of a perfectly diffusing surface varies as the cosine of the angle between that direction and the perpendicular to the surface element.

7.1.1.2 Lambertian surface

A surface that emits or reflects light in accordance with Lambert's cosine law. A Lambertian surface has the same luminance regardless of viewing angle.

7.1.2 complete diffusion

That diffusion in which the diffusing medium completely redirects the incident flux by scattering; i.e., no incident flux can remain in an image-forming state.

7.1.3 incomplete diffusion (partial diffusion)

That diffusion in which the diffusing medium partially redirects the incident flux by scattering while the remaining fraction of incident flux is redirected without scattering; i.e., a fraction of the incident flux can remain in an image-forming state.

7.1.4 wide-angle diffusion

That diffusion in which flux is scattered at angles far from the direction that the flux would take by regular (specular) reflection (see **Section 7.3.1**) or regular transmission.

7.1.5 narrow-angle diffusion

That diffusion in which flux is scattered at angles near the direction that the flux would take by regular (specular) reflection or regular transmission.

7.1.6 partial diffusion

[See **Section 7.1.3 incomplete diffusion (partial diffusion)**.]

7.2 redirecting surfaces and media

Those surfaces and media that change the direction of the flux without scattering the redirected flux.

7.2.1 lens

A transmitting element used to change the direction and control the distribution of light rays.

Also, the part of the eye that allows objects at different distances to be focused onto the retina.

7.2.2 bidirectional scatter distribution function (BSDF)

The general mathematical function or matrix of data that describes the directional scattering of light by a surface or complex glazing system through both reflection and transmission across input and output angles.

In practice, this phenomenon is usually split into the reflected and transmitted components, which are quantified in a bidirectional reflectance distribution function (BRDF) and bidirectional transmittance distribution function (BTDF). (See **Sections 7.3.3.17** and **7.5.3.16**, respectively.)

7.3 reflection

A general term for the process by which the incident flux leaves a (stationary) surface or medium from the incident side, without change in frequency.

Note: Reflection is usually a combination of regular (specular) and diffuse reflection (see **Sections 7.3.1** and **7.3.2**).

7.3.1 specular reflection

That process by which incident flux is redirected at the specular angle (see **Section 7.3.3.1**). Also called *regular reflection*.

7.3.1.1 specular angle

That angle between the perpendicular to the surface and the reflected ray that is numerically equal to the angle of incidence, and that lies in the same plane as the incident ray and the perpendicular but on the opposite side.

7.3.2 diffuse reflection

That process by which incident flux is redirected over a range of angles (see also **Section 7.1**).

7.3.3 reflectance, ρ (of a surface or medium)

$$\rho = \Phi_r / \Phi_i$$

The ratio of the reflected flux to the incident flux. Reflectance is a function of:

- Geometry:
 - of the incident flux
 - of collection for the reflected flux
- Spectral distribution:
 - characteristic of the incident flux
 - weighting function for the collected flux
- Polarization:
 - of the incident flux
 - component defined for the collected flux

Notes:

- (i) Unless the state of polarization for the incident flux and the polarized component of the reflected flux are stated, it shall be considered that the incident flux is unpolarized and that the total reflected flux (including all polarizations) is evaluated.
- (ii) Spectral reflectance depends only on the beam geometry and the character of the reflecting surface (and on polarization). Luminous reflectance also is a function of the spectral distribution of the incident flux.
- (iii) If no qualifying geometric adjective is used, *reflectance for hemispherical collection* is meant (see **Sections 7.3.3.1 to 7.3.3.9** for other modifying adjectives).
- (iv) Certain of the reflectance terms are theoretically imperfect and are recognized only as practical concepts to be used when applicable. Physical measurements of the incident and reflected flux are always biconical in nature. Directional reflectances (see **Sections 7.3.3.3 and 7.3.3.6 through 7.3.3.9**) cannot exist since one component is finite while the other is infinitesimal; here the bidirectional reflectance-distribution function (see **Section 7.3.3.17**) is required. However, the concepts *directional reflectance* and *hemispherical reflectance* have practical application in instrumentation, measurements, and calculations when including the aspect of the nearly zero or nearly 2π conical angle would increase complexity without appreciably affecting the immediate results.
- (v) In each case of conical incidence or collection, the solid angle is not restricted to a right cone, but may be of any cross-section including rectangular, a ring, or a combination of two or more solid angles.
- (vi) For the following geometrical reflectance properties (see **Sections 7.3.3.1 to 7.3.3.10**,

inclusive) it is assumed that the radiance (luminance) is isotropic over the specified solid angle of incidence. Otherwise, the property is a function of the directional distribution of incident radiant (luminous) flux as well as the beam geometry and the character of the reflecting surface.

7.3.3.1 bihemispherical reflectance

$$\rho(2\pi; 2\pi)$$

The ratio of reflected flux collected over the entire hemisphere (see **Figure 5**) to the flux incident from the entire hemisphere. (See **Section 7.3.3.10.**)

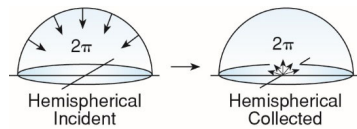


Figure 5.

7.3.3.2 hemispherical-conical reflectance

$$\rho(2\pi; \omega_r)$$

The ratio of reflected flux collected over a conical solid angle to the incident flux from the entire hemisphere. (See **Figure 6.**)

Note: The direction and extent of the cone must be specified.

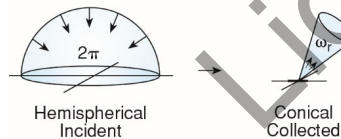


Figure 6.

7.3.3.3 hemispherical-directional reflectance

$$\rho(2\pi; \theta_r, \phi_r)$$

The ratio of reflected flux collected over an element of solid angle surrounding the given direction to the incident flux from the entire hemisphere. (See **Figure 7.**)

Note: The direction of collection and the size of the solid angle “element” must be specified.

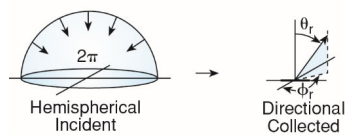


Figure 7.

7.3.3.4 conical-hemispherical reflectance

$$\rho(\omega_i; 2\pi)$$

The ratio of reflected flux collected over the entire hemisphere to the incident flux limited to a conical solid angle. (See **Figure 8**).

Note: The direction and extent of the cone must be specified.

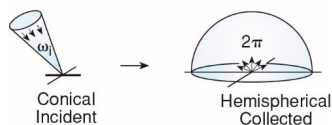


Figure 8.

7.3.3.5 biconical reflectance

$$\rho(\omega_i; \omega_r)$$

The ratio of reflected flux collected through a conical solid angle to the incident flux limited to a conical solid angle (see **Figure 9**).

Note: The direction and extent of each cone must be specified.

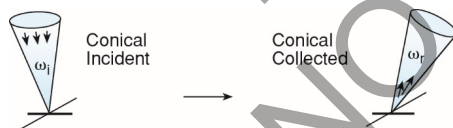


Figure 9.

7.3.3.6 conical-directional reflectance

$$\rho(\omega_i; \theta_r, \phi_r)$$

The ratio of reflected flux collected over an element of solid angle surrounding the given

direction to the incident flux limited to a conical solid angle. (See **Figure 10**.)

Note: The direction and the extent of the cone must be specified, and the direction of collection and size of the solid angle "element" must be specified.



Figure 10.

7.3.3.7 directional-hemispherical reflectance

$$\rho(\theta_i, \phi_i; 2\pi)$$

The ratio of reflected flux collected over the entire hemisphere to essentially collimated incident flux. (See **Figure 11**.)

Note: The direction of incidence must be specified.

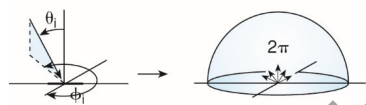


Figure 11. [Annotations missing. Label for left: Directional Incident. Label for right: Hemispherical Collected.]

7.3.3.8 directional-conical reflectance

$$\rho(\theta_i, \phi_i; \omega_r)$$

The ratio of reflected flux collected through a conical solid angle to essentially collimated incident flux. (See **Figure 12**.)

Note: The direction of incidence must be specified, and the direction and extent of the cone must be specified.

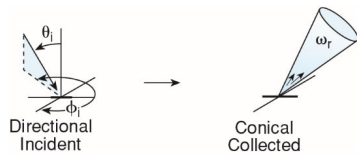


Figure 12.

7.3.3.9 bidirectional reflectance

$$\rho(\theta_i, \phi_i; \theta_r, \phi_r)$$

The ratio of reflected flux collected over an element of solid angle surrounding the given direction to essentially collimated incident flux. (See **Figure 13**.)

Note: The directions of incidence and collection and the size of the solid angle "element" of collection must be specified. In each case of conical incidence or collection, the solid angle is not restricted to a right circular cone, but may be of any cross-section, including rectangular, a ring, or a combination of two or more solid angles.

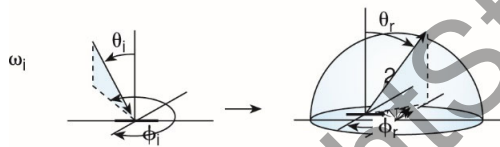


Figure 13.

7.3.3.10 hemispherical reflectance*

The ratio of all the flux leaving a surface or medium by reflection to the incident flux.

7.3.3.11 specular reflectance

The ratio of the flux leaving a surface or medium by specular reflection to the incident flux. Also called *regular reflectance*.

*This term has been deprecated; it is retained here for reference of literature.

7.3.3.11.1 specularly reflective

A material that reflects light in a mirror-like manner, according to Fresnel's Law of reflection, where the angle of reflection of a ray from the surface equals the angle of incidence on that surface.

7.3.3.12 diffuse reflectance

The ratio of the flux leaving a surface or medium by diffuse reflection to the incident flux.

Note: Provision for the exclusion of regularly (specularly) reflected flux, which is nearly always present, must be clearly described.

7.3.3.12.1 diffusely reflective

Able to scatter incident light in a Lambertian, or near-Lambertian, manner.

7.3.3.13 spectral reflectance, $\rho(\lambda)$

$$\rho(\lambda) = \Phi_{r\lambda} / \Phi_{i\lambda}$$

The ratio of the reflected flux to the incident flux at a particular wavelength, λ , or within a small band of wavelengths, $\Delta\lambda$, about λ .

Note: The various geometrical aspects of reflectance may each be considered restricted to a specific region of the spectrum and may be so designated by the addition of the adjective "spectral."

7.3.3.14 luminous reflectance

Any of the geometric aspects of reflectance in which both the incident and transmitted flux are weighed by the luminous efficiency of radiant flux, $V(\lambda)$.

Note: Unless otherwise qualified, the term *reflectance* is interpreted as *luminous reflectance*.

7.3.3.15 effective ceiling cavity reflectance, ρ_{cc}

A number giving the combined reflectance effect of the wall and ceiling reflectances of the ceiling cavity. (See Section 9.4.5.1).

7.3.3.16 effective floor cavity reflectance, ρ_{fc}

A number giving the combined reflectance effect of the wall and floor reflectances of the floor cavity. (See Section 9.4.5.2).

7.3.3.17 bidirectional reflectance-distribution function (BRDF), f_r .¹⁴

The ratio of the differential luminance of a ray $dL_r(\theta_r, \phi_r)$ reflected in a given direction (θ_r, ϕ_r) to the differential luminous flux density $dE_i(\theta_i, \phi_i)$ incident from a given direction (θ_i, ϕ_i) that produces it. Units: sr^{-1} . (See Figure 14.)

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) \equiv [dL_r(\theta_r, \phi_r)]/[dE_i(\theta_i, \phi_i)]$$

$$= [dL_r(\theta_r, \phi_r)]/[L_i(\theta_i, \phi_i)d\Omega_i] ,$$

where:

$$d\Omega \cdot d\omega = \cos \theta$$

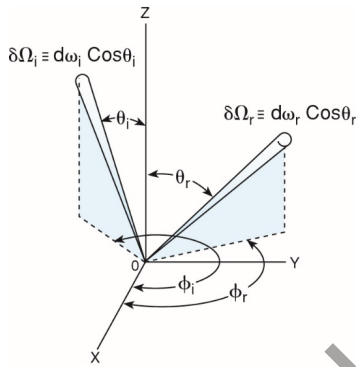


Figure 14.

Notes:

- (i) This distribution function is the basic parameter for describing (geometrically) the reflecting properties of an opaque surface element (negligible internal scattering).
- (ii) It may have any positive value and will approach infinity in the specular direction for ideally specular reflectors.
- (iii) The spectral and polarization aspects must be defined for complete specification, since the BRDF as given above only defines the geometric aspects.

7.3.3.18 reflectance factor, R

The ratio of the flux actually reflected by a sample surface to that which would be reflected into the same reflected-beam geometry by an ideal (glossless), perfectly diffuse (Lambertian), completely reflecting standard surface irradiated in exactly the

same way as the sample.

The analogies to reflectance may be noted in the fact that: nine canonical forms are possible paralleling **Sections 7.3.3.1 to 7.3.3.9**; *spectral* may be applied as a modifier; and it may be a luminous or a radiant reflectance factor. It should also be noted that reflectance cannot exceed unity, but reflectance factor may have any value from zero to values approaching infinity.

7.3.3.19 reflectivity

Reflectance of a layer of a material of such a thickness that there is no change of reflectance with increase in thickness.

7.3.3.20 regular (specular) reflectance

[See **Section 7.3.3.11 specular reflectance**.]

7.3.4 total internal reflection (TIR)

Total reflection of a light ray at the boundary between two media occurring when the angle of incidence in medium 2 exceeds the angle whose sine equals n_1/n_2 , the ratio of indices of refraction when the ray is propagating in the direction from medium 2 toward medium 1.

7.3.5 regular (specular) reflection

[See **Section 7.3.1 specular reflection**.]

7.4 refraction

The process by which the direction of a ray of light changes as it passes obliquely from one medium to another in which its speed is different.

7.5 transmission

A general term for the process by which incident flux leaves a surface or medium on a side other than the incident side, without change in frequency.

Note: Transmission through a medium is often a combination of regular transmission and diffuse transmission. (See **Sections 7.4.1 and 7.4.2**.)

7.5.1 specular transmission

That process by which incident flux passes through a surface or medium without scattering. Sometimes called *regular transmission*.

7.5.2 diffuse transmission

That process by which the incident flux passing through a surface or medium is scattered.

7.5.3 transmittance, τ (of a medium)

$$\tau = \Phi_t / \Phi_i$$

The ratio of the transmitted flux to the incident flux. It should be noted that *transmittance* refers to the ratio of flux emerging to flux incident; therefore, reflections at the surface as well as absorption within the material operate to reduce the transmittance. Transmittance is a function of:

- Geometry:
 - of the incident flux
 - of collection for the transmitted flux
- (2) Spectral distribution:
 - characteristic of the incident flux
 - weighting function for the collected flux
- (3) Polarization:
 - of the incident flux
 - component defined for the collected flux

Notes:

- (i) Unless the state of polarization for the incident flux and the polarized component of the transmitted flux are stated, it shall be considered that the incident flux is unpolarized and that the total transmitted flux is evaluated.
- (ii) Spectral transmittance depends on the beam geometry, the character of the transmitting surfaces and media, and polarization. In addition, luminous transmittance is a function of the spectral distribution of the incident beam.
- (iii) If no qualifying geometric adjective is used, *transmittance for hemispherical collection* is meant (see **Sections 7.5.3.1 to 7.5.3.9** for other modifying adjectives).
- (iv) In each case of conical incidence or collection, the solid angle is not restricted to a right circular cone, but may be of any cross-section, including rectangular, a ring, or a combination of two or more solid angles.
- (v) These concepts must be applied with care, if the area of the transmitting element is not large compared to its thickness, due to internal transmission across the boundary of the

area.

- (vi) For all of the following geometrical quantities (**Sections 7.4.3.1 to 7.4.3.9**, inclusive) it is assumed that the radiance (luminance) is isotropic over the specified solid angle of incidence. Otherwise, the property is a function of the directional distribution of incident radiant (luminous) flux as well as the beam geometry and the character of the transmitting surfaces and/or media.
- (vii) The following breakdown of transmittance quantities is applicable only to the transmittance of thin films with negligible internal scattering so that the transmitted radiation emerges from a point that is not significantly separated from the point of incidence of the ray that produces the transmitted ray(s). The governing considerations are similar to those for application of the bidirectional reflectance-distribution function (BRDF), rather than the bidirectional scattering-surface reflectance-distribution function (BSSRDF).²²

7.5.3.1 bihemispherical transmittance

$$\tau(2\pi; 2\pi)$$

The ratio of transmitted flux collected over the entire hemisphere to the incident flux from the entire hemisphere. (See **Figure 15**.)

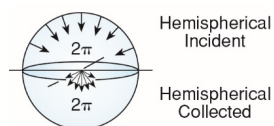


Figure 15.

7.5.3.2 hemispherical-conical transmittance

$$\tau(2\pi; \omega_i)$$

The ratio of transmitted flux collected over a conical solid angle to the incident flux from the entire hemisphere. (See **Figure 16**.)

Note: The direction and extent of the cone must be specified.

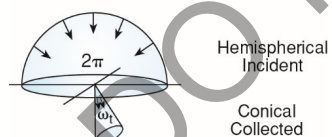


Figure 16.

7.5.3.3 hemispherical-directional transmittance

$$\tau(2\pi; \cos \theta_i; \phi_i)$$

The ratio of transmitted flux collected over an element of solid angle surrounding the given direction to the incident flux from the entire hemisphere. (See **Figure 17.**)

Note: The direction of collection and size of the solid angle "element" must be specified.

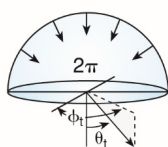


Figure 17.

7.5.3.4 conical-hemispherical transmittance

$$\tau(\omega_i; 2\pi)$$

The ratio of transmitted flux collected over the entire hemisphere to the incident flux limited to a conical solid angle. (See **Figure 18.**)

Note: The direction and extent of the cone must be specified.

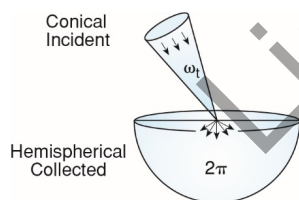


Figure 18.

7.5.3.5 biconical transmittance

$$\tau(\omega_i; \omega_t)$$

The ratio of transmitted flux collected through a conical solid angle to the incident flux limited to a conical solid angle. (See **Figure 19.**)

Note: The direction and extent of each cone must be specified.

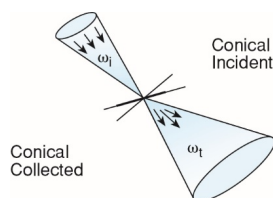


Figure 19.

7.5.3.6 conical-directional transmittance

$$\tau(\omega_i; \theta; \phi)$$

The ratio of transmitted flux, collected over an element of solid angle surrounding the direction, to the incident flux limited to a conical solid angle. (See **Figure 20**.)

Note: The direction and extent of the cone must be specified, and the direction of collection and size of the solid angle "element" must be specified.

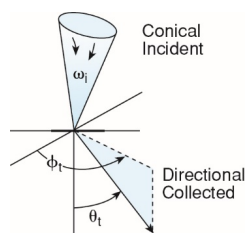


Figure 20.

7.5.3.7 directional-hemispherical transmittance

$$\tau(\theta; \phi; 2\pi)$$

The ratio of transmitted flux collected over the entire hemisphere to essentially collimated incident flux. (See **Figure 21**).

Note: The direction of incidence must be specified.

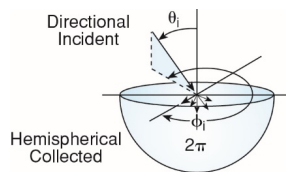


Figure 21.

7.5.3.8 directional-conical transmittance

$$\tau(\theta, \phi; \omega)$$

The ratio of transmitted flux collected through a conical solid angle to essentially collimated incident flux. (See **Figure 22.**)

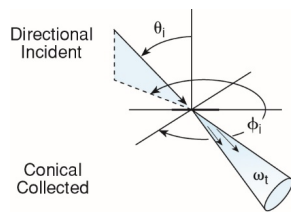


Figure 22.

Note: The direction of incidence must be specified, and the direction and extent of the cone must be specified.

7.5.3.9 bidirectional transmittance

$$\tau(\theta, \phi; \theta_c, \phi_c)$$

The ratio of incident flux collected over an element of solid angle surrounding the given direction to essentially collimated incident flux. (See **Figure 23.**)

Note: The directions of incidence and collection and the size of the solid angle "elements" must be specified.

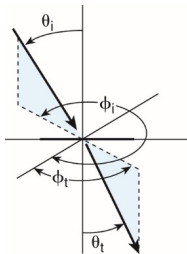


Figure 23.

7.5.3.10 luminous transmittance

Any of the geometric aspects of transmittance in which the incident and transmitted flux are weighed by the luminous efficiency of radiant flux [$V(\lambda)$].

Note: Unless otherwise qualified, the term *transmittance* means *luminous transmittance*.

7.5.3.11 hemispherical transmittance*

The ratio of the transmitted flux leaving a surface or medium to the incident flux.

Note: If *transmittance* is not preceded by an adjective descriptive of the angles of view, *hemispherical transmittance* is implied.

7.5.3.12 specular transmittance

The ratio of the flux passing through a surface or medium without scattering to the incident flux. Also called *regular transmittance*.

7.5.3.12.1 specular angle (direction) for transmission

The exiting angle of a ray that passes through a surface or medium without scattering.

7.5.3.13 diffuse transmittance

The ratio of the diffusely transmitted flux leaving a surface or medium to the incident flux.

Note: Provision for exclusion of regularly (non-diffusely) transmitted flux must be clearly

*This term has been deprecated; it is retained here for reference of literature.

described.

7.5.3.14 spectral transmittance, $\tau(\lambda)$

$$\tau(\lambda) = \Phi_{t\lambda} / \Phi_{i\lambda}$$

The ratio of the transmitted flux to the incident flux at a particular wavelength, λ , or within a small band of wavelengths, $\Delta\lambda$, about λ .

Note: The various geometrical aspects of transmittance may each be considered restricted to a specific region of the spectrum and may be so designated by the addition of the adjective "spectral."

7.5.3.14.1 spectral transmittance distribution

A tabular or graphical representation of the radiant power transmitted through a medium at each wavelength, or within bands of wavelengths, across the electromagnetic spectrum, or some portion of it.

7.5.3.15 filter

A device for changing, by transmission or reflection, the magnitude and/or the spectral composition of the flux incident on it. Filters are called *selective* (or *colored*) or *neutral*, according to whether or not they alter the spectral distribution of the incident flux.

7.5.3.15.1 dichroic filter

A filter that transmits certain wavelengths and reflects those not transmitted; the absorption is small over the wavelength band of interest.

7.5.3.15.2 opaque medium

A medium that transmits no radiation in the spectral range of interest.

7.5.3.15.3 translucent medium

A medium that transmits visible radiation largely by diffuse transmission, so that objects are not seen distinctly through it.

7.5.3.15.4 transparent medium

A medium in which the transmission is mainly regular and which usually has a high regular transmittance in the spectral range of interest.

Note: Objects may be seen distinctly through a medium which is transparent in the visible region, if the geometric form of the medium is suitable.

7.5.3.16 bidirectional transmittance-distribution function (BTDF), f_t

The ratio of the differential luminance $dL_t(\theta_t, \phi_t)$ for a ray transmitted in a given direction (θ_t, ϕ_t) to the differential luminous flux density $dE_i(\theta_i, \phi_i)$, incident from a given direction (θ_i, ϕ_i) , that produces it. Units: sr^{-1} . (See Figure 23.)

$$f_t(\theta_i, \phi_i; \theta_t, \phi_t) \equiv \frac{dL_t(\theta_t, \phi_t)}{dE_i(\theta_i, \phi_i)} = \frac{dL_t(\theta_t, \phi_t)}{L_i(\theta_i, \phi_i) d\Omega_i},$$

where $d\Omega = d\omega \cdot \cos \theta$

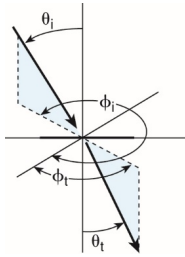


Figure 23.

Notes:

- (i) This distribution function is the basic parameter for describing (geometrically) the transmitting properties of a thin scattering film (with negligible internal scattering) so that the transmitted radiation emerges from a point that is not significantly separated from the point of incidence of the incident ray(s). The governing considerations are similar to those for application of the *bidirectional reflectance-distribution function* (BRDF), rather than the *bidirectional scattering-surface reflectance-distribution function* (BSSRDF). 
- (ii) It may have any positive value and will approach infinity in the direction for regular transmission (possibly refraction but without scattering).
- (iii) The spectral and polarization aspects must be defined for complete specification, since the BTDF as given above only defines the geometrical aspects.

7.5.3.17 visible transmittance (VT, VLT, or T_{vis})

The fraction of incident light that is transmitted through a material.

Also known as *visible light transmittance* (VLT).

7.5.3.18 translucent

Transmitting light with diffusion, so that objects on the opposite side are not seen clearly (or at all).

7.5.4 transillumination

The process whereby a non-opaque surface is illuminated from the side opposite to that from which it is viewed.

7.5.5 attenuation coefficient

(μ)

The decrement in flux per unit distance in a given direction within a medium and defined by the relation:

$$\phi(r) = \phi(0) \cdot e^{-\mu r},$$

where $\phi(r)$ is the flux at a distance r from a reference point having flux $\phi(0)$.

7.6 absorption

A general term for the process by which incident flux is converted to another form of energy, usually and ultimately to heat.

Note: All of the incident flux is accounted for by the processes of reflection, transmission, and absorption.

7.6.1 absorptance, α

$$\alpha = \Phi_a / \Phi_i$$

The ratio of the absorbed flux to the incident flux.

Note: The sum of the hemispherical reflectance, the hemispherical transmittance, and the absorptance is 1.0.

7.6.2 coefficient of attenuation, μ

(at a point in a given direction)

The decrement in flux per unit distance in a given direction within a medium. It is defined by the relationship $\Phi_x = \Phi_0 e^{[\exp(-\mu x)]}$ where Φ_x is the flux at any distance x from a reference point having flux Φ_0 . More generally:

$$\Phi_x = \Phi_0 \exp\left[-\int_0^x \mu(x) dx\right],$$

where the coefficient varies from point to point; $\mu = \mu(x)$ along the path.

7.7 luminance factor, β

The ratio of the luminance of a surface or medium under specified conditions of incidence, observation, and light source, to the luminance of a completely reflecting or transmitting, perfectly diffusing surface or medium under the same conditions.

Note: Reflectance or transmittance cannot exceed unity, but luminance factor may have any value from zero to values approaching infinity.

7.8 polarization

The process by which unpolarized radiation is polarized. It can be accomplished by a reflection and/or transmission processes. (See **Section 7.8.1 polarized radiation.**)

7.8.1 polarized radiation

Radiation whose electromagnetic field, which is transverse, is oriented in defined directions. The polarization can be rectilinear, elliptical, or circular.

7.8.2 linear polarization

The process by which the transverse vibrations of light waves are oriented in or parallel to the same plane. Polarization may be obtained by using either transmitting or reflecting media.

8.0 TESTING PROCEDURES

8.1 luminous standards

8.1.1 primary standard

A light source by which the unit of light is established and from which the values of other standards are derived. This order of standard also is designated as the national standard.

Note: A satisfactory primary (national) standard must be reproducible from specifications (see **Section 3.4.1**). Primary (national) standards usually are found in national physical laboratories such as the U.S. National Institute of Standards and Technology (NIST).

8.1.2 secondary standard

A stable light source calibrated directly or indirectly by comparison with a primary standard. This order of standard also is designated as a reference standard.

Notes:

- (i) National secondary (reference) standards are maintained at national physical laboratories; laboratory secondary (reference) standards are maintained at other photometric laboratories.
- (ii) A self-calibrated detector may be used to derive a secondary standard.

8.1.3 working standard

A standardized light source for regular use in photometry.

8.1.4 comparison lamp

A light source having a constant, but not necessarily known, luminous intensity with which standard and test lamps are successively compared.

8.1.5 national standard of light

(See **Section 8.1.1 primary standard.**)

8.1.6 reference standard

(See **Section 8.1.2 secondary standard.**)

8.2 auxiliary Equipment

The reactor or ballast and, if required, starting device needed for the proper operation of discharge lamps. The auxiliary equipment is used to limit lamp current and provide proper starting voltage and is essential because discharge lamps have a negative volt-ampere characteristic.

8.2.1 reference ballast

A ballast that is specially constructed, having certain prescribed characteristics, and which is used for testing electric-discharge lamps and other ballasts.

8.3 radiometry

The measurement of quantities associated with radiant energy and power.

8.3.1 spectroradiometer

An instrument for measuring radiant flux as a function of wavelength.

8.4 photometry

The measurement of quantities associated with light.

Note: Photometry may be visual, in which the eye is used to make a comparison, or physical, in which

measurements are made by means of physical receptors.

8.4.1 photometer

An instrument for measuring photometric quantities such as luminance, luminous intensity, luminous flux, or illuminance.

8.4.1.1 visual photometer

A device or instrument with which luminance is measured by visually comparing the appearance of two fields.

Note: The two fields are usually viewed simultaneously side-by-side, and a direct brightness match is made. This is satisfactory when the color difference between the test field and the comparison field is small. However, when there is a significant color difference, direct brightness matching does not produce equal luminances of the two fields, and either a flicker photometer or the minimally distinct border (MDB) method of photometry provides more-valid measurements. In flicker photometry, the two fields are viewed alternately at such a rate that the color sensations either nearly or completely blend, and the flicker, due to brightness difference, is minimized by adjusting the comparison field. In the MDB method, two juxtaposed fields are adjusted until the border between the two is least conspicuous, even though the brightness of the two fields may appear unequal; the MDB setting yields equal luminances between the two fields.

8.4.1.2 physical receptor

A device in which incident radiation produces an observable physical change, including but not limited to the generation of electrical current or voltage and the darkening of photographic film. Reversible changes are used in measuring flux and flux-related quantities. Irreversible changes are used in measuring exposure.

8.4.1.3 physical photometer

An instrument containing a physical receptor (see **Section 8.4.1.2**) and associated filters that is calibrated so as to read photometric quantities directly.

8.4.1.4 integrating photometer

A photometer that either enables geometrically total luminous flux to be determined by a single measurement, or integrates a photometric quality with respect to time.

8.4.1.5 goniophotometer

A photometer for measuring the directional light distribution characteristics of sources,

luminaires, media, and surfaces.

8.4.1.6 spectrophotometer

An instrument for measuring the transmittance and reflectance of surfaces and media as a function of wavelength.

8.4.1.7 integrating sphere

A hollow sphere whose internal surface is a diffuse reflector and is as spectrally non-selective as possible.

Notes:

- (i) Two principle uses are: a) to geometrically sum the spatially total flux emitted by a radiator, and b) to eliminate the spatial and polarization inhomogeneities of radiation at the input to optical instruments.
- (ii) This sphere is sometimes known as an Ulbricht sphere.

8.4.1.8 flicker photometer

(See Section 8.4.1.1 visual photometer.)

8.4.2 reflectometer

A photometer for measuring reflectance.

Note: Reflectometers may be visual or physical instruments.

8.4.3 transmissometer

A photometer for measuring transmittance.

Note: Transmissometers may be visual or physical instruments.

8.4.4 densitometer

A photometer for measuring the optical density (common logarithm of the reciprocal of the transmittance or reflectance) of materials.

8.4.5 gloss meter

An instrument for measuring gloss as a function of the directionally selective reflecting properties of a material in angles near to and including the direction giving specular reflection.

Also called a *glossmeter* or *glossometer*.

8.4.5.1 glossometer

[See *gloss meter*.]

8.4.6 visual task evaluator, (VTE)

A form of visibility meter that measures the level of contrast of a given visual display above the threshold of visibility. The ratio of the contrast of a display to its threshold contrast represents its level of visibility. Used to evaluate the visibility level (VL). (See **Sections 5.9.14.3** and **5.9.14.4.**)

8.4.7 illuminance (lux or footcandle) meter

An instrument for measuring illuminance on a plane. Instruments that accurately respond to more than one spectral distribution are referred to as *color corrected*; i.e., the spectral response is balanced to $V(\lambda)$ or $V'(\lambda)$. Instruments that accurately respond to more than one spatial distribution of incident flux are referred to as *cosine corrected*; i.e., the response to a source of unit luminous intensity that is illuminating the detector from a fixed distance and from different directions decreases as the cosine of the angle between the incident direction and the normal to the detector surface. The instrument consists of some form of photodetector with or without a filter driving a digital or analog readout through appropriate circuitry.

8.4.7.1 light meter

A common name for an illuminance meter (see **Section 8.4.7**).

8.4.8 footcandle meter

[See **Section 8.4.7 illuminance (lux or footcandle) meter**.]

8.4.9 lux meter

(See **Section 8.4.7 illuminance (lux or footcandle) meter**.)

8.4.10 absolute photometry

Measurement of the actual photometric quantities produced by a lighting product under test. Also called *direct photometry*. (See also *relative photometry*.)

8.4.11 relative photometry

Measurement of the photometric quantities of a lighting product, which are then scaled to represent performance at the rated lumen or spectral output of a test lamp. (See also *absolute photometry*.)

8.4.12 direct photometry

[See *absolute photometry*.]

8.5 colorimetry

The measurement of color.

8.5.1 colorimeter

An instrument that measures color in units specific to a color space such as the CIE 1931 Chromaticity Diagram, the CIE 1964 Chromaticity Diagram, CIELAB, or CIELUV. (See **Sections 4.6.3.1, 4.10.2, and 4.10.1.**)

8.6 intensity (candlepower) distribution curve

A curve, often polar, that represents the variation in luminous intensity of a lamp or luminaire in a plane passing through the light center.

Note: A vertical candlepower distribution curve is obtained by taking measurements at various angles of elevation about a source in a vertical plane through the light center; unless the plane is specified, the vertical curve is assumed to represent an average such as would be obtained by rotating the lamp or luminaire about its vertical axis. A horizontal intensity distribution curve represents measurements made at various angles of azimuth in a horizontal plane through the light center.

8.6.1 isocandela line

A line plotted on any appropriate set of coordinates to show directions in space about a source of light in which the intensity is the same. A series of such curves, often for equal increments of intensity, is called an *isocandela diagram*.

8.6.2 isolux (isofootcandle) line

A line plotted on any appropriate set of coordinates to show all the points on a surface where the illuminance is the same. A series of such lines for various illuminance values is called an isolux (isofootcandle) diagram.

8.6.3 zonal constant

A factor by which the mean intensity emitted by a source of light in a given angular zone is multiplied to obtain the lumens in the zone.

8.6.4 beam spread (in any plane)

The angle between the two directions in a plane in which the intensity is equal to a stated percentage of the maximum beam intensity.

8.6.4.1 beam angle

The angle between the two directions for which the intensity is 50% of the maximum

intensity as measured in a plane through the nominal beam centerline. For beams that do not possess rotational symmetry, the beam angle is generally given for two planes at 90 degrees, typically the maximum and minimum angles. It should be noted that in certain fields of application, beam angle was formerly measured to 10% of maximum intensity.

8.6.4.1.1 beam lumens

The total flux in that region of space where the intensity exceeds 50% of the maximum intensity.

8.6.4.1.2 beam efficiency

The ratio of beam lumens to the total source lumens.

8.6.4.2 field angle

The angle between the two directions for which the intensity is 10% of the maximum intensity as measured in a plane through the nominal beam centerline. For beams that do not possess rotational symmetry, the field angle is generally given for two planes at 90 degrees, typically the maximum and minimum angles. It should be noted that in certain fields of application the angle between the 10%-of-maximum directions was formerly called *beam angle*.

8.6.4.2.1 field lumens

The total flux in that region of space where the intensity exceeds 10% of the maximum intensity.

8.6.4.2.2 field efficiency

The ratio of field lumens to the total source lumens.

8.6.4.3 center-beam candlepower (CBCP)

The luminous intensity along the geometric centerline of a directional light source. The assumption in application is usually that the centerline candlepower is also the maximum candlepower. Also called *center-beam intensity*.

8.6.5 luminous intensity distribution curve

[See **Section 8.6 intensity (candlepower) distribution curve.**]

8.6.6 isoluminance line

A line on a surface that shows all the points where the luminance is the same for a given

position of an observer and light source(s) in relation to the surface.

8.7 characteristic curve

A curve that expresses the relationship between two variable properties of a light source, such as candlepower and voltage, or flux and voltage.

8.8 life performance curve

A curve that presents the variation of a particular characteristic of a light source (e.g., luminous flux and intensity) throughout the life of the source.

Note: Life performance curves sometimes are called maintenance curves; for example, lumen maintenance curves.

8.9 life test of lamps

A test in which lamps are operated under specified conditions for a specified length of time, for the purpose of obtaining information on lamp life. Measurements of photometric and electric characteristics may be made at specified intervals of time during this test.

8.10 life test racks

A suitable framework or supporting media where multiple lamp testing positions are mounted.

8.11 regulation (of voltage)

The constancy of the voltage applied to the equipment under test.

8.12 seasoning

The process of operating a new lamp for a fixed period of time. This is also sometimes referred to as *aging*. A lamp is typically seasoned prior to measuring its initial photometric characteristics. It is an accepted industry procedure for obtaining the initial rating of lighting sources.

8.13 seasoning time

The length of time a new lamp is operated prior to performing photometric measurements. The photometric data obtained at the end of the seasoning time are referred to as *initial data* or *rated data*. The seasoning time is different depending on the type of lamp. (Refer to the appropriate IES light measurement (LM) document for actual seasoning times.

8.14 aging

A term sometimes used to describe seasoning (see **Section 8.12**).

8.15 stabilization

The operation of the test lamp for a sufficient period of time such that the parameters to be tested are constant. Also sometimes called *warm-up time* or *pre-burning time*. Not to be confused with seasoning; stabilization is of shorter duration.

8.16 stabilization time

The period required for the parameters of the lamps under test to become constant. During this period, the discharge element collects at the coolest zone(s) of the lamp. Because the distribution of the element within the discharge significantly affects the operating characteristics of the lamp, the stabilization time is necessary to obtain repeatable measurements. This is also sometimes referred to as *warm-up time*.

8.17 warm-up time

A term sometimes used to describe stabilization time (see **Section 8.16**).

8.18 pre-burning time

A term sometimes used to describe stabilization time (see **Section 8.16**).

8.19 life of a test lamp

The actual operating time of the lamp until failure, expressed in hours, and which shall not include any off time.

8.20 lamp failure

The point where a lamp stops operating. Lamp failures from manufacturing defects are reported but not included in the calculation of lamp life. For certain lamp types, other criteria than failure to light may be used; e.g., life rating may be based on the average time until the lamp type produces a given fraction of initial luminous flux.

9.0 LIGHTING CALCULATIONS

9.1 laws of illumination

9.1.1 inverse-square law

A law stating that the illuminance E at a point on a surface varies directly with the intensity I of a point source, and inversely as the square of the distance d between the source and the point.

If the surface at the point is normal to the direction of the incident light, the law is expressed by:

$$E = I/d^2$$

Note: For sources of finite size having uniform luminance, this gives results that are accurate within 1% when d is at least 5 times the maximum dimension of the source as viewed from the point on the surface. Even though practical interior luminaires do not have uniform luminance, this distance d is frequently used as the minimum for photometry of such luminaires, when the magnitude of the measurement error is not critical.

9.1.1.1 point source

A source of radiation whose dimensions are sufficiently small, compared with the distance between the source and the irradiated surface, that they can be neglected in calculations and measurements.

9.1.2 cosine law

A law stating that the illuminance on any surface varies as the cosine of the angle of incidence. The angle of incidence is the angle between the normal to the surface and the direction of the incident light. The inverse-square law and the cosine law can be combined as:

$$E = (I \cos \theta) / d^2$$

9.1.3 cosine-cubed law

An extension of the cosine law in which the distance d between the source and surface is replaced by $h/\cos \theta$, where h is the perpendicular distance of the source from the plane in which the point is located. It is expressed by $E = (I \cos^3 \theta) / h^2$. (See Figure 24.)

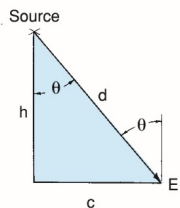


Figure 24.

9.2 work plane

The plane on which a visual task is usually done, and on which the illuminance is specified and measured. Unless otherwise indicated, this is assumed to be a horizontal plane 0.76 meter (30 inches) above the floor.

9.3 initial luminous exitance

This term can be used in two different ways.

In flux transfer, it is the density of luminous flux leaving a surface within an enclosure before interreflections occur. *Note:* For light sources, this is the luminous exitance as defined in **Section 3.3.2**. For non-self-luminous surfaces, it is the reflected luminous exitance of the flux received directly from sources within the enclosure or from daylight.

In lighting calculations, it is the total exitance at time zero before depreciation (light loss) occurs.

9.4 utilization

9.4.1 zonal cavity coefficients

Coefficients used in interior lighting design to determine illuminance and exitance or luminance values from room and luminaire parameters.

9.4.1.1 coefficient of utilization, CU

The ratio of luminous flux (lumens) calculated as received on the work-plane to the total luminous flux (lumens) emitted by the lamps alone. It is equal to the product of room utilization factor (see **Section 9.4.7**) and luminaire efficiency (see **Section 9.4.8**).

Note: For luminaires tested with absolute photometry, use Room Utilization Factor (utilance) and luminaire lumens instead of CU and lamp lumens.

9.4.1.2 exitance coefficient, EC

The ratio of the average (time zero) wall exitance or ceiling cavity exitance to the quotient of the total lamp flux divided by the floor area.

Notes:

- (i) Exitance is in lumens per unit area, where the units of area agree with those of the floor area.
- (ii) Average wall luminance or ceiling cavity luminance can be determined by noting the underlying assumption of Lambertian room surfaces, where $L = M/\pi$; L is in candelas per unit area, and the units of area agree with those of M .
- (iii) Exitance coefficients and former luminance coefficients are numerically identical.

9.4.1.3 luminance coefficient,* (LC)

The ratio of average wall luminance or ceiling cavity luminance to the quotient of the total lamp flux (lumens) divided by the floor area. This requires the luminance in the nonstandard units of footlamberts and floor area in square feet. (See **Section 9.4.1.2 exitance coefficient** for current terminology.)

9.4.1.4 wall direct radiation coefficient, WDRC

The ratio of the wall illuminance caused by that component of light from luminaires in a room cavity that arrives without being reflected by room surfaces to the quotient of the total lamp flux divided by the floor area.

9.4.1.5 wall reflected radiation coefficient, WRRC

The ratio of the wall illuminance caused by the reflected flux within a room cavity to the quotient of the total lamp flux divided by the floor area.

9.4.1.6 reflected radiation coefficient, RRC

The ratio of the floor illuminance caused by the reflected flux within a room cavity to the quotient of the total lamp flux divided by the floor area.

9.4.2 coefficient of beam utilization, CBU

The ratio of the luminous flux (lumens) reaching a specified area directly from a floodlight or projector to the total beam luminous flux (lumens).

9.4.3 room ratio†

A number indicating room proportions, calculated from the length, width, and ceiling height (or luminaire mounting height) above the work-plane. It is used to simplify lighting design tables by expressing the equivalence of room shapes with respect to the utilization of direct or interreflected light.

9.4.3.1 room index*

A letter designation for a range of room ratios.

*This term has been deprecated; it is retained here for reference of literature.

† This term has been deprecated; it is retained here for reference of literature.

9.4.4 room coefficient, K^*

A number computed from wall and floor areas:

$$K_r = \frac{\text{height} \cdot (\text{length} + \text{width})}{2 (\text{length})(\text{width})}$$

9.4.5 cavity ratio, CR

A number indicating cavity proportions.

The cavity ratio is computed from:

$$CR = \frac{5(\text{Height}_{\text{cavity}})(\text{Length}_{\text{cavity}} + \text{Width}_{\text{cavity}})}{(\text{Length}_{\text{cavity}})(\text{Width}_{\text{cavity}})}$$

For cavities of irregular shape:

$$CR = \frac{2.5(\text{Height}_{\text{cavity}})(\text{Perimeter}_{\text{cavity}})}{(\text{Area of Cavity Base})}$$

Note: The relationship between *cavity ratio* and *room coefficient* (K_r) should be noted. If the entire room is considered as a cavity, the room height becomes $\text{Height}_{\text{cavity}}$, and $CR = 10 K_r$.

9.4.5.1 ceiling cavity

The cavity formed by the ceiling, the plane of the luminaires, and the wall surfaces between these two planes.

9.4.5.1.1 ceiling cavity ratio, CCR

A number computed by using the distance from the plane of the luminaire to the ceiling (h_c) as $\text{Height}_{\text{cavity}}$ in the equations given in **Section 9.4.5**.

9.4.5.2 floor cavity

The cavity formed by the work plane, the floor, and the wall surfaces between those two planes.

9.4.5.2.1 floor cavity ratio, FCR

A number computed by using the distance from the floor to the work plane (h_f) as $\text{Height}_{\text{cavity}}$ in the equations given in **Section 9.4.5**.

* This term has been deprecated; it is retained here for reference of literature.

9.4.5.3 room cavity

The cavity formed by the plane of the luminaires, the work plane, and the wall surfaces between these two planes.

9.4.5.3.1 room cavity ratio, RCR

A number computed by using the distance from the work plane to the plane of the luminaires (h_r) as $Height_{cavity}$ in the equations given in **Section 9.4.5**. (See **Figure 25**).

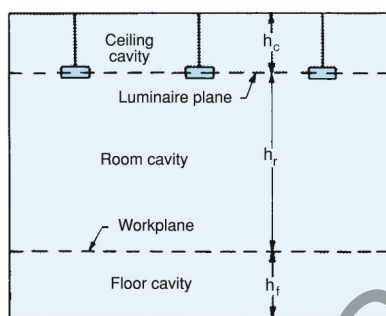


Figure 25.

9.4.6 zonal multipliers

Multipliers for the flux in each ten-degree conical zone from zero degrees (nadir) to 90 degrees (horizontal) from a luminaire, expressing the fraction of that zonal flux that is directly incident on the floor of a room cavity. These multipliers are a function of room cavity ratio and are used to determine the direct ratio.

9.4.7 room utilization factor (utilance)

The ratio of the luminous flux (lumens) received on the work plane to that emitted by the luminaire.

Note: This ratio sometimes is called *interreflectance*. Room utilization factor is based on the flux emitted by a complete luminaire, whereas coefficient of utilization (see **Section 9.4.1**) is based on the total flux generated by the lamps in a luminaire.

9.4.8 luminaire efficiency

The luminous flux emitted by a luminaire, divided by the luminous flux emitted by the source(s). Sometimes called *light output ratio* (LOR).

9.4.9 light loss factor, LLF

The ratio of illuminance (or exitance or luminance) for a given area to the value that would occur if lamps operated at their (initial) rated lumens and if no system variation or depreciation had occurred. Components of this factor may be either initial or maintained. (Formerly called *maintenance factor, MF*.)

Note: The light loss factor is used in lighting calculations as an allowance for lamp(s) or luminaire(s) operating at other than rated conditions (initial) and for the depreciation of lamps, light control elements, and room surfaces to values below the initial or design conditions, so that a minimum desired level of illuminance may be maintained in service. Light loss factors address losses that result in direct changes to lamp lumens, emitted luminaire lumens, or the interreflected light delivered to the space.

9.4.9.1 recoverable light loss factor

A factor that quantifies fractional light delivered by a lighting system where losses can be recovered by cleaning or lamp replacement, and which accounts for depreciation of luminous quantities in a lighting system. Light loss factors of this type are a function of lighting system age and maintenance processes and are applied to lighting calculations for systems after time zero.

9.4.9.1.1 period life

The time interval until lamps are replaced or the luminaires are cleaned.

9.4.9.1.2 lamp lumen depreciation, LLD

The ratio of lamp lumen output after an extended period of operation under rated operating conditions to its initial lumen output under the same operating conditions.

9.4.9.1.3 room surface dirt depreciation, RSDD

The ratio of illuminance (or exitance or luminance) provided with accumulated dirt on room surfaces to that provided with clean room surfaces.

9.4.9.1.4 lamp burnout factor

The ratio of the illuminance (or exitance or luminance) provided when a lighting system is allowed to operate with a set number of lamps burned out to that provided with all lamps operating.

9.4.9.1.5 luminaire dirt depreciation, LDD

The ratio of lumens emitted from a luminaire with dirt accumulated to the lumens emitted from the same luminaire when clean.

9.4.9.2 non-recoverable light loss factor (initial or maintained)

A factor that quantifies the fractional light delivered by a lighting system after depreciation where light loss cannot be recovered by cleaning or lamp replacement. Light loss factors of this type are applied to lighting calculations irrespective of the age of the lighting system.

9.4.9.2.1 luminaire surface depreciation factor

The ratio of lumens emitted from a luminaire following permanent deterioration of luminaire surfaces to the lumens emitted from luminaires with no surface deterioration. This loss may increase with time if a luminaire continues to degrade.

9.4.9.2.2 voltage to luminaire factor

The ratio of lumens emitted from a luminaire operated at other than rated voltage to those emitted at the rated voltage.

9.4.9.2.3 ballast factor

The fractional lumens of a fluorescent lamp(s) operated on a ballast compared to the lumens when operated on the standard (reference) ballasting specified for rating lamp lumens.

Note: The lamp(s) are at specified ambient temperature conditions for photometric testing.

9.4.9.2.4 luminaire ambient temperature factor

The fractional lumen change of a fluorescent luminaire due to internal luminaire temperatures differing from the temperatures at which photometry was performed. This factor would take into consideration a variation in ambient temperature surrounding the luminaire, the means and conditions of mounting the luminaire, and the use of any insulation in conjunction with the application of the luminaire.

9.4.9.2.5 heat extraction thermal factor

The fractional lumen output (loss or gain) due to passage of room air being returned to the plenum through the lamp compartment of the luminaire.

9.4.9.2.6 ballast-lamp photometric factor

The ratio of fluorescent luminaire lumen output using given ballast and lamp types (under photometric test conditions) to the lumen output using the ballast and lamp types used to generate a photometric test report.

Note: This factor is applicable when "energy conserving" lamps and ballasts are used in a luminaire photometered with standard lamps and conventional ballasts; it is also applied in the converse situation.

9.4.9.2.7 lamp position factor

The ratio of the lumens of a high intensity discharge (HID) lamp in a given operating position to the lumens when the lamp is operated in the position at which the lamp lumens are rated.

9.4.9.2.8 equipment operating factor

The ratio of the lumens of a high intensity discharge (HID) lamp-ballast-luminaire combination in a given operating position to the lumens of the lamp-luminaire combination a) operated in the position for rating lamp lumens; and b) using the standard (reference) ballasting specified for rating lamp lumens.

Note: If the given lamp operating position is not the same as the lamp rating position, the lumens ratio for the operating ballast to standard rating ballast is determined in the given operating position. This ratio is multiplied by the lamp position factor to obtain the equipment operating factor.

9.4.9.3 maintenance factor, MF

[See Section 9.4.9 light loss factor, LLF.]

9.4.9.4 unrecoverable light loss factor

[See Section 9.4.9.2 nonrecoverable light loss factor (initial or maintained).]

9.4.10 efficiency

[See Section 9.4.8 luminaire efficiency and Section 3.10.1 radiant efficiency.]

9.4.11 light output ratio

[See Section 9.4.8 luminaire efficiency.]

9.4.12 luminous flux maintenance

The ratio of the luminous flux of an electric light source at a given time in its operational life to its initial luminous flux.

Notes:

1. Luminous flux maintenance is usually expressed as a percentage.
2. For the purpose of measurement, the electric light source is assumed to be operated under specified controlled conditions. In field measurements, the value may vary due to uncontrolled conditions.

9.4.13 lumen maintenance

[See **Section 9.4.12 luminous flux maintenance**.]

9.5 light distribution

9.5.1 direct component

That portion of the light from a luminaire that arrives at the work plane without being reflected by room surfaces.

9.5.2 downward component

That portion of the luminous flux from a luminaire that is emitted at angles below the horizontal.

9.5.3 direct ratio

The ratio of the luminous flux that reaches the floor of a room cavity directly to the downward component from the luminaire.

9.5.4 indirect component

That portion of the luminous flux from a luminaire that arrives at the work plane after being reflected by room surfaces.

9.5.5 upward component

That portion of the luminous flux from a luminaire that is emitted at angles above the horizontal.

9.5.6 ceiling ratio

The ratio of the luminous flux that reaches the ceiling directly to the upward component from

the luminaire.

9.5.7 interreflection

The multiple reflection of light by the various room surfaces before it reaches the work plane or other specified surface of a room.

Also called *interfection*.

9.5.7.1 interreflected component

That portion of the luminous flux from a luminaire that arrives at the work plane after being reflected one or more times from room surfaces, as determined by the flux transfer theory. (See **Section 9.5.7.2**).

Also called *interflectance*.

9.5.7.2 flux transfer theory

A method of calculating the illuminance in a room by taking into account the interreflection of the light flux from the room surfaces, based on the average flux transfer between surfaces.

9.5.7.2.1 adaptive subdivision

A procedure applied to a luminous flux transfer problem where the elements created to exchange flux are further subdivided to better approximate the direct contribution and its variation across that element.

9.5.7.3 form factor, $f_{1 \rightarrow 2}$

The ratio of the flux directly received by surface 2 that is due to Lambertian surface 1 to the total flux emitted by surface 1. It is used in flux transfer theory.

$$f_{1 \rightarrow 2} = (\Phi_{1 \rightarrow 2}) / \Phi_1$$

Also, the ratio of the average illuminance E on surface 1 to the causative exitance M of Lambertian surface 2:

$$f_{1 \rightarrow 2} = E_1 / M_2$$

Note: In the literature, this term is also called *angle factor*, *configuration factor*, *geometrical factor*, *I-factor*, *illumination factor*, and *shape modulus*.

9.5.7.4 configuration factor, $c_{1 \rightarrow 2}$

The ratio of illuminance on a surface at point 1 that is due to the flux directly received

from Lambertian surface 2 to the exitance of surface 2. It is used in flux transfer theory.

$$c_{1 \rightarrow 2} = E_1 / M_2$$

Also, the ratio of the differential flux directly received by surface 2 and due to element 1 to the total differential flux emitted by differential Lambertian surface element 1.

$$c_{1 \rightarrow 2} = (d\phi_{1 \rightarrow 2}) / d\phi_1$$

Note: In the literature, this term is also called *angle factor*, *illumination factor*, *point configuration factor*, and *sky factor*.

9.5.7.5 wrong way "law"

An informal recognition in the equations $E_2 = (c_{2 \rightarrow 1})M_1$ and $E_1 = (c_{1 \rightarrow 2})M_2$ that the configuration factor and form factor flux transfer as indicated by the subscripts are opposite to the actual cause-effect flux transfer.

9.5.7.6 radiosity

An old term for *radiant exitance* (or *radiant flux areal density*; see [Section 2.7](#)). In the computer graphics community, *radiosity* is often used to refer to lighting calculation procedures that apply flux transfer theory.

9.5.8 Abbe's law

The luminance (or radiance) of a ray is proportional to the square of the index of refraction of the medium in which it is measured and is otherwise invariant except for losses.

9.6 lighting calculation methods

9.6.1 lumen (or flux) method

A procedure used to determine the relationship between the number and types of lamps and luminaires, the room characteristics, and the average level of illuminance on the work plane. It takes into account both direct and reflected flux.

9.6.1.1 zonal factor method

A procedure for predetermining, from typical luminaire photometric data in discrete angular zones, the proportion of luminaire output that would be incident initially (without interreflections) on the work plane, ceiling, walls, and floor of a room.

9.6.1.1.1 zonal cavity interreflectance method

A procedure for calculating coefficients of utilization, wall exitance coefficients, and ceiling cavity exitance coefficients, taking into consideration the luminaire

intensity distribution, room size and shape (cavity ratio concepts), and room reflectances. It is based on the flux transfer theory. (See **Section 9.5.7.2**).

*9.6.1.1.2 zonal factor interreflection method**

A procedure for calculating coefficients of utilization based on integral equations that takes into consideration the ultimate disposition of luminous flux from every 10-degree zone from luminaires.

*9.6.1.1.3 interreflectance method**

A procedure for predetermining the luminances of walls, ceiling and floor, and the average illuminance on the work plane based on integral equations. It takes into account both direct and reflected flux.

9.6.2 point method

A procedure for predetermining the illuminance at various locations in lighting installations by use of luminaire photometric data. The direct component of illuminance due to the luminaires and the interreflected component of illuminance due to the room surfaces are calculated separately. The sum is the total illuminance at a point.

Formerly called *point-by-point method*.

9.6.3 fenestra method

A procedure for predicting the interior illuminance received from daylight through windows.

9.6.4 ray tracing

A lighting analysis procedure for analyzing the direct or interreflected light in an environment by casting rays into an environment in a statistical manner to determine the illuminance or luminance at a point. Rays may be recast upon striking a surface in order to consider multiple reflections between different surfaces.

9.6.4.1 forward ray tracing

A ray tracing method that casts rays from the light sources to receiving surfaces, and then recasts additional rays from these receiving surfaces to determine the interreflected light within a space or physical environment.

*Obsolete term. Retained for reference purposes only.

9.6.4.2 backward ray tracing

A ray tracing technique that casts rays from a desired observer position through a viewing plane, or away from a desired analysis point, into the environment. Upon striking a surface, rays may then be recast toward light sources and other surfaces to determine the luminance of the surface. This technique is often used to create images of a lighted space or to determine the illuminance or luminance at a point.

9.6.5 point-by-point method

(See Section 9.6.2 point method.)

10.0 INTERIOR LIGHTING APPLICATIONS

10.1 lighting techniques

10.1.1 general lighting

Lighting designed to provide a substantially uniform level of illuminance throughout an area, exclusive of any provision for special local requirements.

10.1.1.1 localized general lighting

Lighting utilizing luminaires above the visual task and also contributing to the illuminance of the surround.

10.1.2 supplementary lighting

Lighting that is used to provide an additional quantity and quality of illuminance that cannot readily be obtained by a general lighting system and that supplements the general lighting level, usually for specific work requirements.

10.1.3 local lighting

Lighting providing illumination over a relatively small area or confined space without providing any significant general surrounding lighting.

10.1.4 direct lighting

Lighting involving luminaires that distribute 90% to 100% of the emitted light in the general direction of the surface to be illuminated. The term usually refers to light emitted in a downward direction.

10.1.5 semi-direct lighting

Lighting involving luminaires that distribute 60% to 90% of the emitted light downward and the balance upward.

10.1.6 general diffuse lighting

Lighting involving luminaires that distribute 40% to 60% of the emitted light downward and the balance upward, sometimes with a strong component at 90 degrees (horizontal).

10.1.6.1 direct-indirect lighting

A variant of general diffuse lighting in which the luminaires emit little or no light at angles near the horizontal.

10.1.7 semi-indirect lighting

Lighting involving luminaires that distribute 60% to 90% of the emitted light upward and the balance downward.

10.1.8 indirect lighting

Lighting involving luminaires that distribute 90% to 100% of the emitted light upward.

10.1.9 ceiling area lighting

A general lighting system in which the entire ceiling is, in effect, one large luminaire.

Note: Ceiling area lighting includes luminous ceilings (see **Section 10.3.3**) and fouvered ceilings (see **Section 10.3.4**).

10.1.10 cove lighting

Lighting comprising light sources shielded by a ledge or horizontal recess, and distributing light over the ceiling and upper wall.

10.1.11 cornice lighting

Lighting comprising sources shielded by a panel parallel to the wall and attached to the ceiling, and distributing light over the wall.

10.1.12 valance lighting

Lighting comprising light sources shielded by a panel parallel to the wall at the top of a window.

10.1.13 diffused lighting

Lighting provided on the work-plane or on an object and that is not incident predominantly from

any particular direction.

10.1.14 directional lighting

Lighting provided on the work plane or on an object, predominantly from a preferred direction.

10.1.14.1 accent lighting

Directional lighting to emphasize a particular object or surface feature, or to draw attention to a part of the field of view.

10.1.15 emergency lighting

Lighting designed to supply illumination essential to the safety of life and property in the event of failure of the normal supply.

10.1.15.1 emergency exit

A way out of the premises that is intended to be used during an emergency.

10.1.15.2 exit

The portion of a means of egress that segregates all other spaces in the building or structure by fire-resistant construction in order to provide a protected way of travel to the exit discharge. Exits include exterior exit doors, exit passageways, horizontal exits, and separated exit stairs or ramps.

10.1.15.3 exit access

The portion of a means of egress that leads to an exit.

10.1.15.4 exit discharge

The portion of a means of egress between the conclusion of an exit and a public way.

10.1.15.5 exit sign

A graphic device including words or symbols that indicates or identifies an escape route, or the location of, or direction to, an exit or emergency exit.

10.1.15.6 externally illuminated exit sign

An exit sign with an externally mounted light source. The exit legend and background are typically opaque and rely on reflected light for visibility.

10.1.15.7 horizontal exit

An escape route from one building to an area of refuge in another building on approximately the same level.

Also: an escape route through or around a fire barrier to an area of refuge on approximately the same level in the same building.

10.1.15.8 internally illuminated exit sign

A transilluminated exit sign containing its own light source.

10.1.15.9 means of egress

An unobstructed and continuous way of exit from any point in a building or structure to a public way. It consists of three distinct parts: the exit access, the exit, and the exit discharge. A means of egress consists of the vertical and horizontal travel ways, including intervening room spaces, doorways, hallways, corridors, passageways, ramps, stairs, lobbies, horizontal exits, escalators, enclosures, courts, balconies, and yards.

10.1.15.10 panel face exit sign

A transilluminated sign where both the exit legend and background are translucent

10.1.15.11 public way

A street, alley, or other parcel of land open to the outside air; leading to a street that has been deeded, dedicated, or otherwise permanently appropriated to the public for public use; and having a clear height and width of not less than 3 m (10 ft). ¹⁵

10.1.15.12 self-luminous exit sign

An exit sign consisting of phosphor-coated glass tubes filled with radioactive tritium gas. When the radioactive gas bombards the phosphor, the tube emits light (luminescence) and illuminates the exit legend; typically between 0.2 and 0.7 cd/m².

10.1.15.13 stencil face exit sign

An internally transilluminated sign where either the exit legend or the background is opaque. Usually the exit legend is translucent and the background is die-cut from an opaque medium such as plastic or metal.

10.1.15.14 egress

(See Section 10.1.15.9 means of egress.)

10.1.16 portable lighting

Lighting involving equipment designed for manual portability.

10.1.17 task lighting

Lighting directed to a specific surface or area, providing illumination for visual tasks.

10.1.18 ambient lighting

Lighting throughout an area that produces general illumination.

10.1.19 task-ambient lighting

A combination of task lighting and ambient lighting within an area such that the general level of ambient illumination is lower than and complementary to the task lighting.

10.1.20 criteria rating

A technique that determines the probability that a specific criterion will be met anywhere in a defined area. The name of the criteria rating assumes the name of the criterion being rated. It is expressed in shorthand notation by listing the rating in percent followed by the criterion itself and separated by "@". For example, a lighting system producing a luminance of 20 cd/m² over 60% of the specified area could have its luminance rating expressed as 60% @ 20 cd/m².

10.2 light control

10.2.1 bare (exposed) lamp

A light source with no shielding.

10.2.2 diffuser

A device to redirect or scatter light from a source, primarily by the process of diffuse transmission.

10.2.2.1 globe (for light control)

A transparent or diffusing enclosure intended to protect a lamp, to diffuse and redirect its light, or to change the color of the light.

10.2.2.2 bowl

An open-top enclosure made of diffusing glass or diffusing plastic, used to shield a light source from direct view and to redirect or scatter the light.

10.2.2.3 diffusing panel

A translucent material covering the lamps in a luminaire in order to reduce the brightness by distributing the flux over an extended area.

10.2.3 baffle

A single opaque or translucent element to shield a source from direct view at certain angles, to absorb or block unwanted light, or to reflect and redirect light.

10.2.4 louver (or louver grid)

A series of baffles used to shield a source from view at certain angles, to absorb or block unwanted light, or to reflect and redirect light. The baffles are usually arranged in a geometric pattern.

10.2.4.1 louver shielding angle, θ

The angle between the horizontal plane of the baffles or louver grid and the plane at which the louver conceals all objects above. **Figure 26** shows louver shielding angles θ and θ' .

Note: The planes usually are so chosen that their intersection is parallel with the louvered blade.

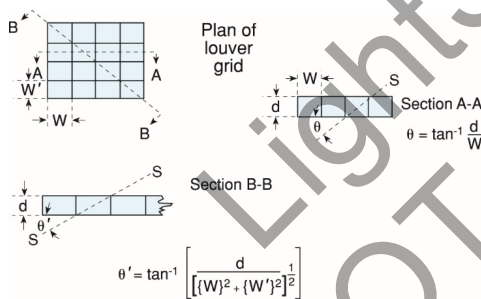


Figure 26.

10.2.4.2 lamp shielding angle, ϕ

The angle between the plane of the baffles or louver grid and the plane most nearly horizontal that is tangent to both the lamps and the louver blades. (See **Figure 27**.)

Notes:

- (i) In **Figure 27**, the lamp shielding angle is formed by a sight line tangent to the lowest

part of the brightness area to be shielded. H is the vertical distance from the brightness source to the bottom of the shielding element. D is the horizontal distance from the brightness source to the shielding element. Lamp shielding angle $\phi = \tan^{-1} H/D$.

(ii) The lamp shielding angle is frequently larger than the louver shielding angle, but never smaller.

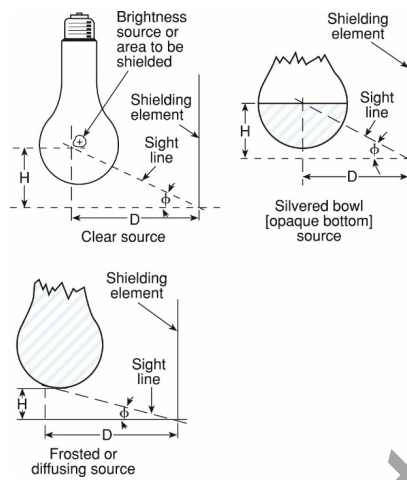


Figure 27.

10.2.5 shielding angle (of a luminaire)

The angle between a horizontal line through the light center and the line of sight at which the bare source first becomes visible.

10.2.5.1 cut-off angle (of a luminaire)

The angle, measured up from nadir, between the vertical axis and the first line of sight at which the bare source is not visible.

10.2.6 shade (daylighting)

A screen made of opaque or diffusing material that is designed to prevent a light source from being directly visible at normal angles of view.

10.2.7 matte surface

A surface from which the reflection is predominantly diffuse, with or without a negligible specular component. (See also **Section 7.3.2.**)

10.2.8 specular surface

A surface from which the reflection is predominantly regular (specular). (See **Section 7.3.1.**)

10.2.9 reflector

A device used to redirect the flux from a source by the process of reflection.

10.2.10 refractor

A device used to redirect the flux from a source primarily by the process of refraction.

10.2.11 acceptance half-angle

The acceptance half-angle is set by the refractive indices of the core and cladding of fiber optics. It is expressed in terms of the numerical aperture (*NA*). The numerical aperture is given as:

$$NA = n_0 \sin \theta_{\max},$$

where:

$\theta_{\max}$ is the greatest angle from the optical axis of a circular fiber where the external meridional plane ray will undergo total internal reflection within the fiber.

n_0 is the refractive index of the medium outside the fiber (for air, $n_0 = 1$).

10.3 lighting equipment

10.3.1 luminaire

A complete lighting unit consisting of a light source(s) and ballast(s) or driver(s) (when applicable), together with the parts designed to distribute the light, to position and protect the sources, and to connect the sources to the power supply. Also known as a *light fixture*.

10.3.1.1 narrow-angle luminaire

A luminaire that concentrates the light within the cone of a comparatively small solid angle.

10.3.1.2 wide-angle luminaire

A luminaire that concentrates the light within the cone of a comparatively large solid angle.

10.3.1.3 luminaire spacing criterion (SC)

A classification parameter for indoor luminaires relating to the distribution of the direct illuminance component produced on the work plane. The SC of a luminaire is an estimated maximum ratio of the luminaire spacing to the luminaire mounting height

above the work plane for a regular array of that luminaire such that the work plane illuminance will be acceptably uniform.

Note: SC in 0-degree and 90-deg horizontal directions can be different.

10.3.1.4 mounting height MH (interior)

The distance from a reference plane (e.g., floor or work place) to a luminaire, or to the plane of the ceiling for recessed luminaires. The default reference plane is the work plane. The luminaire reference location should be stated (e.g., bottom or light center) to avoid ambiguities.

10.3.1.5 spacing-to-mounting-height ratio, S/MH

The ratio of the actual distance between luminaire centers to the mounting height (see **Section 10.3.1.4**). (See also **Section 10.3.1.3**.)

10.3.2 downlight

A small direct-lighting unit that directs the light downward and can be recessed, surface mounted, or suspended.

10.3.3 luminous ceiling

A ceiling area lighting system consisting of a continuous surface of transmitting material of a diffusing or light controlling character with light sources mounted above it.

10.3.4 louvered ceiling

A ceiling area lighting system consisting of a wall-to-wall installation of multi-cell louvers shielding the light sources mounted above it.

10.3.5 coffer

A recessed panel or dome in the ceiling.

10.3.6 troffer

A long recessed lighting unit usually installed with the opening flush with the ceiling. The term is derived from "trough" and "coffer."

10.3.7 valance

A longitudinal shielding member mounted across the top of a window or along a wall (and is usually parallel to the wall) to conceal light sources, giving both upward and downward light distribution.

10.3.8 channel

An electrical enclosure within in a luminaire that may contain ballasts or LED drivers, starters, and wiring, depending on the source type.

10.3.9 projector

A lighting unit that, by means of mirrors and lenses, concentrates the light into a limited solid angle so as to obtain a high value of luminous intensity.

10.3.10 portable luminaire

A lighting unit that is not permanently fixed in place.

10.3.10.1 table lamp

A portable luminaire with a short stand suitable for standing on furniture.

10.3.10.2 floor lamp

A portable luminaire on a high stand, suitable for standing on the floor.

10.3.10.3 torchère (or torchiere)

An indirect floor lamp that sends all or nearly all of its light upward.

10.3.11 hazardous location

An area where ignitable vapors or dust may cause a fire or explosion created by energy emitted from lighting or other electrical equipment or by electrostatic generation.

10.3.11.1 explosion-proof luminaire

A luminaire that is completely enclosed and capable of withstanding an explosion of a specific gas or vapor that may occur within it, and preventing the ignition of a specific gas or vapor surrounding the enclosure by sparks, flashes, or explosion of the gas or vapor within. It must operate at such an external temperature that a surrounding flammable atmosphere will not be ignited thereby.

10.3.11.2 dust-tight luminaire

A luminaire that is so constructed that dust will not enter the enclosing case.

10.3.11.3 dust-proof luminaire

A luminaire so constructed or protected that dust will not interfere with its successful operation.

10.3.11.4 vapor-tight luminaire

A luminaire designed and approved for installation in damp or wet locations. It also is described as *enclosed and gasketed*.

10.3.11.5 enclosed and gasketed luminaire

(See Section 10.3.11.4 vapor-tight luminaire.)

10.3.12 low-bay lighting

Interior lighting where the roof trusses or ceiling height is approximately 6.1 meters (20 feet) or less above the floor.

10.3.13 high-bay lighting

Interior lighting where the roof trusses or ceiling height is greater than approximately 6.1 meters (20 feet) above the floor.

10.3.14 fixture

[See Section 10.3.1 luminaire.]

10.3.15 track lighting

A lighting equipment system consisting of an electrified power channel (track) and removable luminaires (lamp holders, track heads) that can be mechanically attached anywhere along the power channel. The luminaires can be repositioned and re-aimed as desired.

10.3.16 reflected direct luminaire

A class of luminaires that make use of light reflecting off the interior surface of a recessed housing from concealed or obscured sources.

Typically, these luminaires, which are similar in performance to lensed or diffused luminaires, have perforated metal baskets backed with diffuse acrylic to hide the lamps from direct view and to direct light onto the upper reflector surface. The upper reflector is commonly a smooth surface with a matte, high-reflectance finish.

Also called *recessed indirect*, *direct-indirect troffer*, or *basket fixture*.

10.4 luminaire mounting

10.4.1 suspended (pendant)

A luminaire that is hung from a ceiling by supports.

10.4.2 surface mounted

A luminaire that is mounted directly on a wall or on the ceiling.

10.4.3 flush mounted or recessed

A luminaire that is mounted above the ceiling (or behind a wall or other surface) with the opening of the luminaire level with the surface.

10.4.4 regressed

A luminaire that is mounted above the ceiling (or behind a wall or other surface) with the opening of the luminaire above the ceiling line (or behind the face of the wall or other surface).

10.4.5 pendant luminaire

[See **Section 10.4.1 suspended (pendant).**]

10.5 daylighting¹

The use of controlled daylight to illuminate architectural space.

10.5.1 daylight

Direct, diffuse, and/or reflected light that originates at the sun.

10.5.1.1 daylit

Illuminated by daylight. (Preferred over *daylighted*.)

10.5.1.2 daylighted

[See *daylit*.]

10.5.2 Astronomical and Geographical Solar Radiation

10.5.2.1 global solar radiation

The combination of direct solar radiation and diffuse sky radiation on a horizontal plane under a given sky condition.

10.5.2.2 direct solar radiation

Radiation received directly from the collimated sun beam.

10.5.2.2.1 direct normal solar radiation

Solar radiation received directly from the sun on a plane perpendicular to the sun's rays at the earth's surface.

10.5.2.3 diffuse sky radiation

That part of solar radiation that reaches the Earth as a result of being scattered by air molecules, moisture, or other aerosol particles in the atmosphere.

10.5.2.4 total (solar) radiation

Summation of the direct solar radiation and the sky radiation.

10.5.2.4.1 ground light

Visible radiation from the sun and sky reflected by exterior surfaces that lie below the plane of the horizon.

10.5.2.5 extraterrestrial solar irradiance constant

Extraterrestrial beam solar irradiance on a plane perpendicular to a line through the centers of the sun and earth at the mean distance of the earth from the sun.

Units: W/m^2 . The currently accepted default value is $1,366 \text{ W/m}^2$.

Also called *solar constant*.

10.5.2.5.1 extraterrestrial solar illuminance constant

The extraterrestrial solar beam illuminance on a plane perpendicular to a line between the centers of the Sun and Earth at the mean distance of Earth from the Sun.

Units: lm/m^2 . The currently accepted value is 133.1 klux (12,370 footcandles).

10.5.2.5.2 solar efficacy

The ratio of the solar illuminance to the solar irradiance.

10.5.2.6 direct sunlight

The contribution from the collimated solar beam that is neither reflected nor diffused.

10.5.2.7 extraterrestrial solar radiation

Solar radiation (from ultraviolet through infrared) that reaches the top of Earth's atmosphere.

10.5.2.8 global illuminance

The combination of direct solar illuminance and diffuse sky illuminance on a horizontal plane under a given sky condition.

10.5.2.9 sky illuminance

Illuminance from the atmosphere that reaches a surface on Earth as a result of scattering by the air molecules, moisture, or other aerosol particles in the atmosphere. It may include light originating from the sun or from electric sources.

10.5.2.10 sky light

The visible part of diffuse sky radiation. (See also *skylight*.)

10.5.2.11 sunlight

[See *direct sunlight*.]

10.5.2.12 solar constant

[See *extraterrestrial solar irradiance constant*.]

10.5.2.13 solar illuminance constant

[See *extraterrestrial solar irradiance constant*.]

10.5.3 Solar Position

10.5.3.1 altitude angle

The angular distance of a heavenly body measured from the horizon toward the zenith.

10.5.3.2 azimuth

The angular distance between the vertical plane through a direction to a distant object or sky position, or the facing direction of a building façade, and a reference direction (typically North or South). (See also *solar azimuth*.)

10.5.3.3 solar azimuth

The angular distance between the vertical plane through the sun and a vertical (reference) plane aligned with North (or South, in the case of older IES reference material – prior to 2017). (See also *azimuth*.)

10.5.3.4 zenith

The point on the celestial sphere, directly above the observer, corresponding to an angle of 90 degrees above the horizontal plane.

10.5.3.5 solar altitude

The vertical angle of the sun above the horizon.

10.5.3.6 solar declination angle

The angle between the rays of the sun and the plane of the earth's equator, which varies with time of year and ranges from -23.4 degrees to +23.4 degrees.

10.5.4 [heading unassigned]

10.5.5 Sky Characteristics

10.5.5.1 Sky Classifications

10.5.5.1.1 clear sky

A sky that has less than 30% cloud cover.

10.5.5.1.2 partly cloudy sky

A sky that has 30% to 70% cloud cover.

10.5.5.1.3 cloudy sky

A sky that has more than 70% cloud cover.

10.5.5.1.4 overcast sky

A sky that has 100% cloud cover; the sun is not visible.

10.5.5.1.5 sky luminance distribution function

For a specified sky condition, the luminance of each position in the sky (azimuth and altitude) relative to a defined zenith luminance.

10.5.5.2 sky luminance distribution models

Published equations that define patterns of direct solar illuminance and sky luminance over the entire sky dome under specific sky conditions and time of day.

10.5.5.2.1 CIE sky models

A set of sky luminance distributions that model the sky from heavily overcast to cloudless, published by the International Commission on Illumination (CIE).

10.5.5.2.2 Perez Sky Model

Sky luminance distribution models based on the work of Richard Perez derived from measured irradiance data for the direct solar and sky components.

Ref.: Perez R, Seals R, Michalsky J. All-weather model for sky luminance

distribution – Preliminary configuration and validation. Solar Energy. 1993;50(3).

10.5.5.3 air mass (optical air mass)

The ratio of the path length of solar radiation through the atmosphere to a given point, relative to the path length from sea level to the zenith.

10.5.6 Building Position and Geometry

10.5.6.1 orientation

The position of a building with respect to compass directions.

10.5.6.2 façade orientation

The clockwise angle between solar (true) North (or South, in the case of older IES reference material – prior to 2017) and the normal vector to a vertical building façade.

10.5.6.3 vertical shadow angle

The solar altitude angle projected onto a vertical plane perpendicular to a building facade or shading element edge. Alternatively, the vertical angle from a point to the linear edge of a shading element (e.g., overhang, light shelf, blind edge or adjacent building, rooftop) measured from the horizon in a plane perpendicular to that edge.

Also called *profile angle*.

10.5.6.4 horizontal shadow angle

Angle which measures the difference in azimuth between the sun's position and a line normal to a building face.

Formerly: *solar elevation azimuth angle*.

10.5.6.5 incident angle

The angle between a ray of light arriving at a surface and a line normal (perpendicular) to that surface.

10.5.6.6 profile angle

[See *vertical shadow angle*.]

10.5.7 Glazing

10.5.7.1 fenestration

The collection of all fixed and operable daylight transmitting apertures in a building, including doors, windows, skylights, panels, louvered openings, etc.

10.5.7.2 window

A glazed aperture in a section of a building envelope that is vertical or within 30 degrees of vertical.

10.5.7.2.1 view window

A glazed aperture, a portion of which is at eye level for building occupants, that provides a clear view to the outside of a building.

10.5.7.2.2 daylight window

A glazed aperture, commonly above eye level for building occupants, for the transmission and distribution of daylight into a space. It may incorporate specialized daylight management systems.

10.5.7.2.3 window sash

The operable component of a window frame that surrounds the glazing system and is capable of moving.

10.5.7.3 skylight

A glazed opening on the roof of a building making an angle of 0 to 60 degrees from the horizontal. (See also *sky light*.)

10.5.7.4 insulated glazing units

A glazing assembly with multiple light transmitting layers separated by gas or vacuum filled spaces, for the purpose of reducing heat transmission across the glazing.

10.5.7.5 low-iron glass

Glass with reduced iron content that provides high visible light transmittance with excellent clarity and reduced color distortion.

10.5.7.6 haze factor

The fraction or percentage of light transmitted through a material that is diffused.

10.5.7.7 spectrally selective glazing

A coated or tinted glazing with optical properties that make it more transparent to some wavelengths and more reflective to others. Typical spectrally selective coatings are more transparent in the visible spectrum and less in the UV and IR ranges.

10.5.7.8 low-e coating

A coating applied to one or more surfaces of a glazing system, having low emittance over the long-wavelength (IR) region of the solar spectrum to increase thermal resistance.

10.5.7.9 reflective film

A film with metallic layers, applied to a glazing surface to reflect some portion of the radiation spectrum.

10.5.7.10 solar film

A transparent film applied to a glazing surface for the purpose of reducing solar heat gain. Examples may include tinted, low-e, and reflective films.

10.5.7.11 laminated glass

Two or more sheets of glass bonded together by a transparent material, to increase the unit's resistance to force, hold the glass together when shattered, and reduce sound transmission.

10.5.7.12 translucent insulating panels

A lightweight structural sandwich panel that combines a translucent inner and outer glazing element with an insulated internal cavity.*

10.5.7.13 dynamic glazing

Any glazing or glazing system that has the ability to change its performance properties, including U-factor, solar heat gain coefficient (SHGC), or visible transmittance (VT). This includes, but is not limited to, shading systems between the glazing layers and chromogenic glazing.

10.5.7.14 chromogenic glazing

Dynamic glazing that changes its optical properties, most notably its spectral transmittance, due to changes in temperature (thermochromic glazing), radiation (photochromic glazing), or through an applied voltage (electrochromic glazing).

10.5.7.14.1 electrochromic glazing

A controllable glazing where a change in transmission properties is effected by application of a voltage.

* American Architectural Manufacturers Association (AAMA).

10.5.7.14.2 photochromic glazing

Glazing that includes a layer or embedded material whose spectral transmittance can be changed along a range of values via radiation stimulus.

10.5.7.14.3 thermochromic glazing

Glazing whose spectral transmittance varies depending on its temperature.

10.5.8 Building Design Strategies

10.5.8.1 sidelighting

Daylighting through vertical apertures such as windows, clerestories, or translucent walls around the perimeter of a space. Unilateral: one sided. Bi-lateral: from two opposing sides.

10.5.8.1.1 bilateral sidelighting

Daylighting from two opposite sides of a space.

10.5.8.1.2 unilateral sidelighting

Sidelighting systems in which daylight is introduced through apertures in one wall of a space.

10.5.8.2 clerestory

A window located in a section of wall above the normal view-window height. In some cases this window is located above an adjacent rooftop.

10.5.8.3 shading coefficient

[This term has been deprecated. It has been replaced by solar heat gain coefficient, which is defined differently.]

10.5.8.4 lighting-cooling index

The ratio of the visible transmittance to the shading coefficient for a given window.

10.5.8.5 toplighting

Daylighting systems in which daylight is admitted through apertures in the ceiling or roof, such as skylights or roof monitors.

10.5.8.6 tubular daylighting device (TDD)

High reflectance specular metal tube(s), typically with glazing or optical devices at top and bottom, which act as a light pipe to transfer daylight from a rooftop aperture to a diffuser in a space below.

10.5.8.7 rooftop monitor

A raised portion of the roof that typically contains vertical or steeply sloped glazing on one or more sides.

10.5.8.8 sawtooth roof

A roof that allows the entry of daylight via a series of parallel vertical or inclined glazed apertures, separated by horizontal or sloped opaque planar roof sections. (See Figure 10-1.)

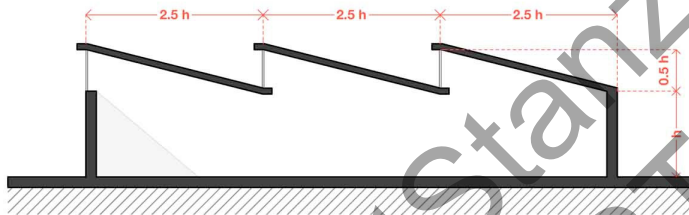


Figure 10-1. Example of a sawtooth roof.

10.5.8.9 daylight harvesting

The reduction of electric lighting power and illumination in a space in direct response to daylight illumination.

10.5.8.10 daylighting system

A single or series of building apertures, along with their associated management components, for the controlled admission and distribution of daylight.

10.5.8.11 daylight management

The control of how daylight is admitted into architectural spaces in quality, quantity and timing. It includes design and/or control of apertures, glazing types, shading, filtration, and light redirection devices, and can be static (i.e., passive) or dynamic (with manual or automated adjustment).

10.5.8.12 window shades

Continuous planar material (translucent or opaque; woven, mesh or film; interior or exterior) that can be deployed/retracted along the plane of the glazing to control solar heat gain, visible transmittance, view, and/or visual privacy.

10.5.8.13 blinds

A light controlling shading device consisting of overlapping thin slats which can be varied in angle and deployment.

10.5.8.13.1 venetian blinds]

Horizontal slatted blinds.

10.5.8.13.2 vertical blinds

Vertical slatted blinds.

10.5.8.13.3 miniblinds

Horizontal slatted blinds 2.5 cm (1 in.) or less in width.

10.5.8.14 daylight louver

A non-retractable, fixed or rotatable series of slatted elements (solid or perforated, specular or diffusing) in a planar relationship used to redirect or shade daylight.

10.5.8.15 light shelf

A horizontal planar element (interior or exterior) along a window that intercepts and redirects sunlight and sky light upward into a space.

10.5.8.16 light well

A rectangular or splayed vertical shaft in a building, either open or roofed with a skylight, that delivers daylight to spaces below.

10.5.9 Daylight Analysis

10.5.9.1 daylight zone

A term used in energy codes to reference an area where separate control must be provided, either manual or automatic depending on the code, for the purpose of daylight harvesting.

10.5.9.1.1 primary daylight zone

A daylight zone close to windows or skylights where a majority of the electric light can be replaced by daylight.

10.5.9.1.2 secondary daylight zone

A daylight zone farther from windows or skylights where some electric light can be replaced by daylight.

10.5.9.2 lighting power density (LPD)

The total input power of the installed lighting system per unit floor area, typically expressed as watts per square meter or per square foot.

10.5.9.3 light-to-solar-gain ratio

Ratio of the visible transmittance (VT or T_{vis}) of a glazing system to its solar heat gain coefficient (SHGC). It is one measure of the degree of a glazing's spectral selectivity.

10.5.9.4 U-factor

The rate of transfer of heat through a building assembly or glazing system per unit area divided by the difference in temperature of the environment on either side.

10.5.9.5 solar heat gain coefficient (SHGC)

The fraction of incident solar radiation that is admitted through a window or skylight, both directly transmitted and absorbed, and subsequently released inward. SHGC can be expressed as a number between 0 and 1 for the glass area alone or for the whole window assembly.

10.5.9.6 critical task location

In daylighting, the work plane location that requires the highest light output from a controlled zone of luminaires to reach a target illuminance in combination with the available daylight.

10.5.9.7 artificial sky

A physically or digitally simulated sky luminance distribution, which may also include the solar disk.

10.5.9.7.1 artificial sky "mirror box"

A physical artificial sky in the form of a box with a luminous ceiling and mirrored walls, used to model uniform or overcast skies.

10.5.9.8 annual sunlight exposure (ASE)

The fraction of the horizontal work plane that exceeds a specified direct sunlight illuminance level more than a specified number of hours per year over a specified daily schedule, with all operable shading devices retracted. [See *ANSI/IES LM-83-23, Approved Method: IES Spatial Daylight Autonomy (sDA) and Annual Sunlight Exposure (ASE)*.]

10.5.9.9 daylight autonomy (DA)

The fraction of time in an annual simulation or study period that an analysis point meets or exceeds a specified illuminance level.

10.5.9.9.1 spatial daylight autonomy (sDA)

An annual daylighting metric that quantifies the fraction of the area within a space for which the daylight autonomy exceeds a specified value.

10.5.9.9.2 continuous daylight autonomy (cDA)

The fraction of time in an annual simulation that an analysis point meets or exceeds a specified illuminance level, with proportional credit given for daylight contributions that partially meet this level.

10.5.9.10 useful daylight illuminance (UDI)

The fraction of time across a daylighting study period where the illuminance at a point lies between selected minimum and maximum levels (typically 100 and 3,000 lux).

10.5.9.11 daylight factor (DF)

A measure of daylight illuminance at a point on a plane, expressed as the ratio of the illuminance at that point to the simultaneous exterior illuminance on a horizontal plane from an unobstructed sky of assumed or known luminance distribution (typically overcast). Direct sunlight is excluded from both interior and exterior illuminance values.

10.5.9.12 daylight coefficient

A factor for an analysis point, that when multiplied by the luminance of a sky patch or the solar disk at a particular solar position, provides the daylight illuminance at that point from that source, typically including both direct and reflected light.

10.5.9.13 local solar time (LST)

The time used in the calculation of the sun's position wherein the sun is at its highest position in the sky at noon as it crosses the local meridian.

Also called *solar time*.

10.5.9.14 direct sunlight hours

The number of total hours within a specified analysis period for which an analysis point receives direct sunlight greater than or equal to a specified illuminance.

10.5.9.15 equation of time

The equation that provides the difference between standard clock time and solar time at a given location, which varies with Julian day.

10.5.9.16 Julian day

For the purpose of daylight simulations and solar position determination: The integer assigned to a calendar day continuously counted from 1 starting with January 1, for the purpose of daylighting calculations.

10.5.9.17 annual operating hours

The number of hours per annum for which a building, lighting, or other system is operating.

10.5.9.18 sky component (SC)

The daylight contribution of scattered light from the sky (excluding direct sun) in a daylight calculation or measurement.

10.5.9.19 externally reflected component (ERC)

The daylight contribution to an interior surface that results from the sunlight and skylight that has been diffusely or specularly redirected from the ground and other exterior surfaces.

10.5.9.20 internally reflected component (IRC)

The daylight contribution at a point within a space resulting from the interreflections that occur within that space.

10.5.9.21 solar time

[See *local solar time*.]

10.5.10 daylight availability

The luminous flux from sun plus sky at a specific location, time, date, and sky condition.

10.5.11 service period

The number of hours per day for which the daylighting provides a specified illuminance level. It often is stated as a monthly average.

10.5.12 annual sunlight exposure (ASE)

In daylighting: The fraction or percentage of the horizontal work plane that exceeds a specified direct sunlight illuminance level more than a specified number of hours per year over a specified daily schedule with all operable shading devices retracted.

10.6 television, film, and theatre lighting

10.6.1 lighting

10.6.1.1 area lighting (television, film and theater)

The practice of lighting the acting space in separately controlled areas.

10.6.1.2 back light

Illumination from behind (and usually above) a subject to produce a highlight along its edge and consequent separation between the subject and its background. Syn.: *fringe light, halo light, contour light, hair light*.

10.6.1.3 backing lighting

The illumination provided for scenery in off-stage areas visible to the audience.

10.6.1.4 base light

Uniform, diffuse, nearly shadow-free illumination sufficiently intense for a television or film picture of acceptable quality at a desired lens opening. Acceptable base level of unaccented stage illumination.

10.6.1.5 blackout

To switch off all illumination (except exit lights).

10.6.1.6 blending lighting

General illumination used to provide smooth transitions between the specific light areas on a stage.

10.6.1.7 brightness ratio

The ratio of maximum to minimum luminances occurring within a scene.

10.6.1.8 contour light

(See Section 10.6.1.2 back light.)

10.6.1.9 counter-key light

Illumination on a subject from a direction that is opposite to that of the key light.

10.6.1.10 cross-lighting

Illumination from two sources on opposite sides of the subject. Often, different color media are used in the luminaires for a given area to give the illusion of shadow while providing sufficient illumination for good visibility.

10.6.1.11 cue

- (1) An event in a production that is the signal for a specific action.
- (2) A signal given in order to cause such an action.
- (3) The response to such a signal that may include intensity settings for one or more luminaires in use in the production.

10.6.1.12 dim

To reduce the illuminance produced by a luminaire. Though mechanical means may be utilized (see **Section 10.6.2.72 iris**), the usual reference is to the use of an electrical dimmer.

10.6.1.13 eye light

Illumination on a person to produce a specular reflection from items such as eyes, teeth, and jewelry, without significantly increasing the total illumination of the subject.

10.6.1.14 fade

A controlled change from one level or set of levels to another. It may refer to lighting, as a fade from one dimmer setting or set of readings to another, or to video level changes.

10.6.1.15 flood (television, film and theater)

- (1) To adjust a luminaire, usually by moving the lamp closer to the lens, to enlarge the diameter of the beam of light emitted.
- (2) The widest field of illumination from a luminaire.

10.6.1.16 fill light

Illumination added to reduce shadows or contrast range.

10.6.1.17 fringe light

(See Section 10.6.1.2 back light.)

10.6.1.18 front lighting

Light from the front or viewer's side of a performance; front-of-house (FOH) lights.

10.6.1.19 general illumination (television, film and theater)

Diffuse illumination of large areas with floodlights or striplights.

10.6.1.20 halo light

(See Section 10.6.1.2 back light.)

10.6.1.21 hard light

Light that causes an object to cast a sharply defined shadow.

10.6.1.22 high-key lighting

A type of lighting that, applied to a scene, results in a picture having gradations from middle gray to white with comparatively limited areas of dark gray and black.

Also: Intense over-all illumination.

In motion pictures: High-level accent lighting with strong contrast (dark, deep shadows with little or no middle gray).

10.6.1.23 key light

The apparent principal source of directional illumination of a subject.

10.6.1.24 low-key lighting

A type of lighting that, applied to a scene, results in a picture having gradations from middle gray to black with comparatively limited areas of light grays and white.

10.6.1.25 modeling light

Illumination that reveals the depth, shape, and texture of a subject. Types of modeling light include *key light*, *cross-lighting*, *counter-key light*, *side light*, *back light*, and *eye light*.

10.6.1.26 paddle

(See Section 10.6.2.144 stage plug.)

10.6.1.27 patch

- (1) To make connections on the patch panel.
- (2) To make dimmer or channel assignments in the patch section of a computerized lighting control system.

10.6.1.28 preset

A set of pre-determined lighting levels that can be mechanically or electronically set up on a system in advance of need, and to which a console operator may fade when desired, or the provision for such sets of levels.

10.6.1.29 set light

Separate illumination of background or scenic elements.

10.6.1.30 side-back light

Illumination from behind the subject in a direction not parallel to a vertical plane through the optical axis of the cameras.

10.6.1.31 side light

Lighting from the side to enhance subject modeling and place the subject in depth, apparently separated from background.

10.6.1.32 soft light

- (1) Diffuse illumination that produces soft-edged, poorly defined shadows on the background when an object is placed in its path.
- (2) A luminaire designed to produce such illumination.

10.6.1.33 specific illumination

Controlled illumination of localized areas.

10.6.1.34 spot

- (1) To adjust a luminaire, usually by moving the lamp away from the lens, to reduce the diameter of the beam of light emitted.
- (2) The narrowest beam possible from a luminaire that can be focused.
- (3) A spotlight.

10.6.1.35 throw

The distance between a luminaire and the area or subject being lighted.

10.6.1.36 tonal light

(See Section 10.6.1.29 set light.)

10.6.1.37 top light

Illumination of a subject directly from above; employed to outline the upper margin or edge of the subject.

10.6.1.38 wash

Even, over-all light on performance space or background.

10.6.1.39 work light

Illumination provided for specific operational purposes, independent of the production lighting, such as for cleaning and set construction.

10.6.1.40 cross fade

The smooth transition from one lighting cue to another. May involve timed, automatic, or manual operations.

10.6.2 equipment

10.6.2.1 ace

A 1000-W Fresnel spotlight.

10.6.2.2 analog multiplex, (AMX)

An industry standard control protocol for the transmission of dimming level or other data using an analog multiplexer.

10.6.2.3 arc

A discharge light source, usually a carbon arc.

10.6.2.5 asbestos tapoff

(See Section 10.6.2.151 twofer.)

10.6.2.6 automated fixture

A lighting fixture whose color, movement, speed, beam size, beam pattern, or other attributes are controlled remotely from a control console.

10.6.2.7 baby

Once a 250-watt to 300-watt planoconvex spotlight with a lens diameter of 10 to 13 cm (4 to 5 in.). Now usually a 500-W to 1000-W Fresnel spotlight with a lens diameter of 15 cm (6 in.), standard in the motion picture industry.

Also: A small version of any spotlight.

10.6.2.8 balcony lights

Luminaires mounted on the front edge of the auditorium balcony. Syn.: *rail lights*, *FOH spots* (see **Section 10.6.2.63**).

10.6.2.9 bank

A group of switches and/or dimmers that may be controlled together, or a group of luminaires.

10.6.2.10 barn doors

A set of adjustable flaps—usually two, four, or eight—that may be attached to the front of a luminaire (usually a Fresnel spotlight) speed ring, or accessory holder in order to partially control the shape and spread of the light beam.

10.6.2.11 barrel

Either or both of the concentric tubes of a focusing lens system.

In Europe, referred to as the *electrical pipe*.

10.6.2.12 batten

A pipe of diameter 3.1 to 6 cm (1.25 to 2 in.) suspended across the stage or studio, often on wire cable, to permit it to be raised or lowered. Lighting battens usually carry a wireway or outlet boxes with pigtails or receptacles so that luminaires hung on the batten may be connected to controlled circuits. In European practice, referred to as a *barrel*.

10.6.2.13 beam lights

Luminaires mounted in the ceiling of the auditorium, often concealed by a false beam.

Syn.: *front-of-house (FOH) spots* (see **Section 10.6.2.63**).

10.6.2.14 beam projector

A luminaire with the light source at or near the focal point of a paraboloidal reflector producing near-parallel rays of light in a beam of small divergence. Some are equipped with spillings to reduce spill and glare. In most types, the lamp may be moved toward or away from the reflector to vary the beam spread.

10.6.2.15 bed

Platform usually suspended from above to provide lighting positions on motion picture stages.

10.6.2.16 board

A switchboard, control board, or dimmer board.

10.6.2.17 boomerang (boom)

- (1) A vertical spotlight-mounting pipe, usually located at the side of the stage or auditorium (i.e., box boom).
- (2) A device mounted on the front of a spotlight that enables an operator to change the color media rapidly and easily, often from the rear of the unit.

10.6.2.18 box boom

(See **Section 10.6.2.17 boomerang** and **Section 10.6.2.140 splay**.)

10.6.2.19 borderlight

A long continuous striplight hung horizontally above the stage and aimed down to provide general diffuse illumination and/or to light the cyclorama or a drop; usually wired in three- or four-color circuits.

Also available in portable versions.

10.6.2.20 bridge (light bridge)

A suspended mounting position and catwalk located behind the proscenium arch, providing access to and a mounting for stage lighting units.

10.6.2.21 broad

A floodlight with a fixed or variable beam; used as a source of fill light.

10.6.2.22 brute

A 225-ampere, direct-current, high-intensity carbon arc spotlight with a 60-cm (24-in.) diameter Fresnel lens.

10.6.2.23 carbon arc

An alternating-current or direct-current arc light source. (See **Section 10.6.2.64 follow spot** and **Section 10.6.2.22 brute**.)

10.6.2.24 camera light

A lighting instrument mounted on a camera for lighting on the optical axis, usually to provide eye light.

10.6.2.25 color frame

A removable metal frame or fiberboard frame used to support color media at the front of a luminaire.

10.6.2.26 color media

Colored transparent material. Currently in use are glass, gelatin, and various plastics.

10.6.2.27 color wheel

A device holding several different color media that can be rotated by hand or motor to change the color in front of the luminaire.

10.6.2.28 concert border

A borderlight mounted immediately behind the proscenium arch and the act curtain.

10.6.2.29 company switch

A disconnect switch and power source for portable lighting equipment.

10.6.2.30 console

The desk or panel position from which production lighting is controlled, containing all master controls, fader controls, and other components needed in production.

10.6.2.31 control board

An electrical panel containing controls for production lighting, including dimmer controllers, faders, and non-dim switches. (See **Section 10.6.2.46 dimmer board** and **Section 10.6.2.30 console**.)

10.6.2.32 cookie

(See Section 10.6.2.38 cucoloris.)

10.6.2.34 cove base

A seamless base, usually on a hard cyclorama, whose purpose is to form a seamless transition from the studio floor. May hide lighting instruments.

10.6.2.35 cross connect

An assembly of plugs and jacks, sliders, or switches permitting connection of studio or stage outlets in various combinations to the output of the dimmers or non-dim circuits.

Also called a *patch panel* (see Section 10.6.2.111).

10.6.2.37 cross fader

A device that produces a cross fade.

10.6.2.38 cucoloris

An opaque cut-out panel mounted between a light source (sun or arc lamp) and a target surface in order to project a shadow pattern (clouds or leaves are typical) upon scenery, cyclorama, or acting area.

10.6.2.39 cutter

(See Section 10.6.2.71 gobo.)

10.6.2.40 cyclorama ("cyc")

A vertical surface used to form a background of apparent infinite depth. It is evenly illuminated with striplights and/or scoops from above and below in order to produce this effect. Those luminaires below the cyc may be recessed below stage level in a pit designed for the purpose.

10.6.2.41 daylight conversion filter

A filter designed to modify the spectral energy distribution of a light source in order to produce illumination of approximately photographic "daylight."

10.6.2.42 designer's remote

A portable lighting console used in the auditorium seating area to set up or monitor lighting cues during rehearsals.

10.6.2.43 deuce

A 2000-watt Fresnel spotlight.

10.6.2.44 digital multiplex, DMX

An industry standard control protocol for the transmission of dimming level or other data using a digital multiplexer.

10.6.2.45 dimmer

A device used to control the light emitted by a light source or luminaire.

10.6.2.46 dimmer board

A panel board containing switches, dimmers, and outlets for the luminaires to be controlled. A control board; switchboard.

10.6.2.47 dimmer curve

The graphically plotted relationship between incremental scale settings of a dimmer control handle and the light output of a lamp controlled by a dimmer.

10.6.2.48 dimmer per circuit

The dimming system architecture that provides a dimmer for each circuit, without a load patch panel. Patching is done on a control basis. (See **Section 10.6.2.115 patch panel**.)

10.6.2.49 dot

A small round gobo (see **Section 10.6.2.72**).

10.6.2.50 double broad

A two-lamp floodlight, usually of rectangular shape, used as a source of fill light.

10.6.2.51 dowser

A cut-off shutter, usually in a carbon arc spotlight, that protects the lens while the arc is struck. (It closes the small opening remaining with a closed iris.)

10.6.2.52 dummy load

(See **Section 10.6.2.114 phantom**.)

10.6.2.53 Dutchman

A condensing lens assembly designed to fit on the front of a plano-convex spotlight. Contains a slide and supports an objective lens system so that the luminaire can be used as an effect projector (see **Section 10.6.2.54**).

10.6.2.54 effect projector

A luminaire incorporating a light source of high luminance, a lens system to provide an even field of illumination at an aperture, and a second lens to project the image of this aperture onto a planar surface with a minimum amount of geometric distortion. Slides, moving film, or rotating discs may be used in the aperture to provide still or motion effects.

10.6.2.55 ellipsoidal reflector spotlight

A spotlight in which a lamp and an ellipsoidal reflector are mounted in a fixed relationship, directing a beam of light into an aperture where it may be shaped by a pattern, iris, shutter system, or other insertion. The beam then passes through a single or compound lens system that focuses it as required, producing a sharply defined beam with variable edge definition.

10.6.2.56 fader

(See **Section 10.6.2.37 cross fader**.)

10.6.2.57 FAY light

A luminaire that uses incandescent parabolic reflector lamps with a dichroic coating to provide "daylight" illumination.

10.6.2.58 filter

(See **Section 10.6.2.26 color media**.)

Also: A component of an electronic dimmer used to control electromagnetic or radio-frequency interference.

10.6.2.59 fiver (or five K)

A 5000-W Fresnel incandescent spotlight.

10.6.2.60 flag

Opaque material placed in front of a source to produce a shadow in the area illuminated. (See **Section 10.6.2.72 gobo**.)

10.6.2.61 floodlight (television, film, theater)

A luminaire consisting of lamp and spread reflector with resultant field angle greater than 100 degrees. Fixed and variable beam types are available. (See **Section 10.6.2.132 scoop**.)

10.6.2.62 floor pocket

An electrical outlet or group of outlets recessed in the floor beneath a hinged cover so notched that cables plugged into these outlets may pass up through the notches of the closed cover.

10.6.2.63 front of house (FOH) intercept panel

An electrical panel through which circuits used in the auditorium are routed, with interruption capability to divert them from house dimmers to portable dimmers. Found in larger theatres.

10.6.2.64 front of house (FOH) spots

Luminaires mounted in auditorium or front-of-house (FOH) positions.

(See **Section 10.6.2.13 beam lights**, **Section 10.6.2.125 rail lights**, and **Section 10.6.2.8 balcony lights**).

10.6.2.65 follow spot (light)

Any instrument operated so as to follow the movement of an actor. Follow spots are usually high intensity, controlled-beam luminaires.

10.6.2.66 footlights

A set of striplights at the front edge of the stage platform, used to soften face shadows cast by overhead luminaires and to add general toning lighting from below.

10.6.2.67 Fresnel spotlight

A luminaire containing a lamp and a Fresnel (a stepped "flat" lens with a textured back) lens that has variable field and beam angles obtained by changing the spacing between lamp and lens (for flooding and spoting). Produces a smooth, soft-edged, defined beam of light.

10.6.2.68 funnel

A metal tube that can be mounted on the front of a spotlight to control stray light.

Syn: *snoot* (**Section 10.6.2.139**), *high hat* (**Section 10.6.2.80**), *top hat* (**Section**

10.6.2.149).

10.6.2.69 gel

A color medium made from gelatin. Sometimes used as generic name for *color media*.

10.6.2.70 ghost (or ghost load)

(See **Section 10.6.2.114 phantom.**)

10.6.2.71 globe (theater lighting equipment)

A light source containing a decorative spherical cover.

10.6.2.72 gobo

In the theatre, a metal cut-out pattern of suitable size to fit in the holder or aperture of an ellipsoidal reflector spotlight. In motion pictures, a flag, usually opaque, that may be made with diffusing material. It can also be an opaque diffusing “cookie” (see **Section 10.6.2.38 cucoloris**).

10.6.2.73 (grid or gridiron)

A continuous working and mounting floor above a stage from which rigging, including that for lighting, is suspended.

10.6.2.74 grip

In motion pictures, the person in charge of rigging and construction; e.g., the key grip.

10.6.2.75 grip gear

In television and motion pictures, the equipment used to mount or hold various types of production equipment on or around the set.

10.6.2.76 ground row

- (1) Lighting strips aimed up from the base of a cyclorama or drop.
- (2) The masking piece concealing these strips.

10.6.2.77 half scrim

(See **Section 10.6.2.133 scrim.**)

10.6.2.78 handheld remote

A device that controls the lighting console or dimmers remotely; used to turn on and off

selected dimmers or channels while focusing or setting, or checking the lighting.

10.6.2.79 hard cyc

A permanent wall, usually built into a studio, with a cove base and constructed for use with “cyc” lights (see **Section 10.6.2.40**) to create the vision of a flat surface without form. Used with chroma or Ultimatte keys, and other projection.

10.6.2.80 high hat

(See **Section 10.6.2.68 funnel**.)

10.6.2.81 house lights

The general lighting system installed in the audience area (house) of a theater, film or television studio, or arena.

10.6.2.82 inky

A small Fresnel spotlight, with a lens diameter of 3.8 to 7.6 cm (1.5 to 3 in.

Also: Any incandescent luminaire.

10.6.2.83 instrument

(See **Section 10.3.1 luminaire**.)

10.6.2.84 instrument schedule

(See **Section 10.6.2.131 schedule**.)

10.6.2.85 inter-lock

- (1) A mechanical connection between the control handles of a number of dimmers to permit simultaneous operation.
- (2) An electrical safety device that removes power from equipment when access covers are opened.

10.6.2.86 iris

An assembly of flat metal leaves arranged to provide an easily adjustable near-circular opening, placed near the focal point of the beam (as in an ellipsoidal reflector spotlight) or in front of the lens to act as a mechanical dimmer, as in older types of carbon arc follow spotlights.

10.6.2.87 junior

(See Section 10.6.2.43 **deuce**.)

10.6.2.88 kicker

An instrument used to provide an additional highlight or accent on a subject.

10.6.2.89 klieg light

A high intensity carbon-arc spotlight, typically used in motion picture lighting.

10.6.2.90 ladder

A ladder-like luminaire mounting assembly hung in the wings of a theatre, often with a cable-fed outlet box.

10.6.2.91 Leko

(See Section 10.6.2.55 **ellipsoidal reflector spotlight**.)

10.6.2.92 light lift

A pantograph or other adjustable-height device used to support luminaires from the grid.

10.6.2.93 light plot

The complete layout of lighting equipment and its application in a production.

10.6.2.94 Linnebach projector

A lensless scenic projector, using a concentrated source in a black box and a slide or cut-out between the source and the projection surface.

10.6.2.95 loader

Usually a plano-convex spotlight (with the lens removed), used as a phantom load.

10.6.2.96 Martingale

(See Section 10.6.2.152 **twofer**.)

10.6.2.97 master

A control, switch, or dimmer that regulates a number of individual control devices, or a number of sub-masters each of which controls a number of individual devices.

10.6.2.98 matte

Having a dull or diffuse surface finish that reflects light uniformly in all directions.

10.6.2.99 memory

A device for storing intensity information (cues or presets) about combinations of luminaires that are used during the course of a production. Originally accomplished by potentiometer arrays (presets); now usually volatile or non-volatile electronic elements.

10.6.2.100 midget

(See Section 10.6.2.82 inky.)

10.6.2.101 non-lens spotlight

A beam projector light source in fixed relationship to a parabolic reflector; often has a mask to prevent direct illumination from the light source.

10.6.2.102 non-dim circuit

An electric circuit providing an on-off control, rather than dimming.

10.6.2.103 Obie light

(See Section 10.6.1.13 eye light.)

10.6.2.104 olivette

A box-like floodlight.

10.6.2.105 one-K

(See Section 10.6.2.1 ace.)

10.6.2.106 overhead (or butterfly)

Usually refers to a 6.1 m x 6.1 m (20 ft x 20 ft) overhead butterfly silk lighting reflector or diffuser. Used to create a large pool of indirect light on an area.

10.6.2.107 paddle

(See Section 10.6.2.144 stage plug.)

10.6.2.108 pantograph

(See Section 10.6.2.92 light lift.)

10.6.2.109 par light

A luminaire that uses incandescent parabolic reflector lamps.

10.6.2.110 parabolic spotlight

(See **Section 10.6.2.14 beam projector.**)

10.6.2.111 parallel (television, film and theater)

(See **Section 10.6.2.15 bed.**)

10.6.2.112 patch panel

- (1) An assembly of plugs, jacks, and cords (either independent or retracting) serving as a cross connect to permit connection of various load circuits to dimmers or non-dim circuits.
- (2) Functional section of computerized lighting control equipment that assigns dimmers (or attributes of automated luminaires) to channels on the control system. (See **Section 10.6.2.35 cross connect.**)

10.6.2.113 pattern spotlight

An ellipsoidal reflector spotlight, especially one with a pattern or gobo slot at the focal point of the unit to permit insertion of pattern templates for projection.

10.6.2.114 phantom

A load added to a resistance dimmed circuit to permit the load to be dimmed to full out; needed when the load is substantially less than that for which the dimmer is designed.

Also known as a *ghost* or *ghost load* (see **Section 10.6.2.70**).

10.6.2.115 pigtail

A length of cable from a wireway to a load connector. Typically located near instrument mounting positions as a ready means of electrical connection.

10.6.2.116 pin plug or connector

Electrical connector using round cross-section pins.

10.6.2.117 pipe

(See **Section 10.6.2.12 batten.**)

10.6.2.118 plano-convex spotlight

A spotlight embodying a plano-convex lens and a lamp movable within the housing in relation to the lens in order to vary beam angle and field angle.

10.6.2.119 plate

A single resistance-dimmer unit. Also, by analogy, applied to other types.

10.6.2.120 plot

(See Section 10.6.2.93 light plot.)

10.6.2.121 practical lamp

A luminaire used on a setting to provide an apparent source of illumination (e.g., a desk lamp, table lamp, or floor lamp).

10.6.2.122 profile spotlight

(See Section 10.6.2.55 ellipsoidal reflector spotlight.)

10.6.2.123 quartz (quartz iodine and quartz halogen)

(See Section 10.6.2.132 tungsten-halogen.)

10.6.2.124 quartz lights

Generic term for luminaires of various types that utilize tungsten-halogen lamps.

10.6.2.125 rail lights

(See Section 10.6.2.8 balcony lights.)

10.6.2.126 rear screen projector

An effect projector that is designed to project images onto a translucent screen from the rear.

10.6.2.127 rheostat

A resistance dimmer.

10.6.2.128 roundel

A round, cast-glass color medium, often used in borderlights. Also available as a clear cover glass.

10.6.2.129 scene master

A device for controlling a group of dimmers that are assigned to a specific set or scene.
(See **Section 10.6.2.97 master.**)

10.6.2.130 scenic projector

(See **Section 10.6.2.54 effect projector.**)

10.6.2.131 schedule

A tabular list of lighting units to be used on a production, usually including type of unit, hanging position, lamp type and size, color media, accessories, cabling instructions, and use.

10.6.2.132 scoop

A floodlight consisting of a lamp in an ellipsoidal or paraboloidal matte reflector, usually in a fixed relationship, though some types permit adjustment of the beam shape.

10.6.2.133 scrim

Material used 1) in the beam of a luminaire to modify the beam shape, intensity, and/or shadow characteristics (also available as half scrim to cut intensity of only one-half of the beam, usually with wire screen); or 2) as a stage curtain that appears opaque when illuminated from the viewing side and becomes transparent when this illumination is reduced and a higher level illumination is provided behind the scrim.

10.6.2.134 scroller

A device that mounts to the front of lighting instruments and allows the selection and use of gel-type filters.

10.6.2.135 sealed beam spotlight

A light source with integral filament, reflector, and lens.

10.6.2.136 senior

(See **Section 10.6.2.59 fiver.**)

10.6.2.137 shutters

Adjustable beam shaping devices (usually four) in a luminaire.

10.6.2.138 silicon controlled rectifier (SCR)

The most common power device in electronic dimmers. Other power devices include triacs, IGBT, and FETS.

10.6.2.139 snoot

(See **Section 10.6.2.68 funnel**.)

10.6.2.140 spider

An electrical distribution device; usually breaks down the feed from the source to individual circuits.

10.6.2.141 splay

A lighting position on the side of the auditorium.

10.6.2.142 spotlight

- (1) Any of several different types of luminaires with relatively narrow beam angles; designed to illuminate a specifically defined area.
- (2) In motion pictures: Generic for a Fresnel-lens luminaire.

10.6.2.143 stage manager's panel

A panel or console almost always located backstage, where the stage manager can control or monitor activities in the lighting control system and has access to the control of the houselights and work lights; an intercom.

10.6.2.144 stage plug

A flat, two- or three-conductor connector.

10.6.2.145 striplight

Formerly, an open trough reflector containing a series of lamps. Now, usually a compartmentalized luminaire, with each compartment containing a lamp, reflector, and color frame holder; wired in rotation in three or four circuits; and used as borderlights, footlights, or cyclorama lighting from above or below. Often in short 0.9-m to 2.4-m (3-ft to 8-ft) portable sections.

10.6.2.146 sub-master

A control device on lighting consoles for the manipulation of groups of lighting channels together. (See also **Section 10.6.2.97 master**.)

10.6.2.147 switchboard

A switch panel; a control board or console of any type; a portable control board of the "piano-box" type.

10.6.2.148 tenner or ten-K

A ten-kilowatt incandescent Fresnel spotlight.

10.6.2.149 top hat

(See Section 10.6.2.68 funnel.)

10.6.2.150 tormentor lighting

Luminaires mounted directly behind the sides of the stage arch.

10.6.2.151

[Number no longer used.]

10.6.2.152 twofer

A load connector wired in parallel to two line connectors so that two luminaires can be connected to a single electrical circuit.

10.6.2.153 two-K

(See Section 10.6.2.43 deuce.)

10.6.2.154 unit (lighting equipment)

A luminaire (see Section 10.3.1).

10.6.2.155 wall pocket

An electrical outlet or group of outlets mounted on or recessed in the wall, usually beneath a hinged cover that is notched so that cables plugged into these outlets may pass through the notches of the closed cover.

10.6.2.156 Y-cord

(See Section 10.6.2.152 twofer.)

10.6.2.157 butterfly

[See Section 10.6.2.106 overhead (or butterfly).]

11.0 EXTERIOR LIGHTING APPLICATIONS

11.1 signal lighting

11.1.1 visual signal

A means of conveying information by the use of light; the light stimulus rather than the equipment that produces it.

11.1.2 signaling system

The whole complex of lights, colors, shapes, and movements designed to communicate information.

11.1.3 message

The particular meaning to be communicated by a signal containing information, a warning, or both.

11.1.4 code (or character)

The sequence of flashes associated with a particular message; a repeated sequence.

11.1.5 atmospheric transmissivity

The ratio of the directly transmitted flux incident on a surface after passing through unit thickness of the atmosphere to the flux that would be incident on the same surface if the flux had passed through a vacuum.

11.1.5.1 Koschmieder's law

The atmospheric transmittance constant (which is conventionally taken as 5%) over a distance equal to the daylight visual range (the meteorological visibility), whatever the density of a fog or mist. It follows that, if the meteorological visibility distance is V , the atmospheric transmittance for unit distance, or the transmissivity, T_1 is given by:

$$(T_1)^V = 0.05,$$

and the atmospheric transmittance T_d for some other distance d may be conventionally calculated from:

$$T_d = (T_1)^d = 0.05^{d/V}$$

11.1.5.2 Allard's law

The point brilliance (see [Section 5.11.1](#)) from a distant light source depends on the

inverse square law and the atmospheric transmittance according to:

$$Ed^2 = I (T_1)^d$$

where:

E = the point brilliance

d = distance

I = intensity of the light source

T_1 = atmospheric transmittance per unit distance or transmissivity

11.1.5.3 Nominal range of a light signal

The luminous range calculated for dark conditions and with nominal values for the constants in Allard's law (see **Section 11.1.5.2**):

- (1) The visual threshold in point vision in dark conditions is equal to 0.2 microlux or 0.686 nautical mile-candela.
- (2) Atmospheric transmittance corresponding to a meteorological visibility of ten nautical miles is equal to 0.85 per kilometer or 0.74 per nautical mile-candela.

Note: A nautical mile-candela is the illuminance produced at one nautical mile by a source of one candela.

11.1.6 beacon

A light (or mark) used to indicate a geographic location.

11.1.7 fixed light

A light having a constant luminous intensity when observed from a fixed point.

11.1.8 rhythmic light

A light that when observed from a fixed point has a luminous intensity that changes periodically.

11.1.8.1 fixed light equivalent

Usually applied to the apparent intensity of a flashing light in point vision, the intensity of a steady light that appears to have the same point-brilliance as the flashing light.

Also known as *apparent intensity*.

11.1.8.2 flash period

- (1) The time interval after which the same flash character reoccurs.

- (2) The sum of the flash duration (on-period) and the flash interval (off-period) for a repeated single flash.

Unit: second, s.

11.1.8.3 flash frequency

The reciprocal of the flash period, expressed in hertz or, if clearly marked as such, in flashes per minute.

11.1.8.4 flash duration, (on-period)

- (1) The total time from the start to the finish of a flash.
- (2) The effective duration, which is the time interval during which the actual intensity (or the actual point brilliance [see **Section 5.11.1**]) exceeds some specified level above the threshold.

Unit: second, s.

11.1.8.5 flash interval, (off-period)

- (1) The time interval from the finish of one flash to the start of the next.
- (2) The difference between the flash period (for repeated single flashes) and the effective flash duration.

Also known as *eclipse time* or *null period*.

Unit: second, s.

11.1.8.6 on-off ratio (or mark-space ratio) for a repeated single flash

The ratio of the flash duration to the flash interval. Less than unity for relatively short flashes; greater than unity for an occulting light (see **Section 11.1.8.11**).

11.1.8.7 duty cycle

The ratio of the total of the flash durations to the flash period.

11.1.8.8 flashing light

A rhythmic light in which the periods of light are of equal duration and are clearly shorter than the periods of darkness.

11.1.8.8.1 group flashing light

A flashing light in which the flashes are combined in groups, each including the same number of flashes, and in which the groups are repeated at regular

intervals. The duration of each flash is clearly less than the duration of the dark periods between flashes, and the duration of the dark periods between flashes is clearly less than the duration of the dark periods between groups.

11.1.8.9 equal interval (isophase) light

A rhythmic light in which the light and dark periods are equal.

11.1.8.10 quick-flashing light

A single flashing light repeated at a frequency equal to or greater than one Hz. There is no agreed-upon verbal differentiation between lights that flash at one Hz and those that flash more rapidly.

A quick flashing light may be a sequence of single flashes or a sequence of multiflick flashes, at one-second intervals. There is no restriction on the ratio of the duration of the light to that of the dark period.

11.1.8.10.1 interrupted quick-flashing light

A quick flashing light in which the rapid alternations are interrupted by periods of darkness at regular intervals.

11.1.8.11 occulting light

A rhythmic light in which the periods of light are clearly longer than the periods of darkness.

11.1.9 very short flashes

In signal lighting: Flashes of millisecond or microsecond duration.

11.1.10 Blondel-Rey law

The ratio of the thresholds of a square-form flashing light (E_a) and of a steady light (E_0), in point vision conditions at night (see **Sections 5.11** and **5.11.1**). The ratio depends on the duration in seconds of the flash (t) according to:

$$E_0/E_a = t/(0.21 + t)$$

11.1.10.1 effective intensity

The effective intensity I_e of a flashing light is, conventionally:

$$I_e = I_0 t / (t + 0.2),$$

for a source of constant magnitude I_0 over time duration t .

For a time-varying source $I(t)$, expressed in candelas:

$$I_e = \left[\int_{t_1}^{t_2} I(t) dt \right] / [(t_2 - t_1) + 0.2]$$

The items t_1 and t_2 are so chosen as to maximize the calculated effective intensity. It is then found that the actual intensity I_t at time t_1 or t_2 is equal to the calculated equivalent intensity I_e .

11.1.10.2 integrated flash intensity

The product of intensity and flash duration. More precisely, the time integral of luminous intensity, in a given direction, integrated over the whole duration of a single flash (or over a stated shorter interval). Evaluated as:

$$\int I(t) dt,$$

and expressed in candela-seconds (cd·s).

11.1.11 linear light

A luminous signal having a perceptible physical length.

11.1.12 conspicuity

The capacity of a signal to stand out in relation to its background so as to be readily discovered by the eye.

11.1.13 visibility (meteorological)

A term that denotes the greatest distance that selected objects (visibility markers) or lights of moderate intensity of the order of 25 candelas can be seen and identified under specified conditions of observation.

The distance may be expressed in kilometers (or miles in the U.S.).

11.1.13.1 runway visibility

The meteorological visibility (see **Section 11.1.13**) along an identified runway.

Where a transmissometer is used for measurement, the instrument is calibrated in terms of a human observer; e.g., the sighting of dark objects against the horizon sky during daylight and the sighting of moderately intense unfocused lights on the order of 25 candelas at night.

11.1.14 meteorological optical range

The length of light-path in the atmosphere that has a transmittance of 5%.

11.1.15 luminous range

The maximum distance at which a light can be seen, as calculated from the luminous intensity, the atmospheric transmissivity, and the visual threshold in point vision.

11.1.16 visual range (of a light or object)

The maximum distance at which a particular light (or object) can be seen and identified.

11.1.16.1 runway visual range, RVR

In the United States, an instrumentally derived value, based on standard calibrations, that represents the horizontal distance a pilot will see down the runway from the approach end; it is based on the sighting of either high-intensity runway lights or on the visual contrast of other targets, whichever yields the greater visual range.

11.1.17 night

The hours between the end of evening civil twilight and the beginning of morning civil twilight.

Note: Civil twilight ends in the evening when the center of the sun's disk is six degrees below the horizon, and begins in the morning when the center of the sun's disk is six degrees below the horizon.

11.1.18 signaling light

A projector used for directing light signals toward a designated target zone.

11.2 aeronautical lighting

11.2.1 aeronautical light

Any luminous sign or signal that is specially provided as an aid to air navigation.

11.2.2 aeronautical ground light

Any light, other than a light displayed on an aircraft, that is specially provided as an aid to air navigation.

11.2.2.1 bar (of lights)

A group of three or more aeronautical ground lights placed in a horizontal line transverse to the axis, or extended axis, of the runway.

11.2.2.1.1 barrette

A short bar in which the lights are closely spaced so that from a distance they appear to be a linear light.

Note: Barrettes are usually less than 4.6 m (15 ft) in length.

11.2.2.2 aeronautical beacon

An aeronautical ground light visible at all azimuths, either continuously or intermittently, whose purpose is to designate a particular location on the surface of the earth.

11.2.2.2.1 aerodrome beacon

An aeronautical beacon used to indicate the location of an aerodrome.

Note: An aerodrome is any defined area on land or water—including any buildings, installations, and equipment—intended to be used either wholly or in part for the arrival, departure and movement of aircraft.

11.2.2.2.2 airway beacon

An aeronautical beacon used to indicate a point on the airway.

11.2.2.2.3 course light

An aeronautical ground light supplementing an airway beacon, for indicating the direction of the airway and to identify by a coded signal the location of the airway beacon with which it is associated.

11.2.2.2.4 hazard or obstruction beacon

An aeronautical beacon used to designate a danger to air navigation.

11.2.2.2.5 identification beacon

An aeronautical beacon emitting a coded signal by means of which a particular point of reference can be identified.

11.2.2.2.6 landmark beacon

An aeronautical beacon used to indicate the location of a landmark used by pilots as an aid to enroute navigation.

11.2.2.2.7 obstruction beacon

(See **Section 11.2.2.2.4 hazard or obstruction beacon.**)

11.2.2.3 angle-of-approach lights

Aeronautical ground lights arranged so as to indicate a desired angle of descent during an approach to an aerodrome runway. (Also called *optical glide path lights*.)

11.2.2.3.1 Visual Approach Slope Indicator System (VASIS)

The system of angle-of-approach lights accepted as a standard by the International Civil Aviation Organization, consisting of two bars of lights located at each side of the runway near the threshold and showing red or white, or a combination of both (pink) to the approaching pilot, depending on his or her position with respect to the glide path.

11.2.2.3.2 VASIS

[See **Section 11.2.2.3.1 Visual Approach Slope Indicator System (VASIS)**.]

11.2.2.4 approach lights

A configuration of aeronautical ground lights located in extension of a runway or channel before the threshold to provide visual approach and landing guidance to pilots.

11.2.2.5 approach-light beacon

An aeronautical ground light placed on the extended centerline of the runway at a fixed distance from the runway threshold to provide an early indication of position during an approach to a runway.

11.2.2.6 runway-end identification lights

A pair of flashing aeronautical ground lights symmetrically disposed on each side of the runway at the threshold to provide additional threshold conspicuity.

11.2.2.6.1 runway threshold

The beginning of the part of the runway usable for landing.

11.2.2.7 runway alignment indicator

A group of aeronautical ground lights arranged and located to provide early direction and roll guidance on the approach to a runway.

11.2.2.8 obstruction lights

Aeronautical ground lights provided to indicate obstructions.

11.2.2.9 boundary lights

Aeronautical ground lights delimiting the boundary of a land aerodrome without runways.

11.2.2.9.1 perimeter lights

Aeronautical ground lights provided to indicate the perimeter of a landing pad for helicopters.

11.2.2.9.2 range lights

Groups of color-coded boundary lights provided to indicate the direction and limits of a preferred landing path, normally on an aerodrome without runways but also occasionally on an aerodrome with runways.

11.2.2.10 runway lights

Aeronautical ground lights arranged along or on a runway.

11.2.2.10.1 runway edge lights

Lights installed along the edges of a runway, marking its lateral limits and indicating its direction.

11.2.2.10.2 threshold lights

Runway lights so placed as to indicate the longitudinal limits of that portion of a runway, channel, or landing path usable for landing.

11.2.2.10.3 runway centerline lights

Runway lights installed in the surface of the runway along the centerline, indicating the location and direction of the runway centerline. They are of particular value in conditions of very poor visibility.

11.2.2.10.4 touchdown zone lights

Barrettes of runway lights installed in the surface of the runway between the runway edge lights and the runway centerline lights to provide additional guidance during the touchdown phase of a landing in conditions of very poor visibility.

11.2.2.11 circling guidance lights

Aeronautical ground lights provided to supply additional guidance during a circling

approach when the circling guidance furnished by the approach and runway lights is not adequate.

11.2.2.12 channel lights

Aeronautical ground lights arranged along the sides of a channel of water aerodrome.

11.2.2.13 taxi-channel lights

Aeronautical ground lights arranged along a taxi channel of a water airdrome to indicate the route to be followed by taxiing aircraft.

11.2.2.14 taxiway lights

Aeronautical ground lights provided to indicate the route to be followed by taxiing aircraft.

11.2.2.14.1 taxiway-edge lights

Taxiway lights placed along or near the edges of a taxiway.

11.2.2.14.2 taxiway centerline lights

Taxiway lights placed along the centerline of a taxiway except, that on curves or corners having fillets these lights are placed a distance equal to half the normal width of the taxiway from the outside edge of the curve or corner.

11.2.2.14.3 taxiway holding-post light

A light or group of lights installed at the edge of a taxiway near an entrance to a runway, or to another taxiway, to indicate the position at which the aircraft should stop and obtain clearance to proceed.

11.2.2.15 runway-exit lights

Lights placed on the surface of a runway to indicate a path of the taxiway centerline.

11.2.2.16 landing direction indicator

A device to indicate visually the direction currently designated for landing and take-off.

11.2.3 portable traffic control light

A signaling light designed for manual portability that produces a controllable distinctive signal for purposes of directing aircraft operations in the vicinity of an aerodrome.

11.2.4 aircraft aeronautical light

Any aeronautical light specially provided on an aircraft.

11.2.4.1 navigation light system

A set of aircraft aeronautical lights provided to indicate the position and direction of motion of an aircraft to pilots of other aircraft or to ground observers.

11.2.4.1.1 position lights

The aircraft aeronautical lights that form the basic, or internationally recognized, navigation light system.

Note: The system is composed of a red light showing from dead ahead to 110 degrees to the left, a green light showing from dead ahead to 110 degrees to the right, and a white light showing to the rear through 140 degrees.

Also called *navigation lights*.

11.2.4.1.2 anti-collision light

A flashing aircraft aeronautical light or system of lights designed to provide a red signal throughout 360 degrees of azimuth for the purpose of giving long-range indication of an aircraft's location to pilots of other aircraft.

11.2.4.1.3 fuselage lights

Aircraft aeronautical lights mounted on the top and bottom of the fuselage; used to supplement the navigation lights.

11.2.4.1.4 formation light

A navigation light especially provided to facilitate formation flying.

11.2.4.1.5 wing clearance lights

A pair of aircraft lights provided at the wing tips to indicate the extent of the wingspan when the navigation lights are located an appreciable distance inboard of the wing tips.

11.2.4.2 anchor light

An aircraft light designed for use on a seaplane or amphibious aircraft to indicate its position when at anchor or moored.

11.2.4.3 landing light

An aircraft aeronautical light designed to illuminate a ground area from the aircraft.

11.2.4.4 taxi light

An aircraft aeronautical light designed to provide necessary illumination for taxiing.

11.2.4.5 ice detection light

An inspection light designed to illuminate the leading edge of an aircraft wing to check for ice formation.

11.3 floodlighting

A system designed for lighting a scene or object to a luminance greater than its surroundings. It may be for utility, advertising, or decorative purposes.

11.3.1 floodlight (exterior lighting)

A projector designed for lighting a scene or object to a luminance considerably greater than its surroundings. It usually is capable of being pointed in any direction and is of weatherproof construction.

Note: The beam spread of floodlights may range from relatively narrow field angles (10 degrees) to wide field angles (more than 100 degrees). (See also **Section 8.6.4.1 beam angle** and **8.6.4.2 field angle**.)

11.3.1.1 beam axis of a projector

A line midway between two lines that intersect the candlepower distribution curve at points equal to a stated percent of its maximum (usually 50%).

11.3.1.2 heavy duty floodlight (HD)

A weatherproof unit having a substantially constructed metal housing into which is placed a separate and removable reflector. A weatherproof hinged door with cover glass encloses the assembly but provides an unobstructed light opening at least equal to the effective diameter of the reflector.

11.3.1.3 general purpose floodlight (GP)

A weatherproof unit so constructed that the housing forms the reflecting surface. The assembly is enclosed by a cover glass.

11.3.1.4 ground-area open floodlight (O)

A unit providing a weatherproof enclosure for the lamp socket and housing. No cover glass is required.

11.3.1.5 ground-area open floodlight with reflector insert (OI)

A weatherproof unit so constructed that the housing forms only part of the reflecting surface. An auxiliary reflector is used to modify the distribution of light. No cover glass is required.

11.3.1.6 spotlight

A form of floodlight usually equipped with lens and reflectors to give a fixed or adjustable narrow beam.

11.3.2 protective lighting

A system intended to facilitate the nighttime policing of industrial and other properties.

11.3.3 spill light

In floodlighting: The light emitted by a floodlight that is outside the floodlight distribution as defined by the field angle classification.

11.4 search lighting

11.4.1 searchlight

A projector designed to produce an approximately parallel beam of light.

Note: The optical system of a searchlight has an aperture greater than 20 cm (8 in.).

11.4.2 signal shutter

A device that modulates a beam of light by mechanical means for the purposes of transmitting intelligence.

11.4.3 elevation

The angle between the axis of a searchlight drum and the horizontal. For angles above the horizontal, elevation is positive, and below the horizontal negative.

11.4.4 train

The angle between the vertical plane through the axis of the searchlight drum and the plane in which this plane lies when the searchlight is in a position designated as having zero train.

11.4.5 horizontal plane of a searchlight

The plane that is perpendicular to the vertical plane through the axis of the searchlight drum and in which the train lies.

11.4.6 vertical plane of a searchlight

The plane through the axis of the searchlight drum that contains the elevation angle.

11.4.7 pilot house control

A mechanical means for controlling the elevation and train of a searchlight from a position on the other side of the bulkhead or deck on which it is mounted.

11.4.8 ceiling projector

A device designed to produce a well-defined illuminated spot on the lower portion of a cloud for the purpose of providing a reference mark for the determination of the height of that part of the cloud.

11.5 roadway lighting

11.5.1 street lighting luminaire

A complete lighting device consisting of a light source and ballast, where appropriate, together with its direct appurtenances such as globe, reflector, refractor, housing, and such support as is integral with the housing. The pole, post, or bracket is not considered part of the luminaire.

Note: Modern street lighting luminaires contain the ballasts for high intensity discharge lamps where such lamps are used; a photocontrol may be mounted on the luminaire.

11.5.2 street lighting unit

The assembly of a pole or lamp post with a bracket and a luminaire.

11.5.2.1 spacing (roadway)

The distance between successive lighting units, measured along the centerline of the street.

11.5.2.2 mounting height, MH (roadway)

The vertical distance between the roadway and the center of the apparent light source of a luminaire.

11.5.3 bracket (mast arm)

An attachment to a lamp post or pole, to which a luminaire is attached.

11.5.4 lamp post

A standard support provided with the necessary internal attachments for wiring and external attachments for the bracket and luminaire.

11.5.5 pole

A standard support generally used where overhead-lighting distribution circuits are employed.

11.5.6 constant current transformer

A device that automatically maintains a constant current in its secondary circuit under varying conditions of load impedance when its primary is supplied from a constant-potential source.

11.5.6.1 cutout

A device located in parallel with a series lamp that shorts when the lamp fails. The objective is to maintain circuit continuity.

11.5.7 hand hole

An accessory hole with removable external cover in a pole or standard support and utilized for accessing wiring during construction and repair.

11.5.8 isolation transformer

An auxiliary device, for use with lamps in series circuits (e.g., with street lighting) to isolate the lamp from the high voltage circuit. It may also transform current and limit peak open circuit voltage across the lamp socket.

11.5.9 lamp control

A device to turn roadway lighting on and off, either locally or remotely.

11.5.9.1 time switch

A clock-operated switch programmed to turn lighting on and off for selected periods of time. Sometimes called a *timeclock*.

11.5.9.2 astronomical time switch

A time switch control that automatically compensates for local sunrise and sunset times across the year.

11.5.9.3 photocontrol (also called light activated switch)

A photoelectric switch that controls lighting by the level of daylight illuminance. Also called a *light activated switch*.

11.5.9.4 remote control switch

A switch for roadway lighting that is controlled by a signal from a remote location.

11.5.9.5 group control

An arrangement in which a group of lamps are switched as a unit.

11.5.9.6 timeclock

[See Section 11.5.9.1 time switch.]

11.5.9.7 astronomical timeclock

[See Section 11.5.9.2 astronomical time switch.]

11.5.10 overhang

In roadway lighting: The distance between a vertical line passing through a specified point (often the photometric center) of a luminaire and the curb or edge of a roadway.

11.5.11 reference line

In roadway lighting: Either of two radial lines where the surface of the cone of maximum candlepower is intersected by a vertical plane parallel to the curb line and passing through the light-center of the luminaire.

11.5.12 width line

In roadway lighting: The radial line (the one that makes the larger angle with the reference line) that passes through the point of one-half maximum candlepower on the lateral candlepower distribution curve plotted on the surface of the cone of maximum candlepower.

11.5.13 lateral width of a light distribution

In roadway lighting: The lateral angle between the reference line and the width line, measured in the cone of maximum candlepower. This angular width includes the line of maximum candlepower.

11.5.14 longitudinal roadway line (LRL)

Any line along a roadway that is parallel to the curb line.

11.5.15 transverse roadway line (TRL)

Any line across a roadway that is perpendicular to the curb line.

11.5.16 transition lighting

In roadway lighting: Lighting gauged to compensate for visual adaptation between regions of high and low light level, such as when entering tunnels.

11.5.17 high-mast lighting

Illumination of a large area by means of a group of luminaires that are designed to be mounted in fixed orientation at the top of a high mast, generally 20 m (65 ft) or higher.

11.5.18 bollard

A luminaire having the appearance of a short, thick post; used for walkway and grounds lighting. The optical components are usually top-mounted.

11.5.19 luminance coefficient, q

In roadway lighting:

At a surface of a medium, in a given direction, under specified conditions of illumination: The quotient of the luminance of the surface element in the given direction by the illuminance on the medium, expressed by:

$$q = L/E,$$

where:

L is the luminance in cd/m^2

E is the illuminance in lx

Unit: sr^{-1}

11.5.20 reduced luminance coefficient, r

In roadway lighting:

The luminance coefficient, q , multiplied by the cube of the cosine of the angle of incidence, $(\cos \gamma)^3$, of the light on the point (see figure):

$$r = q(\cos \gamma)^3,$$

where:

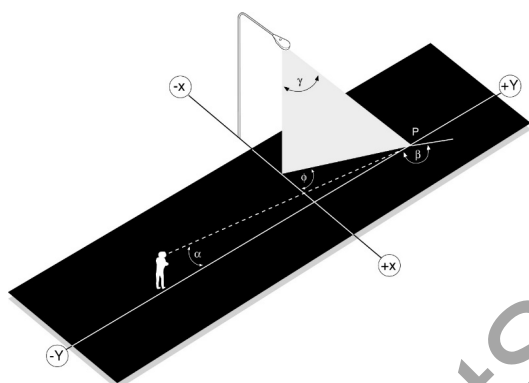
q is the luminance coefficient in sr^{-1}

γ is the angle of incidence in degrees (the road surface is flat and not sloped up-down or right-left)

Unit: sr^{-1}

Note 1: The angle of observation α , affects the value of r . By convention, this angle is fixed at 1° for roadway lighting calculations.

Note 2: The reduced luminance coefficient is used for determining the luminance of a point of infinitesimal size in the roadway luminance formula.

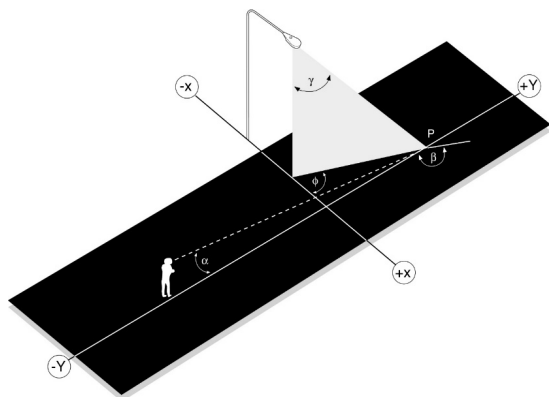


11.5.21 r -table

In roadway lighting:

A table of reduced luminance coefficient values (see Section 11.5.20) for a category of pavement. Values in the table are dependent on the variables β and $\tan \gamma$, where β is 180 minus the angle (as seen in plan view) between the line from the observer to the point of infinitesimal size and the line from the luminaire to the point, in degrees. The value of $\tan \gamma$ is the distance along the plane of the road under the luminaire to the discrete point, divided by the mounting height of the luminaire. (See figure.)

The factors given in the table have been multiplied by 10^4 . This multiplying factor, MF, is divided out in the formula for roadway luminance at the point.



11.6 vehicle lighting

11.6.1 headlamp

A major lighting device mounted on a vehicle and used to provide illumination ahead of it.

Also called a *headlight*.

11.6.1.1 multiple-beam headlamp

A headlamp so designed as to permit the driver of a vehicle to use any one of two or more distributions of light on the road.

11.6.1.2 sealed-beam headlamp

A sealed-beam lamp specifically designed for a transportation application

Note: Examples of sealed beam headlamps are automotive headlights, airplane mounted landing lights and train engine headlights.

11.6.1.3 upper (driving) beams

One or more beams intended for distant illumination and for use on the open highway when not meeting other vehicles.

Often referred to as *high beams*.

Formerly: *country beam*.

11.6.1.4 lower (passing) beams

One or more beams directed low enough on the left to limit glare in the eyes of

oncoming drivers; intended for use in congested areas and on highways when meeting other vehicles within a distance of 300 m (1,000 ft).

Often referred to as *low beams*.

Formerly: *traffic beam*.

11.6.1.5 dual headlighting system

Two double headlighting units, one mounted on each side of the front end of a vehicle. Each unit consists of two sealed beam lamps mounted in a single housing. The upper or outer lamps have two filaments, which supply the lower beam and part of the upper beam. The lower or inner lamps have one filament, which provides the primary source of light for the upper beam.

11.6.1.6 driving light

A lower-power headlamp used for daytime driving, principally as a front-viewed locator, to improve visibility in daytime driving conditions.

11.6.1.7 country beam

[See Section 11.6.1.3 upper (driving) beams.]

11.6.1.8 driving beam

[See Section 11.6.1.3 upper (driving) beams.]

11.6.1.9 passing beams

[See Section 11.6.1.4 lower (passing) beams.]

11.6.1.10 traffic beams

[See Section 11.6.1.4 lower (passing) beams.]

11.6.2 fog (adverse weather) lamps

Units that may be used in lieu of headlamps or in connection with the lower-beam headlights to provide road illumination under conditions of rain, snow, dust, or fog.

11.6.3 tail lamp

A lighting device used to designate the rear of a vehicle by a warning light.

11.6.4 stop lamp

In vehicle lighting: A lighting device giving a steady warning light to the rear of a vehicle or train

of vehicles, to indicate the intention of the operator to diminish speed or to stop. Also known as a *brake light*.

11.6.4.1 center high-mounted stop lamp (CHMSL)

An automotive stop lamp so mounted in order to alert the following vehicle. It is used in addition to other stop lamps in the brake system.

11.6.5 parking lamp

A lighting device placed on a vehicle to indicate its presence when parked.

11.6.6 turn-signal operating unit

That part of a signal system by which the operator of a vehicle indicates the direction a turn will be made, usually by a flashing light.

11.6.7 back-up lamp

A lighting device mounted on the rear of a vehicle for illuminating the region near the back of the vehicle while moving in reverse. It normally can be used only while backing up.

11.6.8 clearance lamps

Lighting devices for the purpose of indicating the width and height of a vehicle.

11.6.8.1 side marker lamps

Lamps indicating the presence of a vehicle when seen from the front and sometimes serving to indicate its width. When seen from the side they may also indicate its length.

11.6.9 retro-reflector (reflex reflector)

A device designed to reflect light in a direction close to that at which it is incident, whatever the angle of incidence.

11.6.10 adverse-weather lamp

[See **Section 11.6.2 fog (adverse weather) lamps**.]

11.6.11 headlight

(See **Section 11.6.1 headlamp**.)

11.6.12 reflex reflector

[See **Section 11.6.9 retro-reflector (reflex reflector)**.]

11.7 Obtrusive Light

Direct or reflected light that, because of quantitative, directional or spectral attributes in a given context, causes annoyance, discomfort, distraction or a reduction in the ability to see.

11.7.1 light trespass

The encroachment of light, typically across property boundaries, causing annoyance, loss of privacy, or other nuisance.

11.7.2 lighting zone

In exterior lighting: One of several categories (descriptive and/or prescriptive) that designate outdoor ambient light levels. (Refer to lighting zone guidelines in the Model Lighting Ordinance User Guide, <http://www.darksky.org/our-work/public-policy/mlo/>.)

11.7.3 sky glow

The brightening of the night sky that results from the scattering and reflection of light from the constituents of the atmosphere (gaseous molecules and aerosols), in the direction of the observer. It has two separate components: natural sky glow and artificial sky glow.

11.7.3.1 natural sky glow

That part of sky glow which is attributable to natural sources. It is composed of starlight, zodiacal light (scattering of sunlight from dust in the solar system), and airglow (radiation from luminescent processes in the earth's upper atmosphere).

11.7.3.2 artificial sky glow

That part of sky glow which is attributable to scattering of light from human-made sources of radiation (e.g., outdoor electric lighting), including radiation that is emitted directly upward and radiation that is reflected from surfaces.

12.0 NONLIGHTING APPLICATIONS OF RADIANT POWER

12.1 ultraviolet radiation

Radiant energy within the wavelength range 10 to 400 nm. (See also **Section 2.15 regions of electromagnetic spectrum** for nominal subdivisions of the ultraviolet spectrum.)

12.1.1 ozone-producing radiation

Ultraviolet energy shorter than about 220 nm, which decomposes oxygen (O₂) thereby producing ozone (O₃). Some ultraviolet sources generate energy using the 184.9-nm mercury

discharge line, which is particularly effective in producing ozone.

12.1.2 erythema

The abnormal redness due to an over-accumulation of blood in the capillaries of the skin. Ultraviolet-induced erythema is due to actinic action (see **Section 12.3.1**) and is a delayed effect occurring several hours after exposure. This differs from infrared-induced erythema, a thermal effect occurring only for the duration of time that the skin temperature is elevated.

Note: The degree of erythema is sometimes used as a guide to dosages applied in ultraviolet therapy.

12.1.2.1 erythema radiation

Optical radiation effective actinically in causing skin reddening.

12.1.2.2 minimal perceptible erythema, MPE

The least erythema that can be detected (under specified viewing conditions).

12.1.2.3 erythema effectiveness

The capacity of various portions of the ultraviolet spectrum to produce erythema.

12.1.2.4 erythema action spectrum

$S_{\text{er}}(\lambda)$

The standardized relative spectral effectiveness in producing erythema (evaluated 24 hour post-exposure) on sensitive human skin of type 1 (always burn, never tan) and type 2 (usually burn, sometimes tan).

$$S_{\text{er}}(\lambda) = 1.0 \quad \text{for } 250 \leq \lambda \leq 298 \text{ nm}$$

$$S_{\text{er}}(\lambda) = 10^{0.094(298-\lambda)} \quad \text{for } 298 < \lambda \leq 328 \text{ nm}$$

$$S_{\text{er}}(\lambda) = 10^{0.015(140-\lambda)} \quad \text{for } 328 < \lambda \leq 400 \text{ nm}$$

Note: The absolute dose required to produce a given degree of erythema varies enormously in the range of 250 nm to 298 nm, especially below 280 nm. Consequently, for the sake of simplicity and as a "reference" to compare erythema efficacy of different UV sources, a uniform spectral sensitivity was adopted for wavelengths below 298 nm.

12.1.2.5 erythema flux

Radiant power evaluated according to its capacity to produce erythema of the untanned

human skin; i.e., in accordance with the erythral effectiveness of the flux.

Unit: watt, W. (See also *erythral action spectrum*.)

Note: An obsolete term (retained for reference purposes only) of erythral flux is the erythral unit (EU) or E-viton (erythme) which is equal to the amount of radiant flux that will produce the same erythral effect as ten microwatts of radiant flux at wavelength 296.7 nm.

12.1.2.6 -erythral flux density

Erythral flux per unit area.

Unit: W/m².

Note: An obsolete unit (retained for reference purposes only) of erythral flux density is the finsen which is equal to one E-viton (see **Section 12.1.2.5**) per square centimeter.

12.1.2.7 erythral dose

Time integrated erythral flux density.

Unit: J/m².

Note: *Erythral exposure* is an obsolete term (retained for reference purposes only) for *erythral dose*.

12.1.2.7.1 minimum erythral dose (MED)

A subjective measure based on the reddening of the skin; it depends on many variables including individual sensitivity to UV, radiometric characteristics of the source, skin pigmentation, anatomic site, and the elapsed time between irradiation and the observation of reddening (typical value is for 24 hours post-exposure). MED is reserved solely for observational studies in humans and other animals.

Unit: J/m².

(See also **Section 12.1.2.7.2 standard erythral dose**.¹¹⁾

12.1.2.7.2 standard erythral dose (SED)

A standardized measure of erythemogenic UV radiation. One SED is equivalent to an erythemally effective radiant exposure of 100 J/m² when the radiation is weighted by the erythra action spectrum normalized to its maximum, $S_{er}(\lambda)$.

(See also **Section 12.1.2.4 erythral action spectrum**.¹¹⁾

12.1.2.8 erythema threshold

(See Section 12.1.2.2 minimal perceptible erythema.)

12.1.2.9 sunburn

Inflammation with reddening (*erythema*) of the skin of variable degree caused by exposure to direct or diffuse solar radiation or other sources of ultraviolet radiation.

12.1.2.10 suntan

A darkening of the skin due to an increase of melanin pigmentation above constitutive level and induced by ultraviolet radiation.

12.1.2.10.1 solarium

A room (as in a hospital) or device for exposure of the body to natural or simulated solar radiation (as in sunbathing).

12.1.2.11 erythema unit (EU)

(See Section 12.1.2.5 erythema flux.)

12.1.2.12 erythema exposure

(See Section 12.1.2.7 erythema dose.)

12.1.2.13 E-viton (erythema)

(See note under Section 12.1.2.5 erythema flux.)

12.1.2.14 finsen

(See Section 12.1.2.6 erythema flux density.)

12.1.3 germicidal functions

Strictly, those functions relating to inactivation of pathogens, including bacteria, fungi, protozoa, and viruses.

12.1.3.1 ultraviolet germicidal irradiation (UVGI)

The use of ultraviolet radiant energy to inactivate bacteria, mold spores, fungi, protozoa, and/or viruses.

12.1.3.1 germicidal radiation

Optical radiation capable of inactivating pathogens.

12.1.3.2 germicidal effectiveness

The capacity of optical radiation to inactivate bacteria, fungi, protozoa, and/or viruses.

12.1.3.3 germicidal action spectrum

The effectiveness of optical radiation as a function of wavelength to inactivate a particular type of pathogen to a specified criterion level under a given set of conditions.

12.1.3.4 germicidal flux

Radiant power evaluated according to its capacity to produce germicidal effects; i.e., in accordance with the germicidal effectiveness of the flux.

Unit: watt, W.

(See also *germicidal action spectrum*.)

Note: Because of the ubiquitous low-pressure mercury germicidal lamp whose principal radiation in the germicidal band is at 253.7 nm, 253.7-nm flux often is described as *germicidal flux*. This should not be confused with the term as defined here, because it has not been weighted in accordance with the germicidal action spectrum.

12.1.3.5 germicidal flux density

Germicidal flux per unit area.

Unit: W/m².

12.1.3.6 germicidal dose

Time integrated *germicidal flux density*.

Unit: J/m².

12.1.4 black light

The lay term for ultraviolet energy near the visible spectrum, typically in the range of 320 to 400 nm.

Note: For engineering purposes, the wavelength range 320 to 400 nm has been found useful for rating lamps and their effectiveness upon fluorescent materials (excluding phosphors used in fluorescent lamps).

12.1.4.1 black light flux

Radiant flux within the wavelength range 320 to 400 nm.

Unit: watt, W.

Note: The *floren* is an obsolete unit (retained for reference purposes only) of black light flux and is equal to one milliwatt of radiant flux in the wavelength range 320 to 400 nm. Because of the variability of the spectral sensitivity of materials irradiated by black light in practice, no attempt is made to evaluate black light flux according to its capacity to produce effects.

12.1.4.2 black light flux density

Black light flux per unit area. Unit: W/m².

12.1.4.3 glow factor

A measure of the visible light response of a fluorescent material to black light. It is equal to π times the luminance in candelas per square meter produced on the material, divided by the incident black light flux density in milliwatts per square meter. It may be measured in lumens per milliwatt.

Note: π was introduced to maintain the same numerical glow factor values when the system involving footlamberts was declared obsolete. π is omitted when luminance is in footlamberts and the area is in square feet.

12.1.4.3 floren

(See **Section 12.1.4.1 black light flux.**)

12.2 infrared radiation

For practical purposes, any radiant energy within the wavelength range 770 to 1,000,000 nm is considered infrared energy. (See **Section 2.15**).

12.3 photochemical radiation

Energy in the ultraviolet, visible, and infrared regions that produce chemical changes in materials.

Note: Examples of photochemical processes are accelerated fading tests, photography, photoreproduction, and chemical manufacturing. In many such applications a specific spectral region is of importance.

12.3.1 actinic

Of or pertaining to radiation capable of producing a photochemical effect.

Also: A term referring to photochemical effects.

12.3.1.1 direct actinic effect (of optical radiation)

A chemical effect of optical radiation directly upon or mediated by the absorbing atom(s) or molecule(s) of a substance or living organism.

12.3.1.2 indirect actinic effect (of optical radiation)

An effect of optical radiation absorbed by a living organism that is mediated by a change in specialized photoreceptor cells.

12.4 law of reciprocity (Bunsen-Roscoe law)

Where a photochemical reaction exhibits reciprocity between irradiance (exposure dose rate, W/m^2) and exposure duration over some range of time. In other words, a given response results from a given radiant exposure (J/m^2) over a wide range of exposure times. Reciprocity may also fail. For very short duration, failure may be due to photon saturation. For long duration, failure may be due to repair mechanisms or recombination.

12.5 action spectrum

A function representing the relative spectral effectiveness of optical radiation, for a specified biological effect, in a specified system.

12.6 dose (optical radiation)

Quantity of actinically effective optical radiant energy effectively absorbed per unit of exposed area.

Unit: J/m^2 .

12.6.1 incident dose (exposure dose)

Quantity per unit exposed area of incident, actinically effective optical radiant energy.

Unit: J/m^2 .

12.6.2 exposure dose rate

Exposure dose per unit time of actinically effective irradiance.

Unit: W/m^2 .

12.7 photosensitization

The quality or state of system being made susceptible to actinic effects of optical radiation through the agency of another substance or system.

12.8 photobiology

A branch of biology that deals with the effects of optical radiation on living systems.

12.8.1 photoperiod

The environmental natural or artificial cycle of light and dark(er) to which living organisms may be exposed.

For example, the natural cycle at the earth's equator of light (L) for 12 hours, and darkness (D) for 12 hours, which would be expressed as LD12:12.

12.8.1.1 photoperiodism

Response of organisms to length of the night and day periods, and changes in these periods with progression of the seasons.

12.8.1.2 diurnal

Pertaining to the daylight hours, as opposed to *nocturnal*.

12.8.1.3 nocturnal

Pertaining to the hours of darkness, as opposed to *diurnal*.

12.8.1.4 circadian

Relating to or showing rhythmic behavior with a period of 24 hours.

12.8.2 phototherapy

The treatment of disease involving the use of optical radiation.

Note: The terms *photochemotherapy* and *photodynamic therapy* often are used when chemical sensitizing agents are used in conjunction with optical radiation.

12.8.2.1 heliotherapy

Treatment of disease by the use of solar irradiation.

12.8.3 biological rhythm

A characteristic period change in a living organism or life-related process.

Note: Some biological rhythms are induced and/or synchronized by light.

12.8.4 photocarcinogenesis

Carcinogenesis resulting from exposure to ultraviolet radiation.

12.8.5 photomorphogenesis

The property of plants that enables them to utilize radiant energy to control the growth and development of their cells, tissues, and organs.

12.8.6 photomutagenesis

The production of mutations induced by exposure to optical radiation.

12.8.7 photosynthesis

Conversion of light energy to energy in chemical bonds in plants, involving assimilation of carbon dioxide into simple sugars.

12.8.7.1 photosynthetic irradiance

Irradiance within the wavelength band 400 to 700 nm.

Unit: W/m^2 .

12.8.7.2 photosynthetically active radiation (PAR)

The spectral range (waveband) of radiation, from 400 to 700 nm, which by definition photosynthetic organisms are able to use in the process of photosynthesis. The measure result of PAR can be reported as photosynthetic photon flux (PPF) or photosynthetic photon flux density (PPFD).

12.8.7.3 photosynthetic photon flux density (PPFD)

The number of photons per unit time and per unit area in the spectral range (waveband) 400 to 700 nm.

Unit: $\mu\text{mol/s}\cdot\text{m}^2$.

Note: There are 6.0222×10^{23} photons (Avogadro's number) in one mole of photons.

12.8.8 phototaxis

The property of mobile organisms to move in response to light. Positive phototaxis involves movement toward a light source, and negative phototaxis away from a light source.

12.8.9 phototropism

A change in direction of growth of plant parts (cells, tissues, and/or organs) in response to a directional source of light.

12.8.10 photochemotherapy

(See **Section 12.8.2 phototherapy**.)

12.8.11 photodynamic therapy

(See **Section 12.8.2 phototherapy**.)

12.9 photobiological hazards

12.9.1 photobiological hazard

The possible risk of biological injury due to optical radiation.

12.9.2 aversion response

The response to a stimulus that causes one to avoid or turn from the stimulus.

12.9.3 retinal hazard region

The spectral region from 400 nm to 1,400 nm, within which the normal ocular media transmit clinically significant optical radiation to the retina.

12.9.3.1 retinal burn

A photochemical or thermal retinal lesion.

12.9.3.2 blue-light hazard

Potential for photochemically induced injury to the retina of the eye resulting from radiant exposure at wavelengths principally between 400 nm and 500 nm.

Note: For non-laser sources, it is primarily due to fixating on the source with high radiance in this spectral band. Such injury commonly leads to permanent loss of vision in the region where the source is focused.

12.9.3.3 photoreinitis

Inflammation of the retina of the eye due to radiant power.

12.9.4 photokeratoconjunctivitis

Inflammatory response of the cornea and conjunctiva of the eye following exposure to ultraviolet radiation. Wavelengths shorter than 320 nm are most effective in causing photokeratoconjunctivitis. The peak of the action spectrum is in the vicinity of 270 nm.

12.9.4.1 photokeratitis

Inflammation of the cornea of the eye caused by exposure to UV.

Also known as *actinic keratitis*.

12.9.4.2 photoconjunctivitis

Inflammation of the conjunctiva of the eye caused by exposure to UV.

Also known as *actinic conjunctivitis*.

12.9.5 photocataractogenesis

Clouding (opacification) of the lens in the eye as a result of lifelong exposure to excessive radiant energy in the ultraviolet or infrared parts of the optical spectrum.

Note: The classic "glass-blower's cataract" is an example of an infrared heat cataract, whereby an individual had been exposed to high-temperature furnaces for years without adequate eye protection.

12.9.6 photobiological (human) hazard function

A relative spectral weighting function proportional to the efficiency of monochromatic radiation for producing photobiological injury.¹⁶

12.9.6.1 ultraviolet hazard function, $S(\lambda)$

A function in the ultraviolet spectrum for evaluating risk of acute injury to the eyes (photokeratoconjunctivitis). For convenience in safety analysis, this function is defined to include acute risk to the skin (erythema); there is a de facto safety factor for skin because of its lower sensitivity.

12.9.6.2 blue-light hazard function, $B(\lambda)$

A function in the visible spectrum for evaluating risk of acute photochemically induced injury to the retina (the blue-light hazard).

12.9.6.3 aphakic hazard function, $A(\lambda)$

An extension of the blue-light hazard function into the 300-nm to 400-nm region of the ultraviolet for evaluating the aphakic (lens-less) eye.

12.9.6.4 retinal burn hazard function, $R(\lambda)$

A function in the visible and near infrared spectrum for evaluating risk of acute thermally induced injury to the retina.

12.9.7 radiant exposure

The time-integrated irradiance at a surface. Units: joules per square meter (J/m^2) or millijoules per square centimeter (mJ/cm^2).

12.9.7.1 radian exposure rate

Irradiance at a surface.

Unit: watts per square meter, W/m^2 .

12.9.7.2 exposure limit, EL

The maximum value of exposure to the eye or skin that is not expected to result in adverse biological effects.

12.9.8 risk group classification (of lamps)

A system of evaluating and labeling lamps in terms of their potential for causing injury due to radiation in the region of the spectrum from 200 nm to 3,000 nm.

12.9.8.1 light source description (for risk group classification)

Four light source descriptions are included: continuous wave (CW), lamp pulsed lamp, general lighting source lamp (GLS), and non-general lighting source lamp.

12.9.8.1.1 continuous wave (CW) lamp

A lamp that is operated with a continuous output for a 0.25-second period or greater.

12.9.8.1.2 pulsed lamp

A lamp that delivers its energy in the form of a single pulse or a train of pulses, where each pulse shall have duration of less than 0.25 s.

12.9.8.1.3 general lighting source lamp (GLS)

A general term for a continuous wave (CW) lamp intended for lighting spaces that are typically occupied or viewed by people.

Used to differentiate from non-general lighting sources (see **Section 12.9.8.1.4**) in the process of determining risk group classification.

12.9.8.1.4 non-general lighting source lamp

A category of lamps for which assumptions about typical conditions of use are uncertain, and which are more likely to represent a significant hazard under a

broad range of use conditions.

12.9.8.2 Exempt Group

The philosophical basis is that the lamp does not pose any photobiological hazard under the classification of general light source lamp (see **Section 12.9.8.1.3**) or of non-general light source lamp (see **Section 12.9.8.1.4**).

12.9.8.3 Risk Group 1 (low risk)

The philosophical basis is that the lamp does not pose a hazard, due to normal behavioral limitations on exposure.

12.9.8.4 Risk Group 2 (moderate risk)

The philosophical basis is that the lamp does not pose a hazard, due to the aversion response to very bright light sources or due to thermal discomfort.

12.9.8.5 Risk Group 3 (high risk)

The philosophical basis is that the lamp may pose a hazard even for momentary or brief exposures.

12.10 photocuring

A technique employing optical (typically ultraviolet) radiation to irradiate a liquid polymer resin and thereby harden it by means of photopolymerization.

12.11 reciprocity

[See **Section 12.4 law of reciprocity (Bunsen-Roscoe law)**.]

13.0 LIGHTING CONTROLS

13.1 lighting zone

In interior lighting controls: Any number of individual luminaires in a space that, when combined, form a logical control group.

Sometimes called a *control channel* or a *lighting control zone*.

13.2 Automatic Controls

13.2.1 adaptive lighting

Lighting with the ability to automatically adjust light level and spectral content based on

environmental conditions in order to optimize space, human, and/or building performance.

13.2.2 occupancy sensor

A control device that detects occupant presence and automatically turns luminaires and/or other equipment on and, after a preset delay during which no presence is detected, turns them off.

Also called a *motion detector*.

13.2.3 vacancy sensor

A type of occupancy sensor that requires the luminaires to be turned on manually. (See also **Section 10.7.3 occupancy sensor**.)

13.2.4 motion detector

[See **Section 13.2.2 occupancy sensor**.]

13.2.5 photosensor

A device that generates an electric signal proportional to the radiation incident on a light-sensitive material.

13.2.6 field of view (FOV)

In lighting control systems: The solid angle from which a sensor receives a signal.

13.2.7 photosensor control algorithm

One or more functions that translate the signal from a photosensor to a control signal that adjusts the electric lighting or shading devices.

13.2.8 bi-level lighting control

A control that includes two different levels of light output, plus off.

13.2.9 multilevel lighting control

Control of light output of a source or luminaire, with ranges that include at least two levels of light output, plus off.

13.2.10 closed-loop control

For daylighting, a photosensor-based method for electric lighting and/or shading system control where the outcome of the controlled system impacts the photosensor signal.

13.2.11 open-loop control

For daylighting, a photosensor-based method for electric lighting and/or shading system control, where the outcome of the controlled system has minimal impact on the photosensor signal.

13.2.12 dual-loop photosensor control

A photosensor-based control of electric lighting or shading devices that uses both closed-loop and open-loop sensing.

13.3 tunable lighting

The ability to alter the luminous flux and/or spectral power distribution of a lamp or luminaire.

13.3.1 spectral tuning

Adjustment of a light source spectrum in response to task, space, or occupant needs. Sometimes referred to as *color tuning*.

13.3.2 dimming

The ability to change the luminous flux of one or more lamps or luminaires.

13.3.3 task tuning

The practice of adjusting the maximum luminous flux and/or the spectral power distribution of the luminaires in a space to a desired target illuminance and/or spectrum in support of the visual or non-visual needs of the occupants and/or to save energy.

13.3.4 personal tuning

The ability of occupants in a space to adjust the luminous flux and/or spectral power distribution of luminaires according to their personal preferences.

13.3.5 color tuning

Color tuning is one use of *spectral tuning*.

13.3.6 institutional tuning

Adjustment of the maximum light level and/or the spectrum in a space, set by the owner, tenant, or designer.

13.3.7 high-end trim

The capability of a lighting control system to limit the maximum light output to a level below its maximum factory setting.

13.3.8 dim to warm

The process of simultaneously dimming a light source and lowering its correlated color temperature (CCT).

Note: This is sometimes referred to as "warm dim" and is done to imitate the spectral characteristics of dimmed incandescent light sources.

13.3.9 tunable white

A form of spectral tuning used to vary the correlated color temperature (CCT) of a white light source.

Note: Other terms used for this concept include "white tunable" and "CCT tunable."

13.4 lighting system

A collection of daylighting and/or electric lighting equipment that function together to generate, deliver and control the light in an application.

13.5 networked lighting control system

A lighting control system with multiple components that are interconnected, permitting one-way or two-way communication with the devices.

13.6 luminous flux maintenance control

A lighting control strategy that increases light source power over time to maintain light levels as sources age, dirt accumulates in luminaires, or both.

Note: This strategy allows for energy savings early in the life of a system.

Also known as *lumen maintenance control*, or *lumen depreciation compensation*, or *constant lumen output*.

13.6.1 lumen maintenance control

(See **Section 13.6 luminous flux maintenance control**.)

13.6.2 constant lumen output

(See **Section 13.6 luminous flux maintenance control**.)

13.6.3 lumen depreciation compensation

(See **Section 13.6 luminous flux maintenance control**.)

13.7 load shedding

A control strategy for selectively reducing the load of a system on a temporary basis to reduce energy usage.

Note: A building manager or utility may utilize load shedding to avoid peak pricing or to avoid a condition where electricity demand exceeds supply.

13.8 photocell

A solid-state device that converts light into electrical energy by producing current, as in a photovoltaic cell, or uses light to regulate the flow of current, as in a photoconductive cell.

T.0 – TABLES

T.1 Luminous Conversion Factors

[insert Table T-1]

T.2 Standard Units, Symbols, and Defining Equations for Fundamental Photometric and Radiometric Quantities*

[insert Table T-2]

T.3 Illuminance Conversion Factors

[insert Table T-3]

T.4 Definitive Values of the Special Luminous Efficiency Function for Photopic Vision $V(\lambda)$

Note: Information in this table was obtained from CIE 18.2 (1983)

[insert Table T-4]

T.5A Color-Matching Functions and Chromaticity Coordinates of CIE 1931 Standard Colorimetric Observer

Note: Information in this table was obtained from Table 1 of ISO/CIE 10527-1991, *Colorimetric Observers*.

[insert Table T-5A]

T.5B Color-Matching Functions and Chromaticity Coordinates of CIE 1964 Supplementary Standard Colorimetric Observer

Note: Information in this table was obtained from Table 2 of ISO/CIE 10527-1991, *Colorimetric*

Observers.

[Insert Table T-5B]

T.6 Relative Spectral Power Distribution of CIE Illuminant “C”

Note: Information in this table was derived from Table 1.1 of CIE 15.2: 2004, *Colorimetry*, 3rd ed.

[Insert Table T-6]

T.7 Scotopic Spectral Luminous Efficiency Values, $V'(\lambda)$ (Unity at Wavelength of Maximum Luminous Efficacy)

[Insert Table T-7]

T.9 CIE 1988 Modified Two-Degree Spectral Luminous Efficiency Function for Photopic Vision

Note: Information in this table was obtained from Appendix of CIE 86 (1990), *CIE 1988 2° Spectral Luminous Efficiency Function for Photopic Vision*.

Note: This function is identical to $V(\lambda)$ at wavelengths of 460 nm and longer.

[Insert Table T-9]

ANNEX A – PHYSICAL RECEPTORS AND DEVICES

Commented [DD1]: Keep and update, or reject??

[A.A*] (This informative annex is not part of ANSI/IES RP-16-17; it is included to provide definitions from other sources that the user may encounter.)

alphanumeric display (digital display). An electrically-operated display of letters and/or digits. Tungsten filaments, gas discharges, light emitting diodes, liquid crystals, projected numerals, illuminated numbers, fluorescent screens, and other principles of operation may be used.

Bolometer. Thermal receptor in which the heating of the part that absorbs the radiation gives rise to a change in its electrical resistance.

light controlled silicon controlled rectifier, SCR (also called photothyristor). Photoelectric receptor utilizing a four-layer p-n-p-n junction rectifier whose "on," or conducting state, is controlled by absorbed radiation.

liquid crystal display, LCD. A display made of material whose reflectance or transmittance changes when an electric field is applied.

numerical display (digital display). An electrically operated display of digits. Tungsten filaments, gas discharges, light-emitting diodes, liquid crystals, projected numerals, illuminated numbers, and other principles of operation can be used.

photoconductive cell. Photoelectric receptor utilizing a semiconductor in which absorption of radiation gives rise to an increase in conductivity by the internal photoelectric effect.

photodiode. Photoelectric receptor in which absorption of radiation in the neighborhood of a p-n junction between two semiconductors, or of a contact between a semiconductor and a metal, produces a change of resistance depending on the current direction.

photoelectric receiver. A device that reacts electrically in a measurable manner in response to incident radiant energy.

Phototransistor. Photoelectric receptor utilizing semiconductors in which the photoelectric effect is produced in the neighborhood of a double p-n junction (p-n-p or n-p-n) possessing amplification properties.

Note: A phototransistor may be combined with other transistors in a single case or on a "chip" in order to increase the overall amplification.

Phototriac. Photoelectric receptor utilizing a symmetrical silicon controlled rectifier (SCR) or double-ended thyristor of five-layer semiconductor construction for AC application whose "on," or

* The section number for this annex in the online version.

conducting, state is controlled by absorbed radiation.

phototube (photoemissive cell). Photoelectric receptor comprising a vacuum or gas-filled electronic tube containing a cathode capable of emitting electrons by the external photoelectric effect.

photovoltaic cell. Photoelectric receptor in which absorption of radiation in the neighborhood of a p-n junction between two semiconductors, or of a contact between a semiconductor and a metal, produces an electromotive force.

physical receptor (of radiation). Instrument in which a measurable physical effect is produced by the action of the radiation it receives.

pyroelectric detector. A radiation detector in which the sensitive element is a thin layer of a pyroelectric material. Such a material changes polarization when heated, and thus develops a charge across its thickness, which causes a current to flow between electrodes on the two surfaces.

quantum efficiency (of a detector) (η). Ratio of the number of elementary events (such as release of an electron) contributing to the detector output, to the number of incident photons.

(radiation) thermocouple. Thermal receptor in which the absorbed radiation produces, by the heating of a contact between two conductors of different kind, an electromotive force by the thermoelectric or Seebeck effect.

(radiation) thermopile. Thermal receptor comprising a number of thermocouples.

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